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Interventions to improve infant and young child feeding practices in Ethiopia: a systematic review

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Interventions to improve infant and young child feeding practices in Ethiopia: a systematic review

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Abstract

Objective: In Ethiopia, despite the implementation of several interventions to improve IYCF practices, no published studies have highlighted the most effective IYCF interventions in the country. This systematic review investigated the impacts of various interventions on IYCF in Ethiopia.

Source of information: A systematic search was conducted on seven computerized bibliographic databases (Embase, ProQuest Central, Scopus, PsycINFO, Web of Science, MEDLINE/PubMed and CINAHL) to locate experimental or quasi-experimental studies.

Study eligibility criteria: Interventional studies that measured IYCF indicators (early initiation of breastfeeding [EIBF], exclusive breastfeeding [EBF], the introduction of complementary foods, minimum dietary diversity [MDD], minimum meal frequency [MMF] and minimum acceptable diet [MAD]) as outcome variables were included.

Study appraisal and synthesis: All included studies examined for biases of interventional studies. Author reports of effect size measures were used to narratively report the findings of each study.

Results: Of the thirteen eligible studies, ten studies were quasi-experimental and three studies were cluster randomised trials. Five studies showed that interventions in the form of policy advocacy, health service strengthening, interpersonal communication, community mobilisation, and mass media campaigns improved EBF. Four studies indicated that community-level and health facility breastfeeding promotions increased EIBF. Three studies for MDD and two studies for MMF indicated significant effects of policy advocacy,

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interpersonal communication, community mobilisation, and mass media campaigns on infants and young children. Interventions that were delivered in combination increased the impacts in improving EIBF, MDD and MMF compared to single intervention.

Conclusion: Our review showed that a combination of community-level interventions improved EIBF, EBF, MDD, and MMF in Ethiopia. The improvement in IYCF in Ethiopia would require multi-prong approaches at the household and community levels along the continuum of care from conception until a child's second birthday.

Keywords: Infant and young child feeding, breastfeeding, nutritional education, intervention, systematic review, Ethiopia

Protocol registration number: PROSPERO CRD42020155519.

Strengths and limitations of the study

- A meta-analysis was not conducted due to the wide heterogeneity of the design and outcome measures.
- This review was limited by the incompleteness of information from the individual studies. Nevertheless, the study authors were contacted for information when it is necessary.
- Despite the limitations, the application of multiple search strategies in consultation with the medical librarian and registration of the systematic review protocol can be considered as strengths of this review.

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Introduction

Appropriate infant and young child feeding (IYCF) improves childhood growth (through reduced risk of diarrhoea and respiratory infections as well as improved nutrition) [1-7], increases the intellectual capacity of children [4 8], and reduces the risk of mothers to experience diabetes mellitus, overweight and obesity [4 9]. Promoting appropriate IYCF using evidence-based interventions is essential to protect and support maternal knowledge, beliefs, and confidence in IYCF practices [10-13]. Community- and facility-based interventions that range from direct support and education to mothers and families on appropriate IYCF behaviours to training of health professionals can significantly improve IYCF behaviours [14-16].

Global efforts to improve maternal IYCF behaviours have increased in the past three decades. These efforts included the International Code of Marketing of Breast-milk Substitutes [17]; the Innocenti Declaration [18]; the Baby-Friendly Hospital Initiative (BFHI) [19]; the Millennium Development Goals (MDGs) [20]; and more recently, the Global Nutrition Targets 2025 [21] and the Sustainable Development Goals (SDGs) [22]. The World Health Organization and United Nations Children's Fund (WHO/UNICEF) have played major roles in driving these initiatives which aimed to protect, promote, and support breastfeeding and child nutrition [19]. For example, the WHO Global Nutrition Targets (WHO GNT, including Target– 1 and 5) aim to reduce the global number of stunted children under-5 by at least 40% and increase the EBF rate to at least 50% by the year 2025, respectively. However, most subnational areas of many African countries (including Ethiopia) are not on track to achieve WHO GNT goal–5 [23].

In Ethiopia, while both government and non-governmental organisations have implemented various maternal and child health-related policy interventions over the last two decades to

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increase infant feeding [24-26], a recent study has revealed that early initiation of breastfeeding (EIBF) prevalence was 75.5% and that for exclusive breastfeeding (EBF) was 59.9% [27], with a wide heterogeneity across subnational areas of the country [23]. These rates are well below the Ethiopian Health Sector Transformation Plan (HSTP) target of 90% and 72%, respectively [25]. Additionally, the proportion of children who met minimum dietary diversity (MDD) and minimum meal frequency (MMF) was 7.1% and 43.6%, respectively [28]. In Ethiopia, more than five million children were stunted in 2016 [29], and this is a major public health issue that is largely attributable to inappropriate IYCF.

To date, no previously published studies have examined the most effective IYCF interventions in the Ethiopian context. A detailed understanding of the specific types of interventions that can be implemented to improve IYCF in Ethiopia is essential, to specifically guide the allocation of scarce resources and the provision of targeted strategies to increase childhood feeding. This information will also be helpful in scaling-up and/or improve upon current and future IYCF interventions in the country. Data from the assessment of relevant IYCF interventions would be crucial to stakeholders as Ethiopia implements national nutrition efforts (e.g., HSTP) [25] and global health programs such as the Sustainable Development Goal, SDG-2.2, to end all forms of malnutrition [22]. Accordingly, this systematic review investigated the impacts of various IYCF interventions on Ethiopian children under two years of age.

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Materials and methods

Our review adhered to the 2009 Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [30], and the protocol for the review was registered in PROSPERO (International Prospective Register of Systematic Reviews, CRD42020155519) [**PRISMA checklist**] [31].

Information sources and search strategy

Consistent with previously published systematic reviews [15 32], a three-stage search strategy was applied to locate both published and unpublished studies. In stage one, a manual search of previously published systematic reviews and the reference lists of all the eligible studies was performed to check for systematic reviews of interventions on IYCF in Ethiopia. In stage two, index terms (e.g., medical subject headings [MeSH] for Medline) and key terms were generated using the identified articles from the initial search for the full search strategy. In stage three, a full search strategy was conducted on seven computerized bibliographic databases (Embase, ProQuest Central, Scopus, PsycINFO, Web of Science, MEDLINE [OVID]/PubMed and Cumulative Index to Nursing and Allied Health Literature, CINAHL) (EBSCO) to locate published studies, and secondary searching of OpenGrey and Grey Literature Report was conducted to locate unpublished works. The Population Intervention Comparator Outcome (PICO) criteria were used to devise the review question and related search terms. A combination of specified medical subject headings (MeSH) terms and keywords was drafted with some assistance from our institution-based medical librarian. The search strategy, including all identified index and keywords terms, was slightly adapted for each of the information sources. The initial and the top-up searches were conducted in November 2019, and June 2020, respectively. The search terms/keywords used included:

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Term 1 (participants): Child OR infant OR female OR caregiver OR women OR community health workers OR health extension worker AND

Term 2 (feeding): Breastfeed OR infant food OR meals OR diet OR feeding behaviour OR healthy diet OR child nutritional physiological phenomena AND

Term 3 (interventions): child health services OR counselling OR dietary supplements OR health education OR food supply OR health promotion OR prenatal care OR postnatal care OR early intervention OR community health planning OR models/educational OR impact OR influence OR effect AND

Term 4 (context): Ethiopia

Eligibility criteria

Studies were included if they meet the following criteria: (i) experimental or quasi-experimental study design (including randomized controlled trials [RCT], non-RCTs, before and after studies, and interrupted time-series studies) with or without comparison groups or cluster; (ii) conducted among pregnant women and mothers living with live children under 2 years to improve IYCF practices; (iii) used any type of IYCF interventions delivered between the antenatal period and 24 months postpartum; (iv) measured at least one core IYCF indicators such as EIBF, EBF, the introduction of solid, semi-solid and soft foods, continued breastfeeding at one year, MDD, MMF, and minimum acceptable diet (MAD); (v) were published in the English language; and (vi) conducted in Ethiopia from the year 2000 to date. We considered studies published over this period to demonstrate the impacts of the specific IYCF interventions in the context of Ethiopia to reflect the implementation of relevant global health programs, including the MDGs and the current SDGs [23].

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The intervention strategies considered in this review focused on those performed in both facility- and community-levels. The interventions included nutritional education and counselling, interpersonal communication, mass media campaigns, nutrition-sensitive agricultural activities, group demonstration sessions, and strengthening of health facilities. Studies conducted among mothers with health problems such as HIV/AIDS and children with cleft palate were excluded.

The IYCF practices included EIBF, EBF, the introduction of solid, semi-solid and soft foods, continued breastfeeding at one year, MDD, MMF, and MAD, measured according to the WHO definitions for assessing IYCF indicators [33]. These indicators were selected based on their effect on the health and development of infants and young children [11, 12, 35] and the evidence from previously published systematic reviews [11 34].

- EIBF was defined as the proportion of children aged 0–23 months who commenced breastfeeding within the first hour of birth.
- EBF was defined as the proportion of infants 0–5 months of age who were fed no other food or drink, not even water, except breast milk (including milk expressed or from a wet nurse), but allows the infant to receive Oral Rehydration Salt (ORS), drops, and syrups (vitamins, minerals, and medicines).
- Continued breastfeeding at one year was defined as the proportion of children 12–15 months of age who are fed breast milk.
- Introduction of complementary foods (solid, semi-solid and soft foods) was defined as the proportion of infants 6–8 months of age who received solid, semi-solid or soft foods in the previous 24 hours, during the day and at night.
- MDD was defined as the proportion of children 6–23 months of age who received foods from four or more of the seven food groups. The seven food groups included grains,

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roots and tubers; legumes and nuts; dairy products (milk, yogurt, cheese); flesh foods (meat, fish, poultry and liver/organ meats); eggs; vitamin-A rich fruits and vegetables; and other fruits and vegetables.

- MMF defined as the proportion of breastfed and non-breastfed children 6–23 months of age, who received solid, semisolid or soft foods (including milk feeds for non-breastfed children) the minimum number of times or more (i.e., two times for breastfed infants aged 6–8 months, three times for breastfed children aged 9–23 months and four times for non-breastfed children aged 6–23 months, on the previous day).
- MAD defined as the proportion of children 6–23 months of age who received both minimum dietary diversity and minimum meal frequency.

Study selection

Following the full search, all articles retrieved from the search databases were exported to the EndNote X9 (Clarivate Analytics, USA) for removing duplicates, screening, and selection. Titles and abstracts were screened by two reviewers (KYA and FAO) independently for assessment against the inclusion criteria. Potentially relevant articles that were selected in the first screening phase were retrieved in full and their citation details imported into the Joanna Briggs Institute System for the Unified Management, Assessment and Review of Information (JBI SUMARI). The full text of the included studies read and checked by two independent reviewers (KYA and FAO) against the eligibility criteria. Studies that did not meet the eligibility criteria were excluded and the reasons for exclusion of studies recorded and reported in the systematic review. Any disagreements between the two independent reviewers were resolved by consensus and arbitration by the third (KEA) author of the review. For studies with uncertainties regarding the eligibility, the authors were contacted to seek additional information. A total of two contact attempts were made, and if no response was received, the

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articles were screened for eligibility based on the information available. The results of the study selection process are presented in a PRISMA flow diagram (Figure 1).

Data collection process and data items

Relevant data were extracted from the eligible studies by KYA, and independently verified by FAO using piloted forms adapted from the Joanna Briggs Institute (JBI) reviewer's manual [35] and the Cochrane Pregnancy and Childbirth Group for Systematic Reviews [36]. Eligible studies were identified using the following information: first author, publication year, study design and setting, study participants, sample size, geographical region, intervention components, location of intervention delivery, outcome measures, statistical analysis, and results. Corresponding authors of eligible studies were contacted for information that was not clear in the data extraction. A total of two contact attempts were made, and if no response was received, only the information available was used.

Risk of bias Assessment

All included studies examined for the five types of biases of interventional studies: (i) selection bias; (ii) performance bias; (iii) attrition bias; (iv) detection bias and; (v) reporting bias. KYA and KEA independently conducted the risk of bias assessment using the revised Cochrane risk of bias tool for clustered randomized trials (ROB 2.0) [37 38] and the risk of bias in non-randomised studies - of interventions (ROBINS-I) tool [39] for assessing the risk of bias in quasi-experimental studies. Cluster randomised trials were assessed for five domains (i.e., randomization and identification or recruitment bias, deviations from intended interventions, missing data, measurement of outcomes, and selection of reported result), and quasi-experimental studies were assessed for seven domains (i.e., confounding, selection of

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participants, classification of interventions, deviations from intended interventions, missing data, measurement of outcomes, and selection of reported result). For cluster randomised trials, the overall risk of bias judgement included low risk, some concerns and high risk [37 38], and for the quasi-experimental studies, the overall risk of bias outcome included low, moderate, serious, and critical risk of bias [39]. There was no disagreement between reviewers in the risk of bias assessment, no third arbitrator was needed.

Data synthesis

Author reports of effect size measures were used in reporting the findings of each study. Effect size measures included a comparison between experimental and control groups (e.g., odds ratio [OR]), as well as the difference in proportion between pre- and post-outcome measures (e.g., change in proportion). Where available, 95% confidence interval (CI) and P-value were obtained from the eligible studies. Narrative methods including text, table, and figure were also used as a tool for data presentation.

Patient and public involvement

There was no direct patient or public involvement in this review

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Results

Characteristics of eligible studies

A total of 1653 articles were found in the database and hand searching. Of these, 1021 were retained in Endnote library after removing duplicates. Titles and abstracts of the articles were screened, and 996 articles were removed. Full-text eligibility checks of the remaining 25 articles excluded 12 studies. The reasons for exclusion of studies included: 2 studies were duplicates [40 41], 1 study was cross-sectional [42 43], 2 studies were based on interventions designed for reproductive-age women [44 45], 1 study used qualitative evaluation [46], 4 studies were abstracts [47-50], and 1 study designed for under-five children [51] (Figure 1).

Among the 13 included studies, ten were quasi-experimental studies [52-61], while the remaining three were cluster randomized trials [62-64]. At least one interventional study was conducted in Sothern Nations Nationalities and Peoples' Region (SNNPR) [55-57 59 60 64], Oromia [52 55 56 61 65], Amhara [54-56 63], Tigray [55 57], and Afar [58] regions, but we did not find published interventional study conducted in Addis Ababa and Dire Dawa city administrations, as well as Harari, Benishangu1, Gambella, or Somali regions.

Risk of bias in included studies

Supplementary file 1 and 2 presents the risk of bias assessment of the included studies. Three cluster randomised trials were assessed for risk of bias using the Cochrane RoB 2.0 tool. Overall, studies by Kang et al and Tariku et al judged as some concerns in the risk of bias [62 64], and the remaining study by Kim et al had a low risk of bias [63]. All cluster randomised trials had a low risk of bias in randomisation, missing data, and reporting of results. Out of ten quasi-experimental studies assessed using the ROBINS-1 tool, six studies had a serious risk of bias

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[52 55 56 59-61], and two studies had a moderate risk of bias [57 58]. The remaining two studies judged as critical risk of bias [53 54]. Seven out of ten studies had a serious or critical risk of bias in confounder adjustment [52-56 59-61], while five out of ten studies had a low risk of bias in outcome measurement and selection of reported result [55-59]

Evidence from reviewed studies

In the eligible studies, the majority of the interventions were provided in the form of behaviour change communication (BCC) using nutrition education [59 60], community mobilisation [54 56 63], health education materials [52 58 60 61], and mass media communication [57 63]. Other interventions were also provided in the form of strengthening of health care system [56 58], training and supervision of health care providers [54 56 61], and food preparation and demonstration sessions [59 60 62]. Additionally, two studies used the health belief model (HBM) as a mode of intervention delivery for BCC [59 64].

Seven studies used control groups for comparison with experimental groups [56 58-61 63 64], while six studies did not use a control or comparison group [52-55 57 62]. One study included two experimental arms (HBM and traditional intervention groups) as an intervention group [64].

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Impact of interventions on IYCF practices in Ethiopia

EIBF

Four studies reported the effect of interventions to improve EIBF [54 55 57 58]. A health facility level intervention in the Afar region that integrated nutrition information into health professional guidelines, and also provided family education and counselling booklets resulted in a significant improvement in EIBF in the experimental group compared to the control group (OR = 1.70; 95% CI: 1.17, 2.46) [58]. Kim et al. showed that interventions delivered through advocacy and policy dialogues, interpersonal communication and community mobilization, and mass communication improved EIBF [57]. This study also showed that the association between exposure to two or more types of interventions and improvement in EIBF was stronger compared to exposure to a single type of intervention (AOR = 2.07; 95% CI: 1.09, 3.91 for multiple intervention vs AOR = 1.88; 95% CI: 1.08, 3.27 for single intervention) [57]. Buchanan et al. [54] and Baxter et al. [55] found an increase in the proportions of EIBF (from 31.0% to 77.0% and 51.4% to 67.9%, respectively), following the implementation of BCC and community-level breastfeeding promotions.

EBF

Eight studies reported the effect of interventions on EBF in Ethiopia. Of these, seven studies measured EBF for up to six months, while one study measured EBF for up to one month. Villadsen et al. found that antenatal care strengthening for pregnant women significantly improved EBF in health facilities under intervention compared to health facilities without intervention (OR = 2.30; 95% CI: 1.4, 3.6) [58 61].

Integrating nutrition into the health workers' guidelines, and providing family education and counselling booklet (OR = 2.62; 95% CI: 0.07, 10.3) [58] and exposure to BCC intervention (i.e.,

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advocacy and policy dialogues, interpersonal communication and community mobilization, and mass communication) did not affect EBF behaviours [57]. Five studies showed that there were increases in EBF during the intervention period [52-56], but out of these studies, two indicated that improvements in EBF were not statistically significant [52 56].

Introduction of solid, semi-solid and soft foods

Kim et al. investigated the effect of exposure to BCC interventions (advocacy and policy dialogues, interpersonal communication and community mobilization, and mass communication) on the introduction of solid, semi-solid and soft foods in two separate arms. The first arm of the study found an increase in the proportion of infants who were introduced to solid, semi-solid and soft foods (from 37.4% to 59.7%, P-value <0.001) [57]. In contrast, the second arm of the study did not find a significant improvement (from 58.7% to 61.3%, P-value >0.05) [63].

Continued breastfeeding at one year

Only one study measured the impact of interventions on continued breastfeeding at one year. Tariku et al. showed that IYCF messages using the health belief model (HBM) did not impact continued breastfeeding rate at one year [64].

Minimum dietary diversity

Five studies reported the effect of interventions on MDD. Of these, three measured according to the WHO definitions for MDD [57 63 64], while the remaining two studies used any four food groups as a cut-off point to measure MDD [60 62]. Two studies by Kim et al. revealed the

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significant impacts of exposure to more than one type of BCC interventions (advocacy and policy dialogues, interpersonal communication and community mobilization, and mass communication) on MDD, but these studies showed that exposure to a single type of intervention did not have an impact on MDD [57 63]. Tariku et al. showed interventions involving complementary feeding messages using the HBM model significantly increased the proportion of MDD (from 37.5% to 67.9%, P-value <0.05) [64]. The remaining two studies indicated that the proportion of MDD increased over the intervention period, but the effect was not significant [60 62].

Minimum meal frequency

MMF was included as an outcome variable in six interventional studies. Two studies used the WHO definitions assessing for MMF [57 63]. Kim et al. 2019 indicated that exposure to two types of BCC interventions (advocacy and policy dialogues, interpersonal communication and community mobilization, and mass communication) had a significant effect on MMF of infants and young children compared to those who did not expose to intervention (OR = 1.69; 95% CI: 1.13, 2.52) [63]. Carnell et al. found that interventions using a community-based participatory nutrition program significantly increased the proportion of MMF over the intervention period [62]. Except for one study, which did not do a significant test [60], the remaining two studies did not show significant increases in MMF [59 64].

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373 **Table 1: Summary of eligible studies**

Author	Design	Participants	Region	Intervention components	Location of intervention delivery	Outcome measures	Statistical analysis	Results
Baxter et al., 2016 [52]	Quasi-experimental	Experimental group: Before intervention=173 After intervention=269	Oromia	EBF promotion (Home visits, educational materials, and training of influential community and family members)	Health facility and community- based	EBF	Percent change in proportion	EBF increased from 69.4% to 75.1%, but not statistically significant
Berti et al., 2010 [53]	Quasi-experimental	No information	No information	Breastfeeding communication through community-based interventions for behaviour change	Community-based	EBF	Percent change in proportion	Higher improvement in exclusive breastfeeding
Buchanan et al., 2012 [54]	Quasi-experimental	No information	No information	Behavioural change communication Community mobilization Capacity building	Health facility-based and community-based	EIBF EBF	Percent change in proportion	EIBF increased from 31.0% to 77.0% EBF increased from 73.0% to 95.0%
Callaghan-Koru et al., 2016 [55]	Quasi-experimental	Experimental group: Before intervention=218 After intervention=215	Tigray Oromia Amhara SNNPR	An integrated package of facility-based kangaroo mother care Community-based promotion of EBF	Health facility and community based	EIBF EBF	Percent change in proportion	EIBF significantly increased from 51.4% to 67.9% (P value <0.01) EBF increased from 86.0% to 95.8% (P value <0.01)
Carnell et al., 2014 [56]	Quasi-experimental	Experimental group: Before intervention=1016 After intervention=911 Control group: Before intervention=1074 After intervention=903	Oromia Amhara SNNPR	Strengthen the health system Improve health workers' performance Engage the community	Health facility and community based	EBF	Difference-in-difference	No significant difference in the improvement of EBF was observed between intervention and control groups
Kang et al., 2017 [62]	Cluster randomized trial	Experimental group: 570 Control group: 629	Oromia	Community-based participatory nutrition promotion program Daily group nutrition sessions Follow-up visits	Community-based	Meal frequency Dietary diversity	Mixed effect multilevel regression	There was a significant improvement in meal frequency No significant improvement and dietary diversity

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Kim et al., 2016 [57]	Quasi-experimental	Experimental group: Before intervention=1481 After intervention=1494	SNNPR Tigray	Behavioural change communication interventions including: Advocacy and policy dialogues Interpersonal communication and community mobilization Mass communication	Community-based	EIBF EBF Introduction of complementary foods MDD MMF MAD	Fixed effect regression modelling	Exposure to one type of intervention: EIBF (OR = 1.88; 95% CI: 1.08, 3.27); No significant effect on EBF, MDD, MMF, and MAD Exposure to two types of interventions: EIBF (OR = 2.07; 95% CI: 1.09, 3.91); MDD (OR = 6.87; 95% CI: 1.27, 37.14); No significant effect on EBF, MMF, and MAD Introduction of complementary foods increased from 37.4% to 59.7% (P-value <0.001)
Kim et al., 2019 [63]	Cluster-randomized trial	Experimental group: Before intervention=1328 After intervention=1360 Control group: Before intervention=1318 After intervention=1360	Amhara region	Behavioural change communication interventions including: Interpersonal communication Nutrition sensitive agricultural activities community mobilization Mass communication	Community-based	Introduction of complementary foods MDD MMF	Fixed effect regression modelling	Exposure to one group of intervention: MDD (OR = 1.05; 95% CI: 0.74, 1.48) MMF (OR = 1.05; 95% CI: 0.90, 1.23) Exposure to two groups of intervention: MDD (OR = 1.82; 95% CI: 1.14, 2.89) MMF (OR = 1.69; 95% CI: 1.13, 2.52) Introduction of complementary foods did not show any improvement
Kung'u et al., 2018 [58]	Quasi-experimental	Experimental group: Before intervention=2570	Afar	Behaviour change interventions	Health facility	EIBF	Logistic fixed-effect regression	EIBF (OR = 1.70; 95% CI: 1.17, 2.46)

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		After intervention=2905 Control group: Before intervention=1670 After intervention=1620		Integrating nutrition to the frontline workers' guideline Family education and counselling booklet		EBF at six months		EBF at six months (OR = 2.62 95% CI: 0.07, 1.03)
Mulualem et al., 2016 [59]	Quasi-experimental	Experimental group: Before intervention=80 After intervention=80 Control group: Before intervention=80 After intervention=80	SNNPR	Nutrition education (individual and group sessions, and recipe demonstration) based on the HBM model	Community-based	Meal frequency	ANOVA	MMF frequency decreased in both experimental and control groups
Negash et al., 2014 [60]	Quasi-experimental	Experimental group: Before intervention=100 After intervention=81 Control group: Before intervention=92 After intervention=76	SNNPR	Nutrition education using visual materials Demonstration of food preparation	Community-based	Dietary diversity Meal frequency	Independent sample t-test	Dietary diversity and meal frequency increased over the intervention period
Tariku et al., 2015 [64]	Cluster randomized trial	HBM intervention group: Before intervention=60 After intervention= 54 Traditional nutrition education group: Before intervention=60 After intervention= 56 Control group: Before intervention=60 After intervention= 56	SNNPR	Complementary feeding messages using the HBM model	Community-based	Continued breastfeeding MDD MMF	ANOVA Kruskal-Wallis test	No significant difference in the improvement of continued breastfeeding and MMF HBM intervention group showed significant improvement in MDD (P<0.001)
Villadsen et al., 2016 [61]	Quasi-experimental	Experimental group: Before intervention=775 After intervention=1434 Control group: Before intervention=350 After intervention=422	Oromia	Training of health professionals Health education materials Supervisions of ANC providers	Health facility	EBF at 1 month	Logistic fixed-effect regression	EBF at 1 month (OR = 2.3; 95% CI: 1.4, 3.6)

EIBF: early imitation of breastfeeding; EBF: exclusive breastfeeding; MDD: minimum dietary diversity; MMF: minimum meal frequency; MAD: minimum acceptable diet; HBM: health belief model; SNNPR: Southern Nations Nationalities and Peoples Region.

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Discussion

A total of thirteen studies were included, and of these, ten were quasi-experimental studies, and three were cluster randomised trials. Five studies for EBF and four studies for EIBF showed that community- and facility-based interventions delivered in the form of BCC (such as policy advocacy, interpersonal communication, community mobilisation, and mass media campaigns) and health service strengthening had an impact on infants. Three studies for MDD and two studies for MMF indicated significant effects of BCC on infants and young children. Two eligible studies showed that combined interventions delivered in the form of interpersonal communication, nutrition-sensitive agricultural activities, community mobilisation, and mass media campaigns had more impacts on EIBF, MDD, and MMF.

Global evidence indicates that BCC interventions (including policy advocacy, interpersonal communication, community mobilisation and mass media campaigns) are essential in promoting IYCF practices. These initiatives also create a supportive environment that helps mothers to adopt and sustain positive child feeding behaviours [11 66]. The main goals of BCC intervention include improving behaviours of a facility- and community-level health workers, mothers and other caregivers, and members of their households [11 66]. Our review showed that BCC interventions in the form of policy advocacy, interpersonal communication, community mobilisation, and mass media campaigns improved EIBF, EBF, MDD and MMF. Our finding is consistent with systematic reviews from LMICs, which indicated that group nutritional education and counselling, family or social support, integrated mass media coverage, and community mobilization were effective in improving IYCF practices [11 67]. Although BCC interventions promote skills and knowledge that are necessary for improving IYCF practices, achieving the intended target is often challenging without complementary interventions that

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provide or facilitate material inputs [66]. Our review suggests that interventions that promote material inputs (such as cash transfer and provision of food materials) and/or non-material inputs (skills and knowledge) are required in achieving appropriate and sustained IYCF behaviours.

Research has shown that exposure to more than one IYCF interventions, delivered through different platforms (e.g., community- or facility-based), were more successful in improving IYCF outcomes compared to a single form of intervention delivered at a single setting [5]. Our review indicated that exposure to more than two types of interventions was associated with improvement in EIBF, MDD, and MMF [57 63]. This result may be due to that intervention delivered through multiple settings may have an increased intensity that can be translated to behavioural changes due to repeated contacts and adequate coverage of interventions [68]. Our review suggests that interventions that aim to improve IYCF behaviours in Ethiopia should be delivered through different and specific types of contexts (including household, health facility and community) and should be provided along the continuum of care from conception until a child's second birthday.

In the past two decades, the Ethiopian government's commitment to improve IYCF has been consistent, and this started with the development of the National Strategy for IYCF in 2004 and followed by National Strategy for Nutrition in 2008 [24 69], in line with the WHO/UNICEF IYCF strategy [70]. In 2015, the Ethiopian government launched the Health Sector Transformation Plan to improve health outcomes, including IYCF practices [30]. Based on this initiative, many facility- and community-based interventions have been implemented to protect, promote and support appropriate IYCF behaviours [10 11]. Nevertheless, our review found

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that there has been no formal IYCF intervention conducted in Addis Ababa and Dire Dawa city administrations, and Harari, Benishangu1, Gambella, and Somali regions. Our review suggests an expansion of additional experimental studies to maximise the opportunities for the prevention of IYCF-related morbidity and mortality in those regions.

Additionally, our review did not find any studies which incorporated the use of mobile health (mHealth) as part of an intervention package for improving IYCF [71]. Future experimental studies that consider the use of mHealth techniques are required, especially that previously published studies have shown the positive impacts of interventions (in form of voice messages, SMS, and phone calls) on IYCF practices [72-74]. Finally, interventional studies that consider the role of fathers/partners as well as grandmothers in improving IYCF behaviours may be warranted in Ethiopia [75-77].

The present systematic review of IYCF outcomes from interventions has strengths and limitations. First, a meta-analysis was not conducted due to the wide heterogeneity of the design and outcome measures. Second, this review was limited by the incompleteness of information from the individual studies. Nevertheless, the study authors were contacted for information when it is necessary. Third, our search was limited to published studies in English, which exclude studies published with other languages. Fourth, lack of evidence and under-representation of some regions may limit the generalizability of our review. However, evidence from this review is potentially relevant for IYCF policy efforts in Ethiopia. Despite the above limitations, the application of multiple search strategies in consultation with the medical librarian and registration of the systematic review protocol can be considered as strengths of this review.

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Conclusion

Our systematic review showed that interventions delivered in the form of policy advocacy, health service strengthening, interpersonal communication, community mobilisation, and mass media campaigns improved EIBF, EBF, MDD, and MMF in Ethiopia. The impacts of current and/or future interventions on IYCF practices in Ethiopia will be greater when interventions are combined, introduced and implemented along the continuum of care from conception until a child's second birthday.

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List of abbreviations

ANC: Antenatal Care; OR: Odds Ratio; BFHI: Baby-Friendly Hospital Initiative; CI: Confidence Interval; EBF: Exclusive Breastfeeding; EIBF: Early Initiation of Breastfeeding; HBM: Health Belief Model; HSTP: Health Sector Transformation Plan; IYCF: Infant and Young Child Feeding; JBI: Joanna Briggs Institute; LMICs: Lower and Middle-Income Countries; MAD: Minimum Acceptable Diet; MDD: Minimum Dietary Diversity; MeSH: Medical Subject Headings; MDG: Millennium Development Goal; MMF: Minimum Meal Frequency; Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA); RCTs: Randomised Controlled Trials; SDGs: Sustainable Development Goals; SNNPR: Southern Nations Nationalities and Peoples Regions; SUMARI: System for the Unified Management, Assessment and Review of Information; UNICEF: United Nation Children's Fund; WHO: World Health Organization

Declarations

Contributors

KYA conceptualised the study idea, performed the literature searching, selected the paper of interests, performed data extraction, and critically revised the manuscript. AP and AA critically revised the manuscript for intellectual content. KEA performed the risk of bias assessment, and critically revised the manuscript. FAO contributed to the conception of the research idea, selected the paper of interests, performed data extraction, and critically revised the manuscript. All authors read and approved the final manuscript.

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"Not applicable"

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Competing interests

The authors declare that they have no competing interests

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Supplementary appendix

Research Checklist: PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) 2009 Checklist.

Supplementary file 1: Risk of bias assessment for quasi-experimental studies

Supplementary file 2: Risk of bias assessment for cluster randomised trial studies

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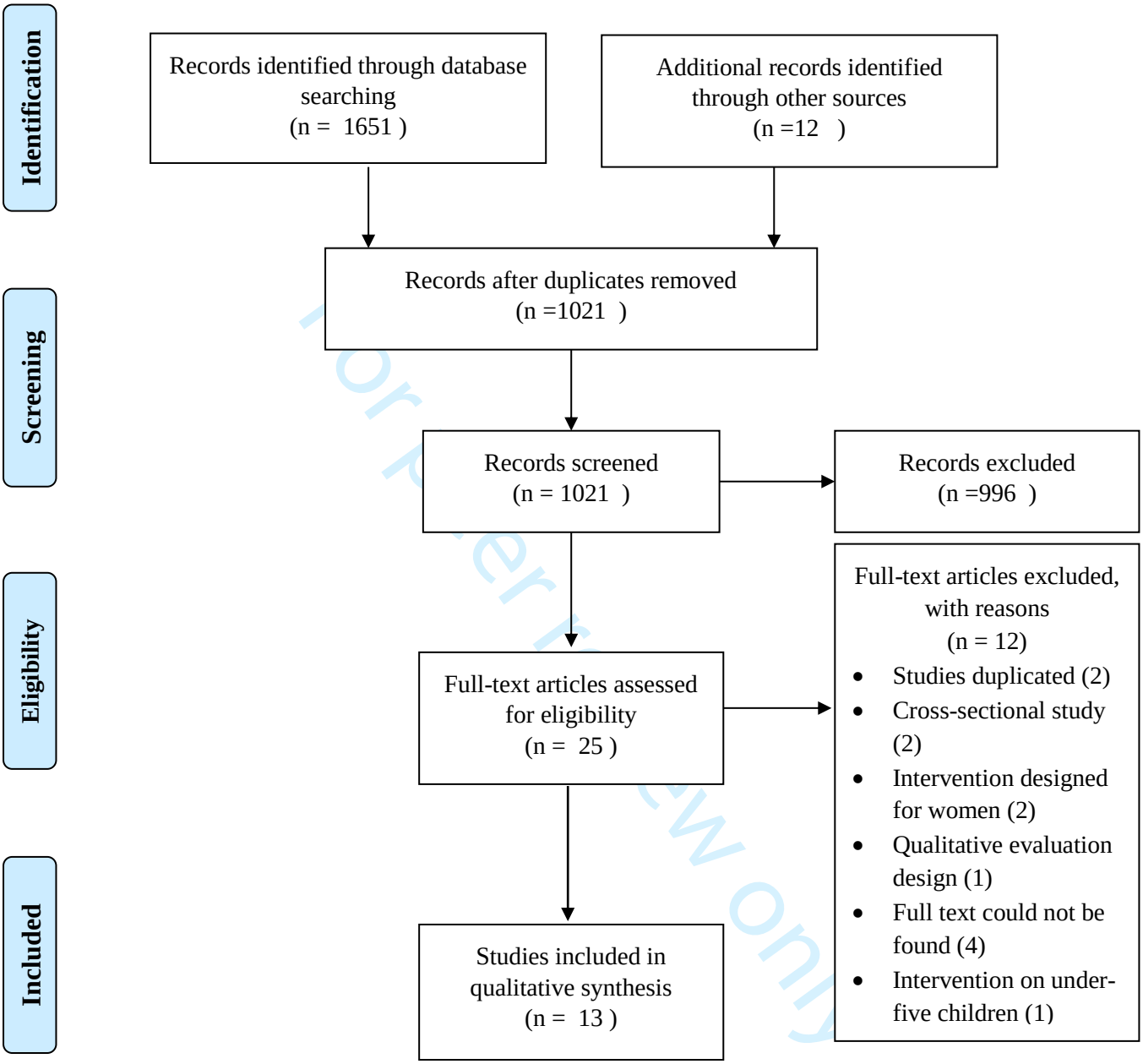


Figure 1: Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flowchart for selection of eligible studies [40]



PRISMA 2009 Checklist

ection/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2, 3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4, 5
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5 Paragraph 2
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	6 Paragraph 1
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	7 Paragraph 5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	6 Paragraph 2
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	6, 7
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	9, 10
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	10 Paragraph 2
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	10 Paragraph 2
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	10 Paragraph 3
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	N/A
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	11 Paragraph 2

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	10 Paragraph 3
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	N/A
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	12 Paragraph 1
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	12 Paragraph 2
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	12 Paragraph 3
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	13–19
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	N/A
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	12 Paragraph 3
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	N/A
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	20–22
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	22 Paragraph 3
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	23
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	23 Paragraph 1

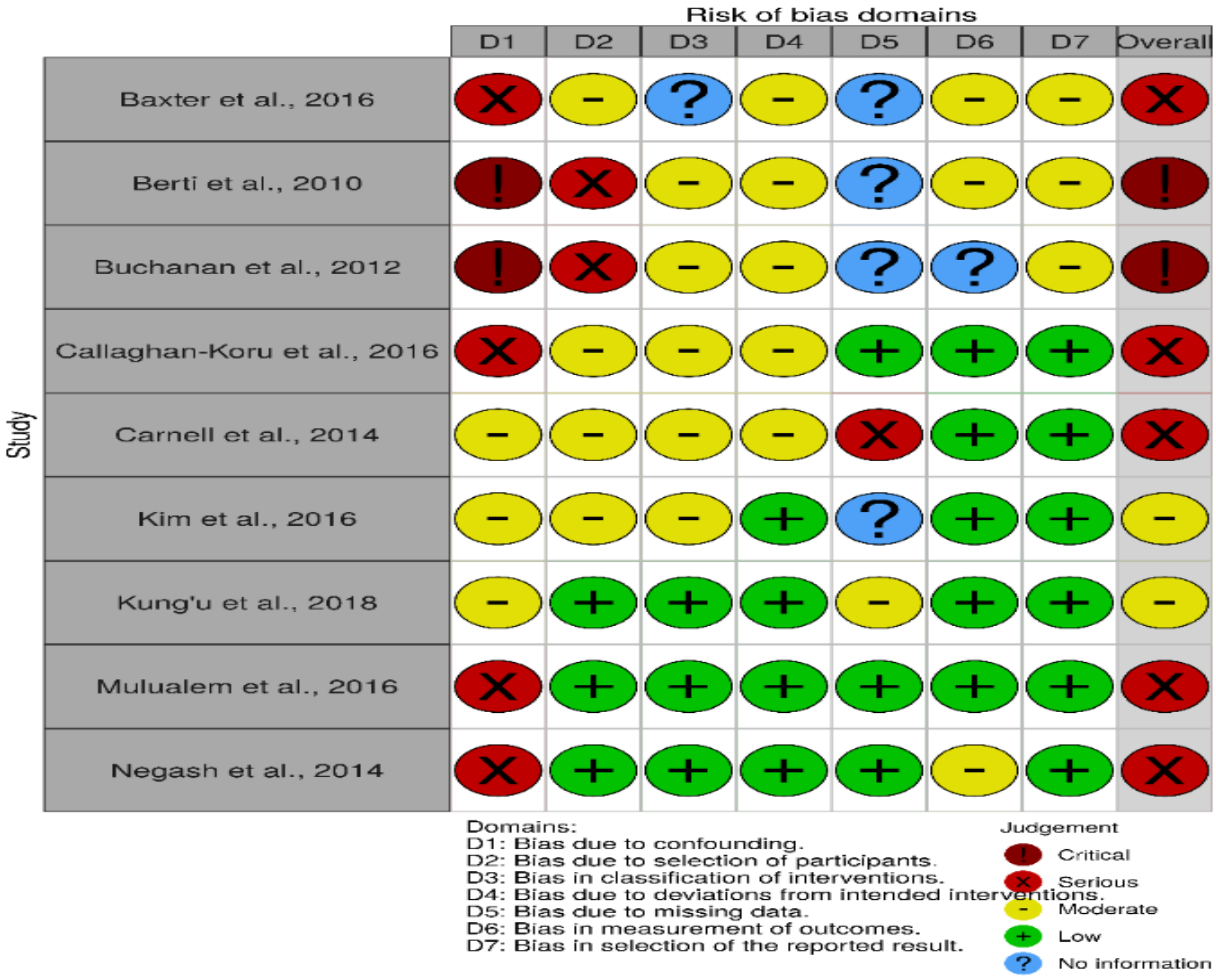
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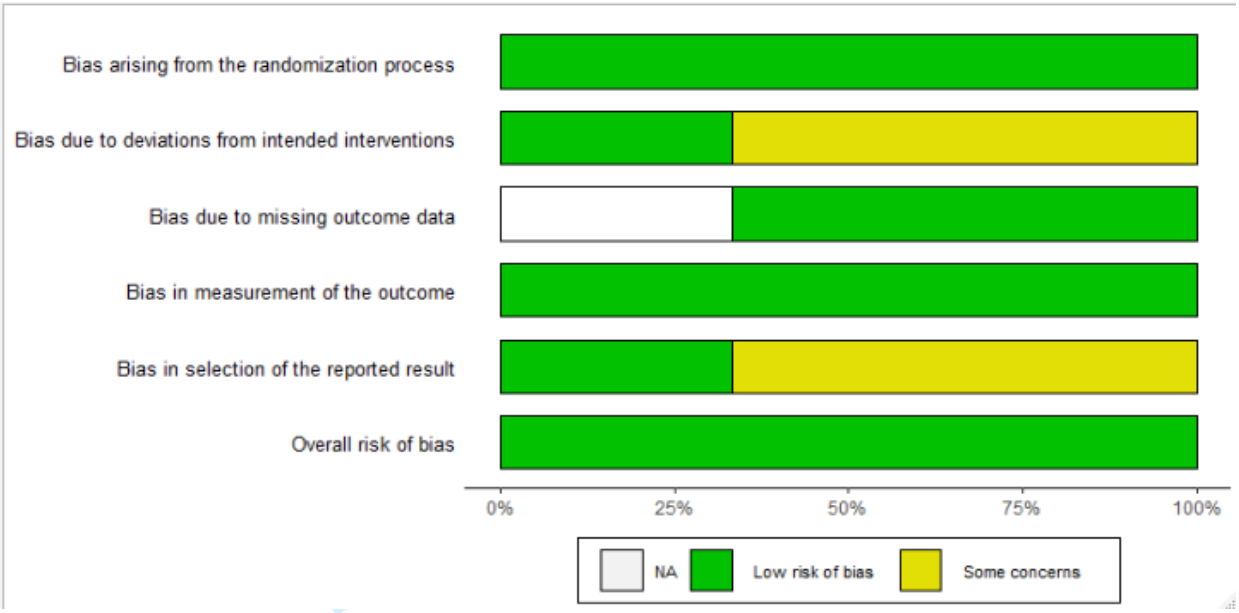
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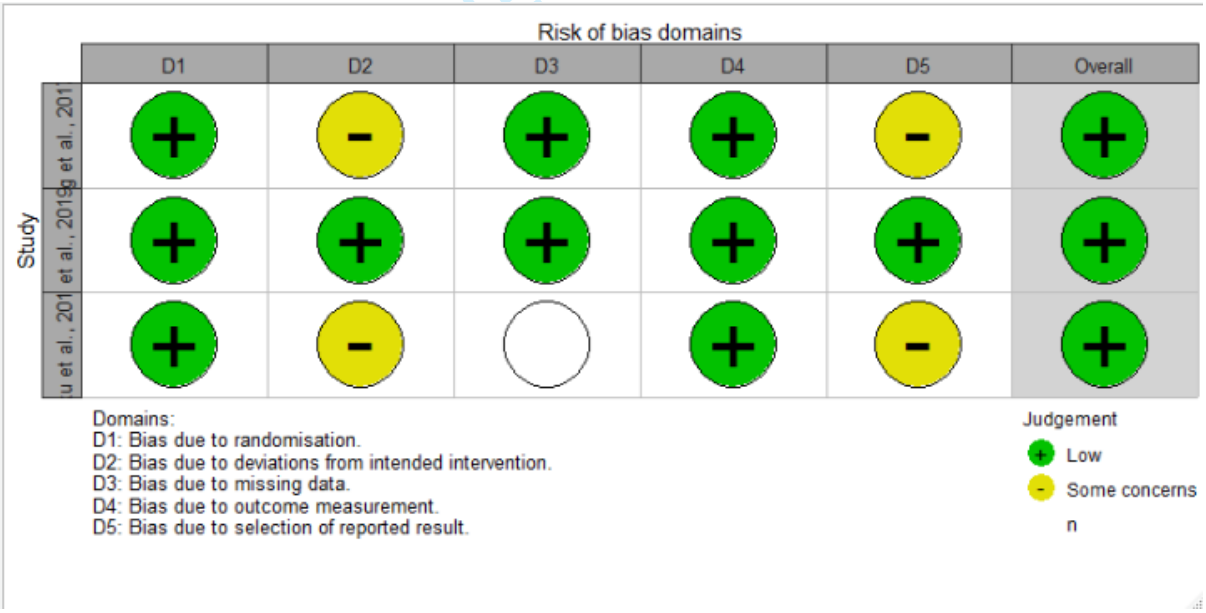
Risk of bias graph: review authors' judgements about each risk of bias item presented as percentages across all included quasi-experimental studies



Risk of bias summary: review authors' judgements about each risk of bias item for each included quasi-experimental study



Risk of bias graph: review authors' judgements about each risk of bias item presented as percentages across all included cluster randomised studies



Risk of bias summary: review authors' judgements about each risk of bias item for each included cluster randomised study

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Interventions to improve infant and young child feeding practices in Ethiopia: a systematic review

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Abstract

Objective: In Ethiopia, despite the implementation of several interventions to improve infant and young child feeding (IYCF) practices, no published studies have highlighted the most effective IYCF interventions in the country. This systematic review investigated the impacts of various interventions on IYCF in Ethiopia.

Source of information: A systematic search was conducted on seven computerized bibliographic databases (Embase, ProQuest Central, Scopus, PsycINFO, Web of Science, MEDLINE/PubMed and CINAHL) to locate experimental or quasi-experimental studies between November 2019 and June 2020.

Study eligibility criteria: Interventional studies that measured IYCF indicators (early initiation of breastfeeding [EIBF], exclusive breastfeeding [EBF], the introduction of complementary foods, minimum dietary diversity [MDD], minimum meal frequency [MMF] and minimum acceptable diet [MAD]) as outcome variables were included.

Study appraisal and synthesis: All included studies were examined for biases related to interventional studies (i.e., selection bias, performance bias, attrition bias, detection bias and reporting bias). Author reports of effect size measures were used to narratively report the findings of each study.

Results: Of the thirteen eligible studies, ten studies were quasi-experimental and three studies were cluster randomised trials (CRTs). All CRTs had a low risk of bias, while six out of ten quasi-experimental studies had a serious risk of bias. Five studies showed that interventions in the form of policy advocacy, health service strengthening, interpersonal communication,

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community mobilisation, and mass media campaigns improved EBF. Four studies indicated that community-level and health facility breastfeeding promotions increased EIBF. Three studies for MDD and two studies for MMF indicated significant effects of policy advocacy, interpersonal communication, community mobilisation, and mass media campaigns on infants and young children. Interventions that were delivered in combination increased the impacts in improving EIBF, MDD and MMF compared to single intervention.

Conclusion: Our review showed that a combination of community-level interventions improved EIBF, EBF, MDD, and MMF in Ethiopia. The improvement in IYCF in Ethiopia would require multi-prong approaches at the household and community levels along the continuum of care from conception until a child's second birthday.

Keywords: Infant and young child feeding, breastfeeding, nutritional education, intervention, systematic review, Ethiopia

Protocol registration number: PROSPERO CRD42020155519.

Strengths and limitations of the study

- A meta-analysis was not conducted due to the wide heterogeneity of the design and outcome measures.
- This review was limited by the incompleteness of information from the individual studies. Nevertheless, the study authors were contacted for information when it is necessary.
- Despite the limitations, the application of multiple search strategies in consultation with the medical librarian and registration of the systematic review protocol can be considered as strengths of this review.

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Introduction

Appropriate infant and young child feeding (IYCF) improves childhood growth (through reduced risk of diarrhoea and respiratory infections as well as improved nutrition) [1-7], increases the intellectual capacity of children [4 8], and reduces the risk of mothers to experience diabetes mellitus, overweight and obesity [4 9]. Promoting appropriate IYCF using evidence-based interventions is essential to protect and support maternal knowledge, beliefs, and confidence in IYCF practices [10-13]. Community- and facility-based interventions that range from direct support and education to mothers and families on appropriate IYCF behaviours to training of health professionals can significantly improve IYCF behaviours [14-16].

Global efforts to improve maternal IYCF behaviours have increased in the past three decades. These efforts included the International Code of Marketing of Breast-milk Substitutes [17]; the Innocenti Declaration [18]; the Baby-Friendly Hospital Initiative (BFHI) [19]; the Millennium Development Goals (MDGs) [20]; and more recently, the Global Nutrition Targets 2025 [21] and the Sustainable Development Goals (SDGs) [22]. The World Health Organization and United Nations Children's Fund (WHO/UNICEF) have played major roles in driving these initiatives which aimed to protect, promote, and support breastfeeding and child nutrition [19]. For example, the WHO Global Nutrition Targets (WHO GNT, including Target-1 and -5) aim to reduce the global number of stunted children under-5 by at least 40% and increase the EBF rate to at least 50% by the year 2025, respectively. However, most subnational areas of many African countries (including Ethiopia) are not on track to achieve WHO GNT goal-5 [23].

In Ethiopia, while both government and non-governmental organisations have implemented various maternal and child health-related policy interventions over the last two decades to increase infant feeding [24-26], a recent study has revealed that early initiation of breastfeeding

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(EIBF) prevalence was 75.5% and that for exclusive breastfeeding (EBF) was 59.9% [27], with a wide heterogeneity across subnational areas of the country [23]. These rates are well below the Ethiopian Health Sector Transformation Plan (HSTP) target of 90% and 72%, respectively [25]. Additionally, the proportion of children who met minimum dietary diversity (MDD) and minimum meal frequency (MMF) was 7.1% and 43.6%, respectively [28]. In Ethiopia, more than five million children were stunted in 2016 [29], and this is a major public health issue that is largely attributable to inappropriate IYCF.

To date, no previously published studies have examined the most effective IYCF interventions in the Ethiopian context. A detailed understanding of the specific types of interventions that can be implemented to improve IYCF in Ethiopia is essential, to specifically guide the allocation of scarce resources and the provision of targeted strategies to increase childhood feeding. This information will also be helpful in scaling-up and/or improve upon current and future IYCF interventions in the country. Data from the assessment of relevant IYCF interventions would be crucial to stakeholders as Ethiopia implements national nutrition efforts (e.g., HSTP) [25] and global health programs such as the Sustainable Development Goal, SDG-2.2, to end all forms of malnutrition [22]. Accordingly, this systematic review investigated the impacts of community- and facility-based interventions in improving IYCF practices of mothers/caregivers with children under two years of age in Ethiopia.

Materials and methods

Our review adhered to the 2009 Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [30] **(Supplementary file 1)**, and the protocol for the review was registered in PROSPERO (International Prospective Register of Systematic Reviews, CRD42020155519) [31].

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Eligibility criteria

Studies were included if they meet the following criteria: (i) experimental or quasi-experimental study design (including randomized controlled trials [RCT], non-RCTs, before and after studies, and interrupted time-series studies) with or without comparison groups or cluster; (ii) conducted among pregnant women and mothers living with live children under 2 years to improve IYCF practices; (iii) used any type of IYCF interventions delivered between the antenatal period and 24 months postpartum; (iv) measured at least one core IYCF indicators such as EIBF, EBF, the introduction of solid, semi-solid and soft foods, continued breastfeeding at one year, MDD, MMF, and minimum acceptable diet (MAD); (v) were published in the English language; and (vi) conducted in Ethiopia from the year 2000 to date. We considered studies published over this period to demonstrate the impacts of the specific IYCF interventions in the context of Ethiopia to reflect the implementation of relevant global health programs, including the MDGs and the current SDGs [23].

The intervention strategies considered in this review focused on those performed in both facility- and community-levels. The interventions included nutritional education and counselling, interpersonal communication, mass media campaigns, nutrition-sensitive agricultural activities, group demonstration sessions, and strengthening of health facilities.

Studies conducted among mothers with health problems such as HIV/AIDS and children with cleft palate were excluded.

The IYCF practices included EIBF, EBF, the introduction of solid, semi-solid and soft foods, continued breastfeeding at one year, MDD, MMF, and MAD, measured according to the WHO definitions for assessing IYCF indicators [32]. These indicators were selected based on their effect on the health and development of infants and young children [11, 12, 35] and the evidence from previously published systematic reviews [11 33].

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- EIBF was defined as the proportion of children aged 0–23 months who commenced breastfeeding within the first hour of birth.
- EBF was defined as the proportion of infants 0–5 months of age who were fed no other food or drink, not even water, except breast milk (including milk expressed or from a wet nurse), but allows the infant to receive Oral Rehydration Salt (ORS), drops, and syrups (vitamins, minerals, and medicines).
- Continued breastfeeding at one year was defined as the proportion of children 12–15 months of age who are fed breast milk.
- Introduction of complementary foods (solid, semi-solid and soft foods) was defined as the proportion of infants 6–8 months of age who received solid, semi-solid or soft foods in the previous 24 hours, during the day and at night.
- MDD was defined as the proportion of children 6–23 months of age who received foods from four or more of the seven food groups. The seven food groups included grains, roots and tubers; legumes and nuts; dairy products (milk, yogurt, cheese); flesh foods (meat, fish, poultry and liver/organ meats); eggs; vitamin-A rich fruits and vegetables; and other fruits and vegetables.
- MMF defined as the proportion of breastfed and non-breastfed children 6–23 months of age, who received solid, semisolid or soft foods (including milk feeds for non-breastfed children) the minimum number of times or more (i.e., two times for breastfed infants aged 6–8 months, three times for breastfed children aged 9–23 months and four times for non-breastfed children aged 6–23 months, on the previous day).
- MAD defined as the proportion of children 6–23 months of age who received both minimum dietary diversity and minimum meal frequency.

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Information sources and search strategy

Consistent with previously published systematic reviews [15 34], a three-stage search strategy was applied to locate both published and unpublished studies. In stage one, a manual search of previously published systematic reviews and the reference lists of all the eligible studies was performed to check for systematic reviews of interventions on IYCF in Ethiopia. In stage two, index terms (e.g., medical subject headings [MeSH] for Medline) and key terms were generated using the identified articles from the initial search for the full search strategy. In stage three, a full search strategy was conducted on seven computerized bibliographic databases (Embase, ProQuest Central, Scopus, PsycINFO, Web of Science, MEDLINE [OVID]/PubMed and Cumulative Index to Nursing and Allied Health Literature, CINAHL) (EBSCO) to locate published studies, and secondary searching of OpenGrey and Grey Literature Report was conducted to locate unpublished works. The Population Intervention Comparator Outcome (PICO) criteria were used to devise the review question and related search terms. A combination of specified medical subject headings (MeSH) terms and keywords was drafted with some assistance from our institution-based medical librarian. The search strategy, including all identified index and keywords terms, was slightly adapted for each of the information sources. The initial and the top-up searches were conducted in November 2019, and June 2020, respectively. A full electronic search strategy developed for the Ovid/Embase is provided as **Supplementary file 2**.

The overall list of search terms/keywords used included:

Term 1 (participants): Child OR infant OR female OR caregiver OR women OR community health workers OR health extension worker AND

Term 2 (feeding): Breastfeed OR infant food OR meals OR diet OR feeding behaviour OR healthy diet OR child nutritional physiological phenomena AND

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Term 3 (interventions): child health services OR counselling OR dietary supplements OR health education OR food supply OR health promotion OR prenatal care OR postnatal care OR early intervention OR community health planning OR models/educational OR impact OR influence OR effect AND

Term 4 (context): Ethiopia

Study selection

Following the full search, all articles retrieved from the search databases were exported to the EndNote X9 (Clarivate Analytics, USA) for removing duplicates, screening, and selection. Titles and abstracts were screened by two reviewers (KYA and FAO) independently for assessment against the inclusion criteria. Potentially relevant articles that were selected in the first screening phase were retrieved in full and their citation details imported into the Joanna Briggs Institute System for the Unified Management, Assessment and Review of Information (JBI SUMARI). The full text of the included studies read and checked by two independent reviewers (KYA and FAO) against the eligibility criteria. Studies that did not meet the eligibility criteria were excluded and the reasons for exclusion of studies recorded and reported in the systematic review. Any disagreements between the two independent reviewers were resolved by consensus and arbitration by the third (KEA) author of the review. For studies with uncertainties regarding the eligibility, the authors were contacted to seek additional information. A total of two contact attempts were made, and if no response was received, the articles were screened for eligibility based on the information available. The results of the study selection process are presented in a PRISMA flow diagram (Figure 1).

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Data collection process and data items

Relevant data were extracted from the eligible studies by KYA, and independently verified by FAO using piloted forms adapted from the Joanna Briggs Institute (JBI) reviewer's manual [35] and the Cochrane Pregnancy and Childbirth Group for Systematic Reviews [36]. Eligible studies were identified using the following information: first author, publication year, study design and setting, study participants, sample size, geographical region, intervention components, location of intervention delivery, outcome measures, statistical analysis, and results. Corresponding authors of eligible studies were contacted for information that was not clear in the data extraction. A total of two contact attempts were made, and if no response was received, only the information available was used.

Risk of bias Assessment

All included studies examined for the five types of biases of interventional studies: (i) selection bias; (ii) performance bias; (iii) attrition bias; (iv) detection bias and; (v) reporting bias. KYA and KEA independently conducted the risk of bias assessment using the revised Cochrane risk of bias tool for clustered randomized trials (ROB 2.0) [37 38] and the risk of bias in non-randomised studies - of interventions (ROBINS-I) tool [39] for assessing the risk of bias in quasi-experimental studies. Cluster randomised trials were assessed for five domains (i.e., randomization and identification or recruitment bias, deviations from intended interventions, missing data, measurement of outcomes, and selection of reported result), and quasi-experimental studies were assessed for seven domains (i.e., confounding, selection of participants, classification of interventions, deviations from intended interventions, missing data, measurement of outcomes, and selection of reported result). For cluster randomised trials, the overall risk of bias judgement included low risk, some concerns and high risk [37 38], and for the quasi-experimental studies, the overall risk of bias outcome included low, moderate,

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serious, and critical risk of bias [39]. There was no disagreement between reviewers in the risk of bias assessment, no third arbitrator was needed.

Data synthesis

Author reports of effect size measures were used in reporting the findings of each study. Effect size measures included a comparison between experimental and control groups (e.g., odds ratio [OR]), as well as the difference in proportion between pre- and post-outcome measures (e.g., change in proportion). Where available, 95% confidence interval (CI) and P-value were obtained from the eligible studies. Narrative methods including text, table, and figure were also used as a tool for data presentation.

Patient and public involvement

There was no direct patient or public involvement in this review

Results

Characteristics of eligible studies

A total of 1653 articles were found in the database and hand searching. Of these, 1021 were retained in the Endnote library after removing duplicates. Titles and abstracts of the articles were screened, and 996 articles were removed. Full-text eligibility checks of the remaining 25 articles excluded 12 studies. The reasons for exclusion of studies included: 2 studies were duplicates [40 41], 1 study was cross-sectional [42 43], 2 studies were based on interventions designed for reproductive-age women [44 45], 1 study used qualitative evaluation [46], 4 studies were abstracts [47-50], and 1 study designed for under-five children [51] **(Figure 1).**

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Among the 13 included studies, ten were quasi-experimental studies [52-61], while the remaining three were cluster randomized trials [62-64]. At least one interventional study was conducted in Sothern Nations Nationalities and Peoples' Region (SNNPR) [55-57 59 60 64], Oromia [52 55 56 61 65], Amhara [54-56 63], Tigray [55 57], and Afar [58] regions, but we did not find published interventional study conducted in Addis Ababa and Dire Dawa city administrations, as well as Harari, Benishangu1, Gambella, or Somali regions.

Risk of bias in included studies

Supplementary files 3 and 4 present the risk of bias assessment of the included studies. Three cluster randomised trials were assessed for risk of bias using the Cochrane RoB 2.0 tool. Overall, studies by Kang et al and Tariku et al judged as some concerns in the risk of bias [62 64], and the remaining study by Kim et al had a low risk of bias [63]. All cluster randomised trials had a low risk of bias in randomisation, missing data, and reporting of results. Out of ten quasi-experimental studies assessed using the ROBINS-1 tool, six studies had a serious risk of bias [52 55 56 59-61], and two studies had a moderate risk of bias [57 58]. The remaining two studies judged as critical risk of bias [53 54]. Seven out of ten studies had a serious or critical risk of bias in confounder adjustment [52-56 59-61], while five out of ten studies had a low risk of bias in outcome measurement and selection of reported result [55-59]

Evidence from reviewed studies

In the eligible studies, the majority of the interventions were provided in the form of behaviour change communication (BCC) using nutrition education [59 60], community mobilisation [54 56 63], health education materials [52 58 60 61], and mass media communication [57 63]. Other interventions were also provided in the form of strengthening of health care system [56 58], training and supervision of health care providers [54 56 61], and food preparation and demonstration sessions [59]

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[60 62]. Additionally, two studies used the health belief model (HBM) as a mode of intervention delivery for BCC [59 64] **(Table 1)**.

Seven studies used control groups for comparison with experimental groups [56 58-61 63 64], while six studies did not use a control or comparison group [52-55 57 62]. One study included two experimental arms (HBM and traditional intervention groups) as an intervention group [64] **(Table 1)**.

Impact of interventions on IYCF practices in Ethiopia

EIBF

Tables 1 and **2** are presented to summarise evidence from eligible studies. Four studies reported the effect of interventions to improve EIBF [54 55 57 58]. A health facility level intervention in the Afar region that integrated nutrition information into health professional guidelines, and also provided family education and counselling booklets resulted in a significant improvement in EIBF in the experimental group compared to the control group (OR = 1.70; 95% CI: 1.17, 2.46) [58]. Kim et al. showed that interventions delivered through advocacy and policy dialogues, interpersonal communication and community mobilization, and mass communication improved EIBF [57]. This study also showed that the association between exposure to two or more types of interventions and improvement in EIBF was stronger compared to exposure to a single type of intervention (AOR = 2.07; 95% CI: 1.09, 3.91 for multiple intervention vs AOR = 1.88; 95% CI: 1.08, 3.27 for single intervention) [57]. Buchanan et al. [54] and Baxter et al. [55] found an increase in the proportions of EIBF (from 31.0% to 77.0% and 51.4% to 67.9%, respectively), following the implementation of BCC and community-level breastfeeding promotions.

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EBF

Eight studies reported the effect of interventions on EBF in Ethiopia. Of these, seven studies measured EBF for up to six months, while one study measured EBF for up to one month. Villadsen et al. found that antenatal care strengthening for pregnant women significantly improved EBF in health facilities under intervention compared to health facilities without intervention (OR = 2.30; 95% CI: 1.4, 3.6) [58 61].

Integrating nutrition into the health workers' guidelines, and providing family education and counselling booklet (OR = 2.62; 95% CI: 0.07, 10.3) [58] and exposure to BCC intervention (i.e., advocacy and policy dialogues, interpersonal communication and community mobilization, and mass communication) did not affect EBF behaviours [57]. Five studies showed that there were increases in EBF during the intervention period [52-56], but out of these studies, two indicated that improvements in EBF were not statistically significant [52 56].

Introduction of solid, semi-solid and soft foods

Kim et al. investigated the effect of exposure to BCC interventions (advocacy and policy dialogues, interpersonal communication and community mobilization, and mass communication) on the introduction of solid, semi-solid and soft foods in two separate arms. The first arm of the study found an increase in the proportion of infants who were introduced to solid, semi-solid and soft foods (from 37.4% to 59.7%, P-value <0.001) [57]. In contrast, the second arm of the study did not find a significant improvement (from 58.7% to 61.3%, P-value >0.05) [63].

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Continued breastfeeding at one year

Only one study measured the impact of interventions on continued breastfeeding at one year. Tariku et al. showed that IYCF messages using the health belief model (HBM) did not impact continued breastfeeding rate at one year [64].

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Table 1: Summary of eligible studies

Authors	Design	Participants	Region	Intervention components	Location of intervention delivery	Outcome measures
Baxter et al., 2016 [52]	Quasi-experimental	Experimental group: • Before intervention = 173 • After intervention = 269	Oromia	EBF promotion strategies including • Home visits, • Educational materials, and • Training of influential community and family members)	Health facility and community- based	EBF
Berti et al., 2010 [53]	Quasi-experimental	No information	No information	• Breastfeeding communication through community-based interventions for behaviour change	Community-based	EBF
Buchanan et al., 2012 [54]	Quasi-experimental	No information	No information	• Behavioural change communication • Community mobilization • Capacity building	Health facility-based and community-based	EIBF EBF
Callaghan-Koru et al., 2016 [55]	Quasi-experimental	Experimental group: • Before intervention=218 • After intervention=215	Tigray Oromia Amhara SNNPR	• An integrated package of facility-based kangaroo mother care • Community-based promotion of EBF	Health facility and community based	EIBF EBF
Carnell et al., 2014 [56]	Quasi-experimental	Experimental group: • Before intervention = 1016 • After intervention = 911 Control group: • Before intervention = 1074 • After intervention = 903	Oromia Amhara SNNPR	• Strengthen the health system • Improve health workers' performance • Engage the community	Health facility and community based	EBF
Kang et al., 2017 [62]	Cluster randomized trial	• Experimental group: 570 • Control group: 629	Oromia	• Community-based participatory nutrition promotion program • Daily group nutrition sessions • Follow-up visits	Community-based	Meal frequency Dietary diversity
Kim et al., 2016 [57]	Quasi-experimental	Experimental group: • Before intervention = 1481 • After intervention = 1494	SNNPR Tigray	BCC strategies including: • Advocacy and policy dialogues • Interpersonal communication and community mobilization • Mass communication	Community-based	EIBF EBF Introduction of complementary foods MDD MMF

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Kim et al., 2019 [63]	Cluster-randomized trial	Experimental group: • Before intervention = 1328 • After intervention = 1360 Control group: • Before intervention = 1318 • After intervention = 1360	Amhara region	BCC strategies including: • Interpersonal communication • Nutrition sensitive agricultural activities • community mobilization • Mass communication	Community-based	Introduction of complementary foods MDD MMF
Kung'u et al., 2018 [58]	Quasi-experimental	Experimental group: • Before intervention = 2570 • After intervention = 2905 Control group: • Before intervention =1670 • After intervention = 1620	Afar	• Behaviour change interventions • Integrating nutrition to the frontline workers' guideline • Family education and counselling booklet	Health facility	EIBF EBF at six months
Mulualem et al., 2016 [59]	Quasi-experimental	Experimental group: • Before intervention = 80 • After intervention = 80 Control group: • Before intervention = 80 • After intervention = 80	SNNPR	Nutrition education based on the HBM model • Individual and group sessions, and • Recipe demonstration	Community-based	Meal frequency
Negash et al., 2014 [60]	Quasi-experimental	Experimental group: • Before intervention = 100 • After intervention = 81 Control group: • Before intervention = 92 • After intervention = 76	SNNPR	• Nutrition education using visual materials • Demonstration of food preparation	Community-based	Dietary diversity Meal frequency
Tariku et al., 2015 [64]	Cluster randomized trial	HBM intervention group: • Before intervention = 60 • After intervention = 54 Traditional nutrition education group: • Before intervention = 60 • After intervention = 56 Control group: • Before intervention = 60 • After intervention = 56	SNNPR	• Complementary feeding messages using the HBM model	Community-based	Continued breastfeeding MDD MMF
Villadsen et al., 2016 [61]	Quasi-experimental	Experimental group: • Before intervention =775 • After intervention = 1434 Control group:	Oromia	• Training of health professionals • Health education materials • Supervisions of ANC providers	Health facility	EBF at 1 month

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		• Before intervention = 350 • After intervention = 422				

EIBF: early imitation of breastfeeding; EBF: exclusive breastfeeding; MDD: minimum dietary diversity; MMF: minimum meal frequency; MAD: minimum acceptable diet; BCC: behaviour change communication; HBM: health belief model; SNNPR: Southern Nations Nationalities and Peoples Region.

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Minimum dietary diversity

Five studies reported the effect of interventions on MDD. Of these, three measured according to the WHO definitions for MDD [57 63 64], while the remaining two studies used any four food groups as a cut-off point to measure MDD [60 62]. Two studies by Kim et al. revealed the significant impacts of exposure to more than one type of BCC interventions (advocacy and policy dialogues, interpersonal communication and community mobilization, and mass communication) on MDD, but these studies showed that exposure to a single type of intervention did not have an impact on MDD [57 63]. Tariku et al. showed interventions involving complementary feeding messages using the HBM model significantly increased the proportion of MDD (from 37.5% to 67.9%, P-value <0.05) [64]. The remaining two studies indicated that the proportion of MDD increased over the intervention period, but the effect was not significant [60 62].

Minimum meal frequency

MMF was included as an outcome variable in six interventional studies. Two studies used the WHO definitions assessing for MMF [57 63]. Kim et al. 2019 indicated that exposure to two types of BCC interventions (advocacy and policy dialogues, interpersonal communication and community mobilization, and mass communication) had a significant effect on MMF of infants and young children compared to those who did not expose to intervention (OR = 1.69; 95% CI: 1.13, 2.52) [63]. Carnell et al. found that interventions using a community-based participatory nutrition program significantly increased the proportion of MMF over the intervention period [62]. Except for one study, which did not do a significant test [60], the remaining two studies did not show significant increases in MMF [59 64].

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Table 2: Summary of evidence from reviewed studies

IYCF outcomes	Studies by authors	Statistical analysis	Results
EIBF	Buchanan et al., 2012 [54]	Percent change in proportion	• EIBF increased from 31.0% to 77.0%
	Callaghan-Koru et al., 2016 [55]	Percent change in proportion	• EIBF significantly increased from 51.4% to 67.9% (P value <0.01)
	Kim et al., 2016 [57]	Fixed-effect logistic regression	• Single type of intervention: EIBF (OR = 1.88; 95% CI: 1.08, 3.27) • Two types of interventions: EIBF (OR = 2.07; 95% CI: 1.09, 3.91)
	Kung'u et al., 2018 [58]	Fixed-effect logistic regression	• EIBF (OR = 1.70; 95% CI: 1.17, 2.46)
EBF	Baxter et al., 2016 [52]	Percent change in proportion	• EBF increased from 69.4% to 75.1%, but not statistically significant
	Berti et al., 2010 [53]	Percent change in proportion	• Higher improvement in exclusive breastfeeding
	Buchanan et al., 2012 [54]	Percent change in proportion	• EBF increased from 73.0% to 95.0%
	Callaghan-Koru et al., 2016 [55]	Percent change in proportion	• EBF increased from 86.0% to 95.8% (P value <0.01)
	Carnell et al., 2014 [56]	Difference-in-difference	• No significant difference in the improvement of EBF was observed
	Kim et al., 2016 [57]	Fixed-effect logistic regression	• No significant effect on EBF
	Kung'u et al., 2018 [58]	Fixed-effect logistic regression	• EBF at six months (OR = 2.62; 95% CI: 0.07, 1.03)
	Villadsen et al., 2016 [61]	Fixed-effect logistic regression	• EBF at 1 month (OR = 2.3; 95% CI: 1.4, 3.6)
The introduction of complementary foods	Kim et al., 2016 [57]	Percent change in proportion	• Introduction of complementary foods increased from 37.4% to 59.7% (P-value <0.001)
	Kim et al., 2019 [63]	Percent change in proportion	• The introduction of complementary foods did not show any improvement
Continued breastfeeding at 1–year	Tariku et al., 2015 [64]	ANOVA and Kruskal-Wallis test	• No significant difference in the improvement of continued breastfeeding
Dietary diversity	Kang et al., 2017 [62]	Mixed-effect logistic regression	• No significant improvement in dietary diversity
	Kim et al., 2016 [57]	Fixed-effect logistic regression	• Single type intervention: No significant effect on MDD • Two types of interventions: MDD (OR = 6.87; 95% CI: 1.27, 37.14)
	Kim et al., 2019 [63]	Fixed-effect logistic regression	• Single type intervention: MDD (OR = 1.05; 95% CI: 0.74, 1.48) • Two type of interventions: MDD (OR = 1.82; 95% CI: 1.14, 2.89)
	Negash et al., 2014 [60]	Independent sample t-test	• Dietary diversity increased over the intervention period
	Tariku et al., 2015 [64]	ANOVA and Kruskal-Wallis test	• Intervention group showed significant improvement in MDD (P<0.001)
Meal frequency	Kang et al., 2017 [62]	Mixed-effect logistic regression	• There was a significant improvement in meal frequency
	Kim et al., 2016 [57]	Fixed-effect logistic regression	• No significant effect on MMF
	Kim et al., 2019 [63]	Fixed-effect logistic regression	• Single type of intervention: MMF (OR = 1.05; 95% CI: 0.90, 1.23) • Two types of interventions: MMF (OR = 1.69; 95% CI: 1.13, 2.52)
	Mulualem et al., 2016 [59]	ANOVA	• MMF frequency decreased in both experimental and control groups
	Negash et al., 2014 [60]	Independent sample t-test	• Meal frequency increased over the intervention period
	Tariku et al., 2015 [64]	ANOVA and Kruskal-Wallis test	• No significance effect on MMF

IYCF: infant and young child feeding; EIBF: early imitation of breastfeeding; EBF: exclusive breastfeeding; MDD: minimum dietary diversity; MMF: minimum meal frequency; MAD: minimum acceptable diet; HBM: health belief model.

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Discussion

A total of thirteen studies were included, and of these, ten were quasi-experimental studies, and three were cluster randomised trials. Five studies for EBF and four studies for EIBF showed that community- and facility-based interventions delivered in the form of BCC (such as policy advocacy, interpersonal communication, community mobilisation, and mass media campaigns) and health service strengthening had an impact on infants. Three studies for MDD and two studies for MMF indicated significant effects of BCC on infants and young children. Two eligible studies showed that combined interventions delivered in the form of interpersonal communication, nutrition-sensitive agricultural activities, community mobilisation, and mass media campaigns had more impacts on EIBF, MDD, and MMF.

Global evidence indicates that BCC interventions (including policy advocacy, interpersonal communication, community mobilisation and mass media campaigns) are essential in promoting IYCF practices. These initiatives also create a supportive environment that helps mothers to adopt and sustain positive child feeding behaviours [11 66]. The main goals of BCC intervention include improving behaviours of a facility- and community-level health workers, mothers and other caregivers, and members of their households [11 66]. Our review showed that BCC interventions in the form of policy advocacy, interpersonal communication, community mobilisation, and mass media campaigns improved EIBF, EBF, MDD and MMF. Our finding is consistent with systematic reviews from LMICs, which indicated that group nutritional education and counselling, family or social support, integrated mass media coverage, and community mobilization were effective in improving IYCF practices [11 67]. Although BCC interventions promote skills and knowledge that are necessary for improving IYCF practices, achieving the intended target is often challenging without complementary interventions that provide or facilitate material inputs [66]. Our review suggests that interventions that promote

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material inputs (such as cash transfer and provision of food materials) and/or non-material inputs (skills and knowledge) are required in achieving appropriate and sustained IYCF behaviours.

Research has shown that exposure to more than one IYCF interventions, delivered through different platforms (e.g., community- or facility-based), were more successful in improving IYCF outcomes compared to a single form of intervention delivered at a single setting [5]. Our review indicated that exposure to more than two types of interventions was associated with improvement in EIBF, MDD, and MMF [57 63]. This result may be due to that intervention delivered through multiple settings may have an increased intensity that can be translated to behavioural changes due to repeated contacts and adequate coverage of interventions [68]. Our review suggests that interventions that aim to improve IYCF behaviours in Ethiopia should be delivered through different and specific types of contexts (including household, health facility and community) and should be provided along the continuum of care from conception until a child's second birthday.

In the past two decades, the Ethiopian government's commitment to improve IYCF has been consistent, and this started with the development of the National Strategy for IYCF in 2004 and followed by National Strategy for Nutrition in 2008 [24 69], in line with the WHO/UNICEF IYCF strategy [70]. In 2015, the Ethiopian government launched the Health Sector Transformation Plan to improve health outcomes, including IYCF practices [30]. Based on this initiative, many facility- and community-based interventions have been implemented to protect, promote and support appropriate IYCF behaviours [10 11]. Nevertheless, our review found that there has been no formal IYCF intervention conducted in Addis Ababa and Dire Dawa city administrations, and Harari, Benishangu1, Gambella, and Somali regions. Our review suggests

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an expansion of additional experimental studies to maximise the opportunities for the prevention of IYCF-related morbidity and mortality in those regions.

Additionally, our review did not find any studies which incorporated the use of mobile health (mHealth) as part of an intervention package for improving IYCF [71]. Future experimental studies that consider the use of mHealth techniques are required, especially that previously published studies have shown the positive impacts of interventions (in form of voice messages, SMS, and phone calls) on IYCF practices [72-74]. Finally, interventional studies that consider the role of fathers/partners as well as grandmothers in improving IYCF behaviours may be warranted in Ethiopia [75-77].

The present systematic review of IYCF outcomes from interventions has strengths and limitations. First, a meta-analysis was not conducted due to the wide heterogeneity of the design and outcome measures. Second, this review was limited by the incompleteness of information from the individual studies. Nevertheless, the study authors were contacted for information when it is necessary. Third, our search was limited to published studies in English, which exclude studies published with other languages. Fourth, lack of evidence and under-representation of some regions may limit the generalizability of our review. However, evidence from this review is potentially relevant for IYCF policy efforts in Ethiopia. Despite the above limitations, the application of multiple search strategies in consultation with the medical librarian and registration of the systematic review protocol can be considered as strengths of this review.

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Conclusion

Our systematic review showed that interventions delivered in the form of policy advocacy, health service strengthening, interpersonal communication, community mobilisation, and mass media campaigns improved EIBF, EBF, MDD, and MMF in Ethiopia. The impacts of current and/or future interventions on IYCF practices in Ethiopia will be greater when interventions are combined, introduced and implemented along the continuum of care from conception until a child's second birthday.

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List of abbreviations

ANC: Antenatal Care; OR: Odds Ratio; BFHI: Baby-Friendly Hospital Initiative; CI: Confidence Interval; EBF: Exclusive Breastfeeding; EIBF: Early Initiation of Breastfeeding; HBM: Health Belief Model; HSTP: Health Sector Transformation Plan; IYCF: Infant and Young Child Feeding; JBI: Joanna Briggs Institute; LMICs: Lower and Middle-Income Countries; MAD: Minimum Acceptable Diet; MDD: Minimum Dietary Diversity; MeSH: Medical Subject Headings; MDG: Millennium Development Goal; MMF: Minimum Meal Frequency; Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA); RCTs: Randomised Controlled Trials; SDGs: Sustainable Development Goals; SNNPR: Southern Nations Nationalities and Peoples Regions; SUMARI: System for the Unified Management, Assessment and Review of Information; UNICEF: United Nation Children's Fund; WHO: World Health Organization

Declarations

Contributors

KYA conceptualised the study idea, performed the literature searching, selected the paper of interests, performed data extraction, and critically revised the manuscript. AP and AA critically revised the manuscript for intellectual content. KEA performed the risk of bias assessment, and critically revised the manuscript. FAO contributed to the conception of the research idea, selected the paper of interests, performed data extraction, and critically revised the manuscript. All authors read and approved the final manuscript.

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The authors declare that they have no competing interests

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Figure 1: Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flowchart for selection of eligible studies

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Supplementary appendix

Supplementary file 1: PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) 2009 Checklist.

Supplementary file 2: Search strategy developed for Ovid/Embase database

Supplementary file 3: Risk of bias assessment for quasi-experimental studies

Supplementary file 4: Risk of bias assessment for cluster randomised trial studies

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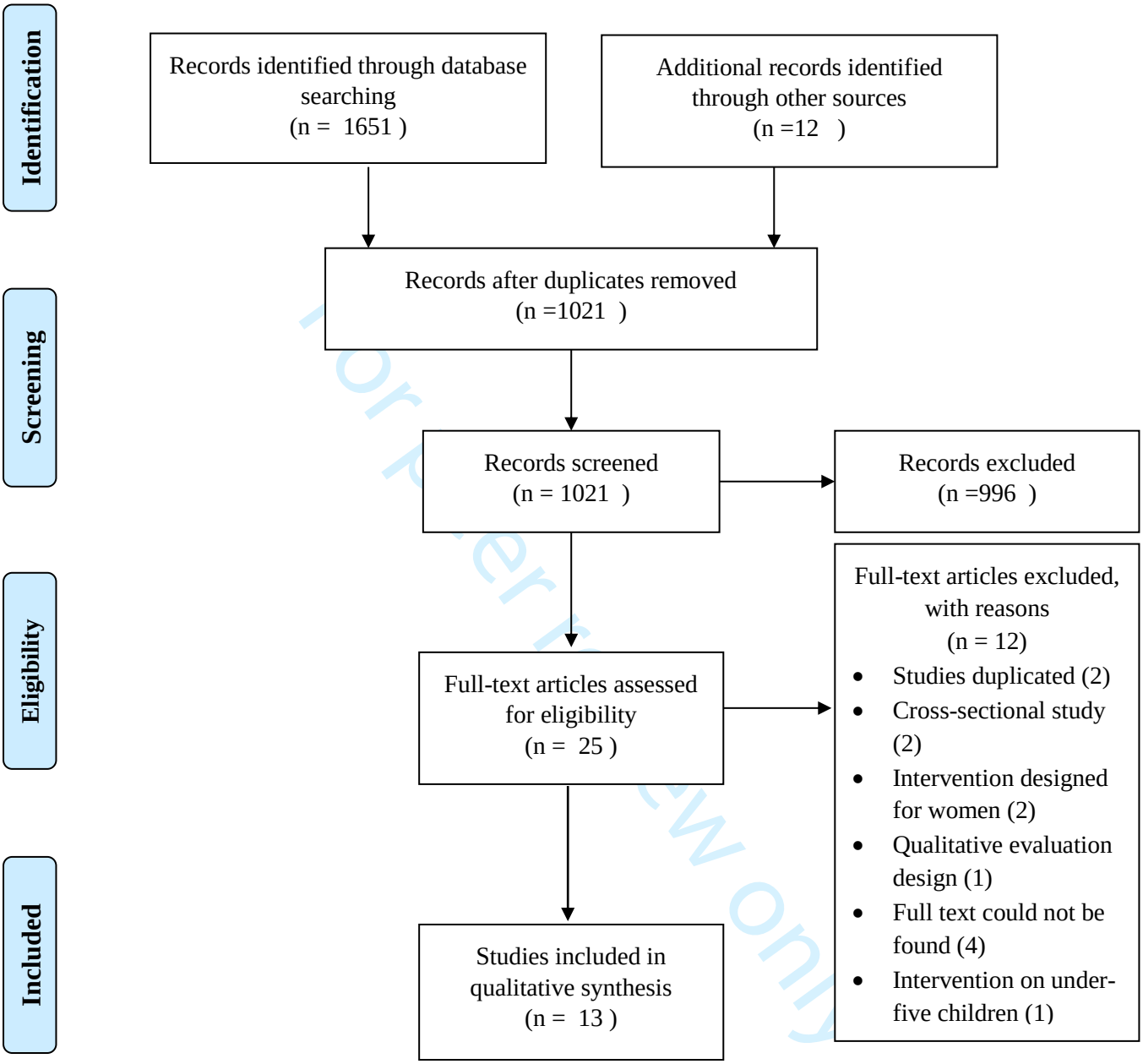


Figure 1: Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flowchart for selection of eligible studies [40]

PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2, 3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4, 5
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5 Paragraph 2
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	5 Paragraph 3
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6, 7
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	8 Paragraph 1
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	8, 9, Supplementary file 2
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	9
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	10 Paragraph 1
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	10 Paragraph 1
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	10 Paragraph 2
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	N/A
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	11 Paragraph 2

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	10 Paragraph 2
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	N/A
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	11 Paragraph 4
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	12 Paragraph 1
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	12 Paragraph 2
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	13–20
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	N/A
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	12 Paragraph 2
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	N/A
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	21–23
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	23 Paragraph 3
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	24
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	26 Paragraph 2

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

For more information, visit: www.prisma-statement.org.

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2	health education/	104392	Advanced	Display Results | More	
3	exp nutritional counseling/	2927	Advanced	Display Results | More	
4	exp child health care/	98696	Advanced	Display Results | More	
5	exp maternal care/	48783	Advanced	Display Results | More	
6	exp community care/	129059	Advanced	Display Results | More	
7	exp health auxiliary/	7604	Advanced	Display Results | More	
8	model/ or Health Belief Model/ or Health Promotion Model/	506507	Advanced	Display Results | More	
9	exp prenatal care/	161968	Advanced	Display Results | More	
10	postnatal care/	7846	Advanced	Display Results | More	
11	exp peer group/	25748	Advanced	Display Results | More	
12	exp nutrition education/	5803	Advanced	Display Results | More	
13	exp crop/	20482	Advanced	Display Results | More	
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23	exp breast feeding/	57955	Advanced	Display Results | More	
24	exp infant feeding/	65117	Advanced	Display Results | More	
25	exp complementary feeding/ or exp feeding/	219788	Advanced	Display Results | More	
26	exp diet supplementation/	90123	Advanced	Display Results | More	
27	exp meal/	24403	Advanced	Display Results | More	
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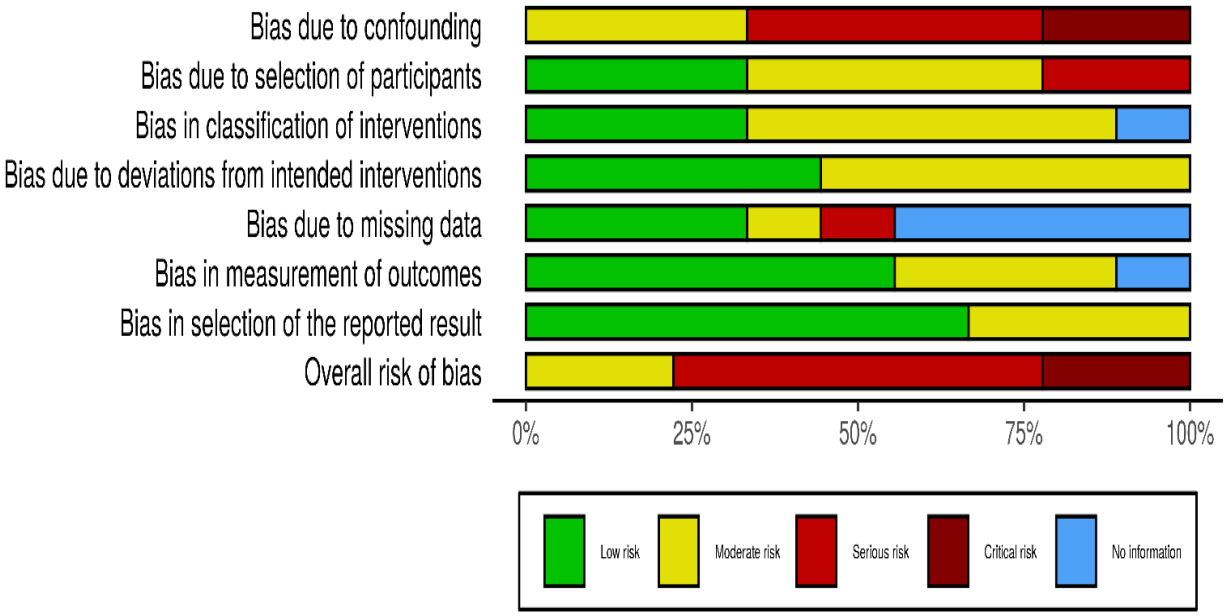
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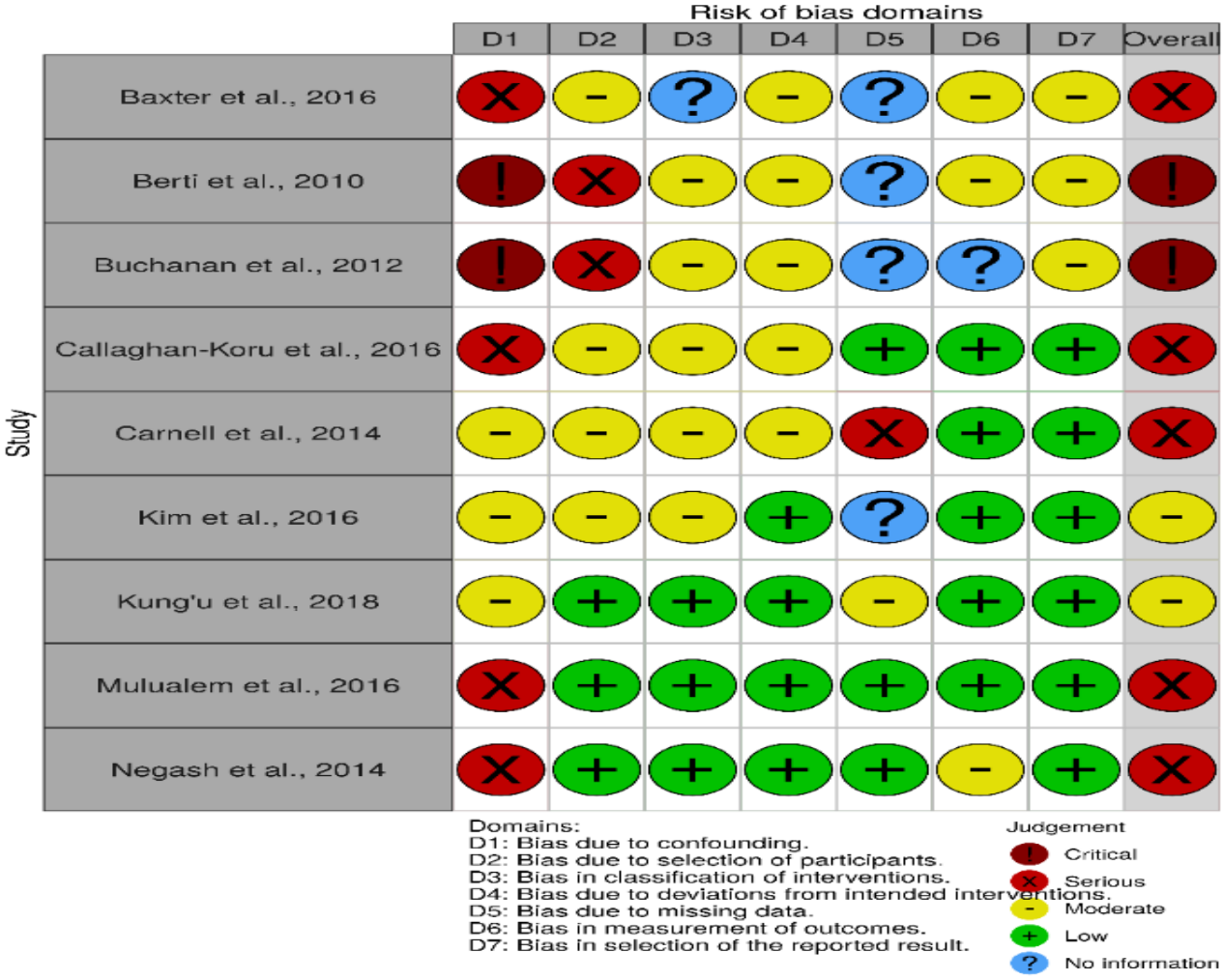
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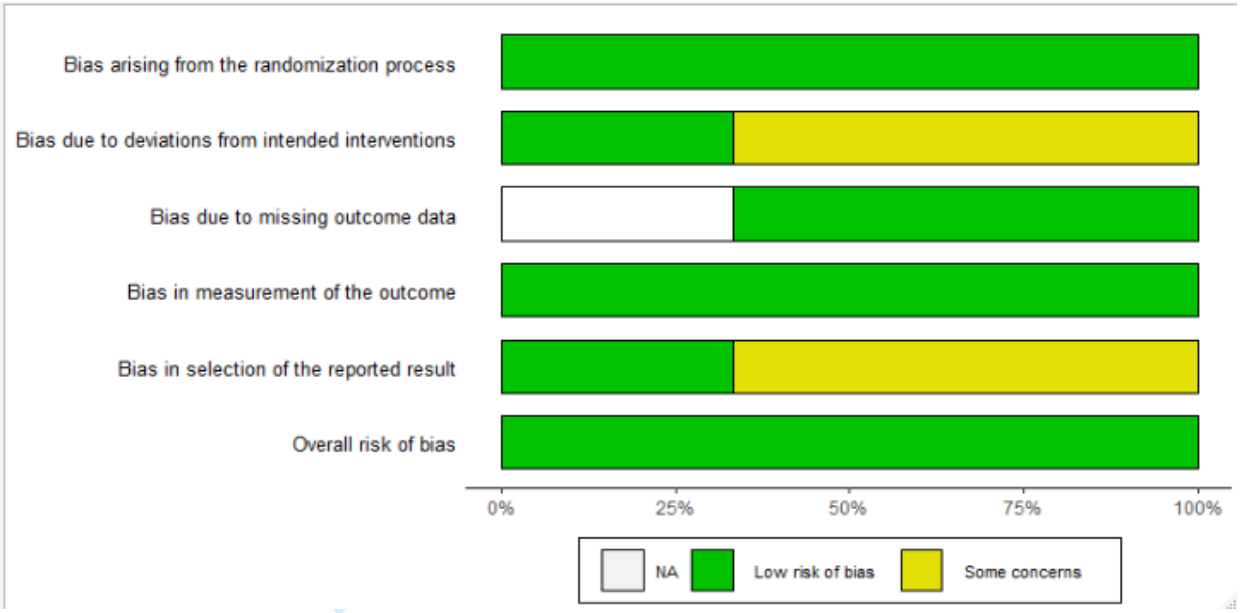
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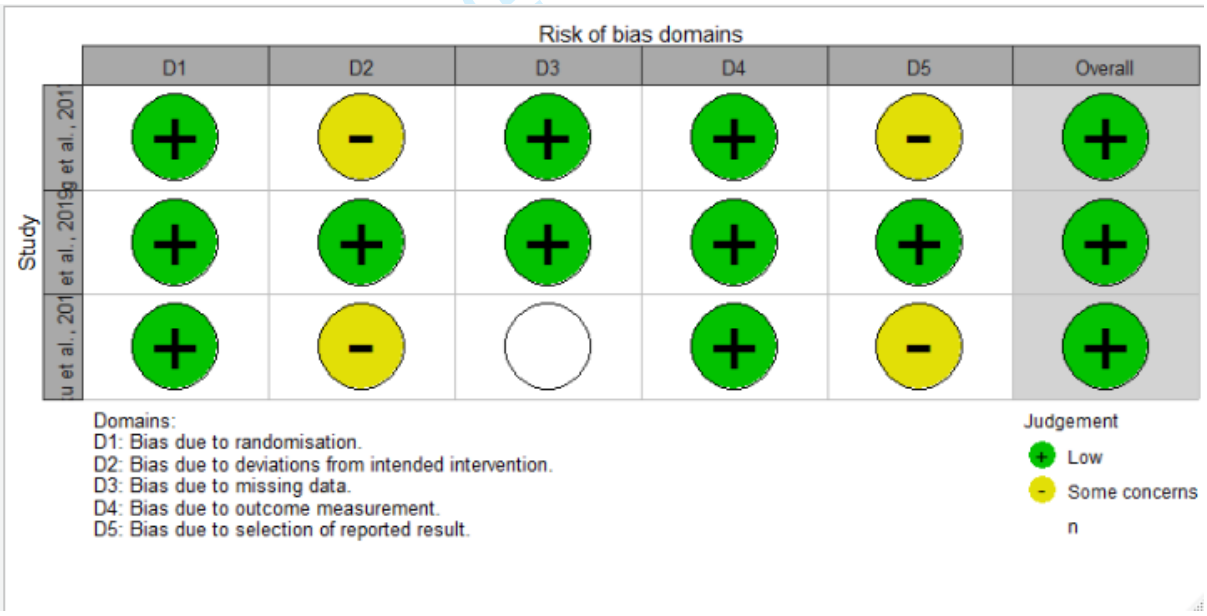
Risk of bias graph: review authors' judgements about each risk of bias item presented as percentages across all included quasi-experimental studies



Risk of bias summary: review authors' judgements about each risk of bias item for each included quasi-experimental study



Risk of bias graph: review authors' judgements about each risk of bias item presented as percentages across all included cluster randomised studies



Risk of bias summary: review authors' judgements about each risk of bias item for each included cluster randomised study

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Interventions to improve infant and young child feeding practices in Ethiopia: a systematic review

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[^]Members of GloMACH are provided in the acknowledgements.

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Abstract

Objective: In Ethiopia, despite the implementation of several interventions to improve infant and young child feeding (IYCF) practices, no published studies have highlighted the most effective IYCF interventions in the country. This systematic review investigated the impacts of various interventions on IYCF in Ethiopia.

Source of information: A systematic search was conducted on seven computerized bibliographic databases (Embase, ProQuest Central, Scopus, PsycINFO, Web of Science, MEDLINE/PubMed and CINAHL) to locate experimental or quasi-experimental studies between November 2019 and May 2021.

Study eligibility criteria: Interventional studies that measured IYCF indicators (early initiation of breastfeeding [EIBF], exclusive breastfeeding [EBF], the introduction of complementary foods, minimum dietary diversity [MDD], minimum meal frequency [MMF] and minimum acceptable diet [MAD]) as outcome variables were included.

Study appraisal and synthesis: All included studies were examined for biases related to interventional studies (i.e., selection bias, performance bias, attrition bias, detection bias and reporting bias). Author reports of effect size measures were used to narratively report the findings of each study.

Results: Of the 23 eligible studies, 14 studies were quasi-experimental and nine studies were cluster randomised trials (CRTs). Eight quasi-experimental studies had a serious risk of bias, while two CRTs had a high risk of bias. Five studies for EBF and six studies for EIBF showed significant impacts of policy advocacy, health service strengthening, interpersonal

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communication, community mobilisation, and mass media campaigns. Six studies for MDD and three studies for MMF indicated significant effects of community-level and health facility complementary feeding promotions on infants and young children. Interventions that were delivered in combination increased the impacts in improving EIBF, MDD and MMF compared to a single intervention.

Conclusion: Our review showed that 13 out of 21 eligible studies that were implemented in the form of community-level and health facility interventions improved EIBF, EBF, and/or MDD in Ethiopia.

Keywords: Infant and young child feeding, breastfeeding, nutritional education, intervention, systematic review, Ethiopia

Protocol registration number: PROSPERO CRD42020155519.

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Strengths and limitations of the study

- A meta-analysis was not conducted due to the wide heterogeneity of the design and outcome measures.
- This review was limited by the incompleteness of information from the individual studies. Nevertheless, the study authors were contacted for information when it is necessary.
- Despite the limitations, the application of multiple search strategies in consultation with the medical librarian and registration of the systematic review protocol can be considered as strengths of this review.

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Introduction

Appropriate infant and young child feeding (IYCF) improves childhood growth (through reduced risk of diarrhoea and respiratory infections as well as improved nutrition) [1-7], increases the intellectual capacity of children [4 8], and reduces the risk of mothers to experience diabetes mellitus, overweight and obesity [4 9]. Promoting appropriate IYCF using evidence-based interventions is essential to protect and support maternal knowledge, beliefs, and confidence in IYCF practices [10-13]. Community- and facility-based interventions that range from direct support and education to mothers and families on appropriate IYCF behaviours to training of health professionals can significantly improve IYCF behaviours [14-16].

Global efforts to improve maternal IYCF behaviours have increased in the past three decades. These efforts included the International Code of Marketing of Breast-milk Substitutes [17]; the Innocenti Declaration [18]; the Baby-Friendly Hospital Initiative (BFHI) [19]; the Millennium Development Goals (MDGs) [20]; and more recently, the Global Nutrition Targets 2025 [21] and the Sustainable Development Goals (SDGs) [22]. The World Health Organization and United Nations Children's Fund (WHO/UNICEF) have played major roles in driving these initiatives which aimed to protect, promote, and support breastfeeding and child nutrition [19]. For example, the WHO Global Nutrition Targets (WHO GNT, including Target-1 and -5) aim to reduce the global number of stunted children under-5 by at least 40% and increase the EBF rate to at least 50% by the year 2025, respectively. However, most subnational areas of many African countries (including Ethiopia) are not on track to achieve WHO GNT goal-5 [23].

In Ethiopia, while both government and non-governmental organisations have implemented various maternal and child health-related policy interventions over the last two decades to increase infant feeding [24-26], a recent study has revealed that early initiation of breastfeeding

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(EIBF) prevalence was 75.5% and that for exclusive breastfeeding (EBF) was 59.9% [27], with a wide heterogeneity across subnational areas of the country [23]. These rates are well below the Ethiopian Health Sector Transformation Plan (HSTP) target of 90% and 72%, respectively [25]. Additionally, the proportion of children who met minimum dietary diversity (MDD) and minimum meal frequency (MMF) was 7.1% and 43.6%, respectively [28]. In Ethiopia, more than five million children were stunted in 2016 [29], and this is a major public health issue that is largely attributable to inappropriate IYCF.

To date, no previously published studies have examined the most effective IYCF interventions in the Ethiopian context. A detailed understanding of the specific types of interventions that can be implemented to improve IYCF in Ethiopia is essential, to specifically guide the allocation of scarce resources and the provision of targeted strategies to increase childhood feeding. This information will also be helpful in scaling-up and/or improve upon current and future IYCF interventions in the country. Data from the assessment of relevant IYCF interventions would be crucial to stakeholders as Ethiopia implements national nutrition efforts (e.g., HSTP) [25] and global health programs such as the Sustainable Development Goal, SDG-2.2, to end all forms of malnutrition [22]. Accordingly, this systematic review investigated the impacts of community- and facility-based interventions in improving IYCF practices of mothers/caregivers with children under two years of age in Ethiopia.

Materials and methods

Our review adhered to the 2009 Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [30] **(Supplementary file 1)**, and the protocol for the review was registered in PROSPERO (International Prospective Register of Systematic Reviews, CRD42020155519) [31].

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Eligibility criteria

Studies were included if they meet the following criteria: (i) experimental or quasi-experimental study design (including randomized controlled trials [RCT], non-RCTs, before and after studies, and interrupted time-series studies) with or without comparison groups or cluster; (ii) conducted among pregnant women and mothers living with live children under 2 years to improve IYCF practices; (iii) used any type of IYCF interventions delivered between the antenatal period and 24 months postpartum; (iv) measured at least one core IYCF indicator such as EIBF, EBF, the introduction of solid, semi-solid and soft foods, continued breastfeeding at one year, MDD, MMF, and minimum acceptable diet (MAD); (v) were published in the English language; and (vi) conducted in Ethiopia from the year 2000 to date. We considered studies published over this period to demonstrate the impacts of the specific IYCF interventions in the context of Ethiopia to reflect the implementation of relevant global health programs, including the MDGs and the current SDGs [23].

The intervention strategies considered in this review focused on those performed at both facility- and community-levels. The interventions included nutritional education and counselling, interpersonal communication, mass media campaigns, nutrition-sensitive agricultural activities, group demonstration sessions, and strengthening of health facilities.

Studies conducted among mothers with health problems such as HIV/AIDS and children with cleft palate were excluded.

The IYCF practices included EIBF, EBF, the introduction of solid, semi-solid and soft foods, continued breastfeeding at one year, MDD, MMF, and MAD, measured according to the WHO definitions for assessing IYCF indicators [32]. These indicators were selected based on their effect on the health and development of infants and young children [11, 12, 35] and the evidence from previously published systematic reviews [11 33].

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- EIBF was defined as the proportion of children aged 0–23 months who commenced breastfeeding within the first hour of birth.
- EBF was defined as the proportion of infants 0–5 months of age who were fed no other food or drink, not even water, except breast milk (including milk expressed or from a wet nurse), but allows the infant to receive Oral Rehydration Salt (ORS), drops, and syrups (vitamins, minerals, and medicines).
- Continued breastfeeding at one year was defined as the proportion of children 12–15 months of age who are fed breast milk.
- Introduction of complementary foods (solid, semi-solid and soft foods) was defined as the proportion of infants 6–8 months of age who received solid, semi-solid or soft foods in the previous 24 hours, during the day and at night.
- MDD was defined as the proportion of children 6–23 months of age who received foods from four or more of the seven food groups. The seven food groups included grains, roots and tubers; legumes and nuts; dairy products (milk, yogurt, cheese); flesh foods (meat, fish, poultry and liver/organ meats); eggs; vitamin-A rich fruits and vegetables; and other fruits and vegetables.
- MMF defined as the proportion of breastfed and non-breastfed children 6–23 months of age, who received solid, semisolid or soft foods (including milk feeds for non-breastfed children) the minimum number of times or more (i.e., two times for breastfed infants aged 6–8 months, three times for breastfed children aged 9–23 months and four times for non-breastfed children aged 6–23 months, on the previous day).
- MAD defined as the proportion of children 6–23 months of age who received both minimum dietary diversity and minimum meal frequency.

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Information sources and search strategy

Consistent with previously published systematic reviews [15 34], a three-stage search strategy was applied to locate both published and unpublished studies. In stage one, a manual search of previously published systematic reviews and the reference lists of all the eligible studies was performed to check for systematic reviews of interventions on IYCF in Ethiopia. In stage two, index terms (e.g., medical subject headings [MeSH] for Medline) and key terms were generated using the identified articles from the initial search for the full search strategy. In stage three, a full search strategy was conducted on seven computerized bibliographic databases (Embase [OVID], ProQuest Central, Scopus, PsycINFO [EBSCO], Web of Science, MEDLINE [OVID]/PubMed and Cumulative Index to Nursing and Allied Health Literature (CINAHL)/[EBSCO] to locate published studies, and secondary searching of OpenGrey and Grey Literature Report was conducted to locate unpublished works. The Population Intervention Comparator Outcome (PICO) criteria were used to devise the review question and related search terms. A combination of specified medical subject headings (MeSH) terms and keywords was drafted with some assistance from our institution-based medical librarian. The search strategy, including all identified index and keywords terms, was slightly adapted for each of the information sources. The initial and the top-up searches were conducted in November 2019, and May 2021, respectively. A full electronic search strategy developed for the Embase (OVID) is provided as **Supplementary file 2**.

The overall list of search terms used included:

Term 1 (population): child, infant, newborn, baby, neonate, perinatal, postnatal, kid, toddler, young child, paediatric, mother, female, women, and caregiver

Term 2 (interventions):

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- ***Education and counselling***: breastfeeding promotion, breastfeeding support, breastfeeding education, health education, health promotion, nutrition education, food education, parent education, mother education, counselling, and nutritional counselling
- ***Social behavioural change communication***: health behaviour, health related behaviour, behaviour change, communication, interpersonal communication, information education communication, behaviour change communication, social change, social movement, social mobilization, social behaviour, social network, peer group, advocacy, advocacy group, mass communication, mass media, print media, mobile phone, mHealth, eHealth, internet, radio, social media, television, text message, and social market
- ***Community based approaches****:* baby friendly community initiative, community program, community project, home visit, community health action, community health service, community health care, community intervention, community engagement, community leader, community mobilization, demonstration, cooking demonstration, community role-play, model breastfeeding community, community health worker, and health extension worker
- ***Facility based approaches****:* maternal care, child health care, child health service, paediatric health care, neonatal care, newborn care, rooming-in-care, newborn nursery, essential nutrition action, caregiver contact, baby-friendly hospital initiative, BFHI, antenatal care, postnatal care, Kangaroo care, and skin-to-skin contact
- ***Intervention designs****:* effectiveness, impact, evaluation study, program evaluation, health care program, project, health project, experimental study, interventional study, quasi-experimental study, randomized controlled trial, clinical trial, cluster randomized trial, time-series study, control, placebo, comparison, and usual care

Term 3 (Context): Ethiopia

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Term 4 (Outcomes): infant and young child feeding, breastfeeding, early initiation of breast feeding, timely initiation of breastfeeding, exclusive breastfeeding, breast milk substitute, complementary food, complementary feeding, weaning, introduction of complementary food, baby food, infant feeding, child feeding, meal, dietary diversity, meal frequency, child nutrition, infant nutrition, nutritional supplement, dietary supplement, lactation, and nursing

Study selection

Following the full search, all articles retrieved from the search databases were exported to EndNote X9 (Clarivate Analytics, USA) for removing duplicates, screening, and selection. Titles and abstracts were screened by two reviewers (KYA and FAO) independently for assessment against the inclusion criteria. Potentially relevant articles that were selected in the first screening phase were retrieved in full and their citation details imported into the Joanna Briggs Institute System for the Unified Management, Assessment and Review of Information (JBI SUMARI). Full texts of the included studies were read and checked by two independent reviewers (KYA and FAO) against the eligibility criteria. Studies that did not meet the eligibility criteria were excluded and the reasons for exclusion of studies recorded and reported in the systematic review. Any disagreements between the two independent reviewers were resolved by consensus and arbitration by the third (KEA) author of the review. For studies with uncertainties regarding the eligibility, the authors were contacted to seek additional information. A total of two contact attempts were made, and if no response was received, the articles were screened for eligibility based on the information available. The results of the study selection process are presented in a PRISMA flow diagram (Figure 1).

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Data collection process and data items

Relevant data were extracted from the eligible studies by KYA, and independently verified by FAO using piloted forms adapted from the Joanna Briggs Institute (JBI) reviewer's manual [35] and the Cochrane Pregnancy and Childbirth Group for Systematic Reviews [36]. Eligible studies were identified using the following information: first author, publication year, study design and setting, study participants, sample size, geographical region, intervention components, location of intervention delivery, outcome measures, statistical analysis, and results. Corresponding authors of eligible studies were contacted for information that was not clear in the data extraction. A total of two contact attempts were made, and if no response was received, only the information available was used.

Risk of bias Assessment

All included studies examined for the five types of biases of interventional studies: (i) selection bias; (ii) performance bias; (iii) attrition bias; (iv) detection bias and; (v) reporting bias. KYA and KEA independently conducted the risk of bias assessment using the revised Cochrane risk of bias tool for clustered randomized trials (ROB 2.0) [37 38] and the risk of bias in non-randomised studies - of interventions (ROBINS-I) tool [39] for assessing the risk of bias in quasi-experimental studies. Cluster randomised trials were assessed for five domains (i.e., randomization and identification or recruitment bias, deviations from intended interventions, missing data, measurement of outcomes, and selection of reported result), and quasi-experimental studies were assessed for seven domains (i.e., confounding, selection of participants, classification of interventions, deviations from intended interventions, missing data, measurement of outcomes, and selection of reported result). For cluster randomised trials, the overall risk of bias judgement included low risk, some concerns and high risk [37 38], and for the quasi-experimental studies, the overall risk of bias outcome included low, moderate,

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serious, and critical risk of bias [39]. There was no disagreement between reviewers in the risk of bias assessment, no third arbitrator was needed.

Data synthesis

Author reports of effect size measures were used in reporting the findings of each study. Effect size measures included a comparison between experimental and control groups (e.g., odds ratio [OR]), as well as the difference in proportion between pre- and post-outcome measures (e.g., change in proportion). Where available, 95% confidence interval (CI) and P-value were obtained from the eligible studies. Based on published methodological studies [40 41], 95% CI and P-value were calculated using the available information where the full text of the eligible studies did not report these estimates. Narrative methods including text, table, and figure were also used as a tool for data presentation.

Patient and public involvement

There was no direct patient or public involvement in this review

Results

Characteristics of eligible studies

A total of 4017 articles were retrieved from the database and hand searching. Of these, 1924 were retained in the Endnote library after removing duplicates. Titles and abstracts of the articles were screened, and 1654 articles were removed. Full-text eligibility checks of the remaining 52 articles excluded 29 studies. The reasons for exclusion of studies included: 4 studies were duplicates[42-44], 1 study was cross-sectional[45], 2 studies used qualitative evaluation [46 47], 10 studies were abstracts [48-57], 4 studies were based on the design and implementation of

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interventions[58-61], 5 studies did not measure IYCF outcomes [62-70], and 3 studies were based on interventions designed for different target groups [71-73] **(Figure 1).**

Among the 23 included studies, 14 were quasi-experimental studies [74-87], while the remaining nine were cluster randomized trials [88-96]. At least one interventional study was conducted in Southern Nations Nationalities and Peoples' Region (SNNPR) [77-79 81 82 84-86 90], Oromia[74 77 78 83-88 91 95], Amhara [76-78 84-87 89 92 93 95], Tigray [77 79 84-87 95], and Afar [80] regions, but we did not find published interventional studies conducted in either Addis Ababa and Dire Dawa city administrations, Harari, Benishangu1, Gambella, or Somali regions.

Risk of bias in included studies

Supplementary files 3 and 4 present the risk of bias assessment of the included studies. Nine cluster randomised trials were assessed for risk of bias using the Cochrane RoB 2.0 tool. Overall, studies by Muluye et. al., 2020 and Passarelli et. al., 2020 judged as high risk of bias[94 95], while studies by Abdulahi et. al., 2021 and Kim et. al., 2019 had a low risk of bias [89 91]. Two cluster randomised studies had a high risk of bias in the domains for randomisation [94 95]. Out of 14 quasi-experimental studies assessed using the ROBINS-1 tool, eight studies had a serious risk of bias [74 77 78 81-84 86], and four studies had a moderate risk of bias [79 80 85 87]. The remaining two studies judged as critical risk of bias [75 76]. Seven out of fourteen studies had a serious or critical risk of bias in the confounder adjustment [74-78 81-83], while eight out of fourteen studies had a low risk of bias in outcome measurement [77-81 84 85 87].

Evidence from reviewed studies

In the eligible studies, n = 23, the majority of the interventions were provided in the form of behaviour change communication (BCC) using nutrition education [81 82 92 93 95], community

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mobilisation [76 78 87 89], health education materials [74 80 82 83 91], and mass media communication [79 89]. Other interventions were also provided in the form of peer/group support and training of mothers [91-93], promotion of chicken husbandry and home gardening [95], productive safety net program (in-cash/food rations) [84], strengthening of health care system [78 80 85-87], training and supervision of health care providers [76 78 83], and food preparation and practical demonstration sessions [81 82 88 91 94 96]. Additionally, two studies used the health belief model (HBM) as a mode of intervention delivery for BCC [81 90] **(Table 1)**.

Sixteen studies used control groups for comparison with experimental groups [78 80-84 87-96], while seven studies did not use a control or comparison group[74-77 79 85 86]. Two studies used two experimental arms as an intervention group [90 95] **(Table 1)**.

Impact of interventions on IYCF practices in Ethiopia

EIBF

Tables 1 and **2** are presented to summarise evidence from eligible studies. Eight studies reported the effect of interventions to improve EIBF [76 77 79 80 85-87 91]. A health facility level intervention in the Afar region that integrated nutrition information into health professional guidelines, and also provided family education and counselling booklets resulted in a significant improvement in EIBF in the experimental group compared to the control group (OR = 1.70; 95% CI: 1.17, 2.46; P-value = 0.005) [80]. Kim et al., 2016 showed that interventions delivered through advocacy and policy dialogues, interpersonal communication and community mobilization, and mass communication improved EIBF [79]. This study also showed that the association between exposure to two or more types of interventions and improvement in EIBF was stronger compared to exposure to a single type of intervention (AOR = 2.07; 95% CI: 1.09, 3.91; P-value < 0.05 for multiple intervention vs AOR = 1.88; 95% CI: 1.08, 3.27; P-

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value < 0.05 for single intervention) [79]. However, Wereta et. al., 2018 reported no significant effect of community-based maternal and child health service quality improvement project on EIBF (average treatment effect [ATE] = 2.8; 95% CI: −4.7, 10.6; P-value = 0.468)[87].

Buchanan et. al., 2012 [76], Callaghan-K. et. al., 2016 [77], and Karim et. al., 2013 [85] found an increase in the proportions of EIBF (from 31.0% to 77.0%, 51.4% to 67.9%, and 46.0% to 54.2%, respectively), following the implementation of BCC and community-level health programs including breastfeeding promotions and family health services.

EBF

Ten studies reported the effect of interventions on EBF in Ethiopia [74-80 83 85 91]. Of these, eight studies measured EBF for up to six months[74-80], while two studies measured EBF for up to one month [83 85]. Villadsen et. al., 2016 found that antenatal care strengthening for pregnant women significantly improved EBF in health facilities under intervention compared to health facilities without intervention (OR = 2.30; 95% CI: 1.40, 3.60; P-value = 0.005) [83]. Integrating nutrition into the health workers' guidelines, and providing family education and counselling booklet (OR = 2.62; 95% CI: 0.95, 7.21; P-value = 0.062) [80], and exposure to BCC intervention (i.e., advocacy and policy dialogues, interpersonal communication and community mobilization, and mass communication) [OR = 1.51; 95% CI: 0.95, 2.40; P-value > 0.050] [79] did not affect EBF behaviours.

Seven studies showed that there were increases in EBF during the intervention period [74-78 85 91], but out of these studies, two indicated that improvements in EBF were not statistically significant [74 78]. Abdulahi et. al., 2021 reported a significant difference in the proportion of EBF between intervention (68.3%) and control groups (54.8%) [P-value <0.001], following the

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interventions of peer support, breastfeeding education, and practical demonstration sessions on position and attachment [91].

Introduction of solid, semi-solid and soft foods

Four studies reported the effect of different type of BCC interventions on the introduction of complementary foods [76 79 89 93]. BCC interventions using complementary feeding messages, training of mothers in group, counselling and support at home level, and facilitating participatory family discussion were effective in improving the timely initiation of complementary foods (RR = 2.60; 95% CI: 1.78, 5.86; P-value = 0.002) [93]. Kim et. al. investigated the effect of exposure to BCC interventions (advocacy and policy dialogues, interpersonal communication and community mobilization, and mass communication) on the introduction of solid, semi-solid and soft foods in two separate arms. The first arm of the study found an increase in the proportion of infants who were introduced to solid, semi-solid and soft foods (from 37.4% to 59.7%, P-value <0.001) [79]. In contrast, the second arm of the study did not find a significant improvement (from 58.7% to 61.3%, P-value >0.05) [89]. The remaining study by Buchanan et. al., 2012 showed improvement in proportion, but no test of significance was reported [76].

Continued breastfeeding at one year

Only one study measured the impact of interventions on continued breastfeeding at one year. Tariku et. al., 2015 showed that IYCF messages using the health belief model (HBM) did not impact continued breastfeeding rate at one year [90].

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417 **Table 1: Summary of eligible studies**

Authors	Design	Participants	Region	Intervention components	Location of intervention delivery	Outcome measures
Abdulahi et. al., 2021[91]	Cluster-randomized trial	Experimental group • Before intervention = 249 • After intervention = 212 Control group (routine care) • Before intervention = 219 • After intervention = 197	Oromia	Breastfeeding education and peer support • Peer support to encourage mothers for health facility birthing • Breastfeeding education including educational materials and • Practical demonstrations on proper positioning and latching	Community based	EIBF EBF
Abiyu et. al., 2020[92]	Cluster-randomized trial	Experimental group • Before intervention = 306 • After intervention = 272 Control group (routine care) • Before intervention = 306 • After intervention = 282	Amhara	Complementary feeding behaviour change communication • Complementary feeding messages • Group training of mothers • Home visits (for counselling and support) • Participatory family level discussion	Community based	MDD MMF MAD
Abiyu et. al., 2020[93]	Cluster-randomized trial	Experimental group • Before intervention = 306 • After intervention = 272 Control group (routine care) • Before intervention = 306 • After intervention = 282	Amhara	Complementary feeding behaviour change communication • Complementary feeding messages • Group training of mothers • Home visits (for counselling and support) • Participatory family level discussion	Community based	Timely initiation of complementary foods
Bahru et. al., 2020[84]	Quasi-experimental	Experimental group • Before intervention = 482 • After intervention = 275 Control group (routine care) • Before intervention = 740 • After intervention = 882	Tigray Oromia Amhara SNNPR	Productive safety net program (PSNP) • Offers employment opportunities for households with able-bodied members to work on community asset building projects and earn a wage either in cash or in-kind (food) • Direct support to households whose breadwinners are the elderly or disabled and hence could not take part in labour-intensive activities	Health facility and community- based	Dietary diversity Meal frequency
Baxter et. al., 2016 [74]	Quasi-experimental	Experimental group: • Before intervention = 173 • After intervention = 269	Oromia	EBF promotion strategies including • Home visits, • Educational materials, and • Training of influential community and family members)	Health facility and community- based	EBF
Berti et. al., 2010 [75]	Quasi-experimental	No information	No information	• Breastfeeding communication through community-based interventions for behaviour change	Community-based	EBF
Buchanan et. al., 2012 [76]	Quasi-experimental	No information	No information	• Behavioural change communication • Community mobilization • Capacity building	Health facility-based and community-based	EIBF EBF Introduction of complementary foods
Callaghan-Koru et. al., 2016 [77]	Quasi-experimental	Experimental group: • Before intervention=218 • After intervention=215	Tigray Oromia Amhara	• An integrated package of facility-based kangaroo mother care • Community-based promotion of EBF	Health facility and community based	EIBF EBF

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			SNNPR			
Carnell et. al., 2014 [78]	Quasi-experimental	Experimental group: • Before intervention = 1016 • After intervention = 911 Control group: • Before intervention = 1074 • After intervention = 903	Oromia Amhara SNNPR	• Strengthen the health system • Improve health workers' performance • Engage the community	Health facility and community based	EBF
Kang et. al., 2017 [88]	Cluster-randomized trial	• Experimental group: 570 • Control group: 629	Oromia	• Community-based participatory nutrition promotion program • Daily group nutrition sessions • Follow-up visits	Community-based	Meal frequency Dietary diversity Dietary adequacy
Karim et. al., 2013[85]	Quasi-experimental	Experimental group: • Before intervention = 1481 • After intervention = 1494	Tigray Oromia Amhara SNNPR	Supporting Ethiopian health extension program in: • Health education and demonstration for families on hygiene and environmental sanitation • Family health (Maternal and child health, reproductive health, immunization, and nutrition) services	Community-based	EIBF EBF at one month
Karim et. al., 2018[86]	Quasi-experimental	Experimental group: • Before intervention = 2124 • After intervention = 2113	Tigray Oromia Amhara SNNPR	Supporting Ethiopian health extension program using community-based data for decision-making • Family Conversation, a forum conducted at the household of the pregnant women • Birth notification to ensure early postnatal care • Establish household level surveillance system • Identify and address barriers to access maternal and newborn health services using generated data	Health facility and community based	EIBF within 30 min of childbirth
Kim et. al., 2016 [79]	Quasi-experimental	Experimental group: • Before intervention = 1404 • After intervention = 1404	SNNPR Tigray	BCC strategies including: • Advocacy and policy dialogues • Interpersonal communication and community mobilization • Mass communication	Community-based	EIBF EBF Introduction of complementary foods MDD MMF MAD
Kim et. al., 2019 [89]	Cluster-randomized trial	Experimental group: • Before intervention = 1328 • After intervention = 1360 Control group: • Before intervention = 1318 • After intervention = 1360	Amhara	BCC strategies including: • Interpersonal communication • Nutrition sensitive agricultural activities • community mobilization • Mass communication	Community-based	Introduction of complementary foods MDD MMF
Kung'u et. al., 2018 [80]	Quasi-experimental	Experimental group: • Before intervention = 2570 • After intervention = 2905 Control group: • Before intervention =1670	Afar	• Behaviour change interventions • Integrating nutrition to the frontline workers' guideline • Family education and counselling booklet	Health facility	EIBF EBF at six months

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		• After intervention = 1620				
Mulualem et. al., 2016 [81]	Quasi-experimental	Experimental group: • Before intervention = 80 • After intervention = 80 Control group: • Before intervention = 80 • After intervention = 80	SNNPR	Nutrition education based on the HBM model • Individual and group sessions, and • Recipe demonstration	Community-based	Meal frequency
Muluye et. al., 2020 [94]	Cluster randomized control trial	Experimental group: • Before intervention = 100 • After intervention = 100 Control group: • Before intervention = 100 • After intervention = 100	SNNPR	Education and practical demonstration on • complementary food recipes and preparation, • amount and frequency of feeding, • personal hygiene and sanitation, and • dietary diversification	Facility-based	MDD
Negash et. al., 2014 [82]	Quasi-experimental	Experimental group: • Before intervention = 100 • After intervention = 81 Control group: • Before intervention = 92 • After intervention = 76	SNNPR	• Nutrition education using visual materials • Demonstration of food preparation	Community-based	Dietary diversity Meal frequency
Passarelli et. al., 2020[95]	Cluster-randomized trial	ACGG intervention group: • Before intervention = 255 • After intervention = 255 ACGG/ATONU intervention group: • Before intervention = 263 • After intervention = 265 Control group: • Before intervention = 311 • After intervention = 311	Tigray Oromia Amhara SNNPR	ACGG • Each farmer received 25 vaccinated chicks that were ~6-week old • Farmers implemented chicken husbandry according to their individual preferences ACGG/ATONU • Additional nutrition sensitive intervention such as promotion of home gardening, provision of fruit and vegetable seeds for the gardens, and behavioural change communication	Community-based	Dietary diversity (7-days)
Teshome et. al., 2020[96]	Cluster-randomized trial	Experimental group: • Before intervention = 386 • After intervention = 307 Control group: • Before intervention = 386 • After intervention = 314	SNNPR	• Nutritional education and counseling on balanced diet, and food groups and benefits of pulse • Recipe demonstrations on preparation of porridge for complementary feeding	Community-based	MDD
Tariku et. al., 2015 [90]	Cluster-rando[91-93]mized trial	HBM intervention group: • Before intervention = 60 • After intervention = 54 Traditional nutrition education group: • Before intervention = 60 • After intervention = 56 Control group:	SNNPR	• Complementary feeding messages using the HBM model	Community-based	Continued breastfeeding MDD MMF

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		• Before intervention = 60 • After intervention = 56				
Villadsen et. al., 2016 [83]	Quasi-experimental	Experimental group: • Before intervention =775 • After intervention = 1434 Control group: • Before intervention = 350 • After intervention = 422	Oromia	• Training of health professionals • Health education materials • Supervisions of ANC providers	Health facility	EBF at 1 month
Wereta et. al., 2018[87]	Quasi-experimental	Experimental group • Before intervention = 408 • After intervention = 990 Control group (routine care) • Before intervention = 408 • After intervention = 990	Tigray Oromia Amhara SNNPR	• Increase access and quality of ANC, delivery and perinatal outcomes • Community engagement in planning (e.g., joint action plan to address these barriers and gaps) and implementation • Enhance interactions between the communities, HEWs, health care providers and local health officials	Health facility and community based	EIBF

EIBF: early imitation of breastfeeding; EBF: exclusive breastfeeding; MDD: minimum dietary diversity; MMF: minimum meal frequency; MAD: minimum acceptable diet; BCC: behaviour change communication; HBM: health belief model; SNNPR: Southern Nations Nationalities and Peoples Region; ACGG: African Chicken Genetic Gains; ATONU: Agriculture to Nutrition.

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Minimum dietary diversity

Ten studies reported the effect of interventions on MDD[79 82 84 88-90 92 94-96]. Of these, six measured according to the WHO definition for MDD [79 89 90 92 94 96], while three studies used any four food groups as a cut-off point to measure MDD [82 84 88]. A study by Abiyu et. al., 2020 using BCC interventions including complementary feeding messages, mothers group training, home-based counselling and support, and participatory family discussion improved (RR = 3.00; 95% CI: 1.34, 7.39; P-value = 0.012)[92]. Two studies by Kim et. al. revealed the significant impacts of exposure to more than one type of BCC interventions (advocacy and policy dialogues, interpersonal communication and community mobilization, and mass communication) on MDD, but these studies showed that exposure to a single type of intervention did not have an impact on MDD [79 89]. Tariku et. al., 2015 showed interventions involving complementary feeding messages using the HBM model significantly increased the proportion of MDD (from 37.5% to 67.9%; P-value <0.05) [90].

End line survey results following the intervention of nutritional education and counselling, and recipe demonstrations by Muluye et. al., 2020 (67.0% vs 33.0%, P-value = 0.034)[94] and Teshome et. al., 2020 (51.5% vs 43.3%; P-value = 0.001)[96] showed significant difference in the proportion of MDD between intervention and control groups. The remaining four studies indicated that the proportion of MDD increased over the intervention period, but the effect was not significant [82 84 88 95].

Minimum meal frequency

MMF was included as an outcome variable in eight interventional studies[79 81 82 84 88-90 92]. Four studies used the WHO definition for assessing MMF [79 89 90 92]. Complementary feeding BCC interventions using nutritional education and counselling, and recipe demonstration sessions improved MMF practice of mothers (RR = 2.40; 95% CI: 1.37, 4.29; P-value = 0.003)[92]. Kim

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et. al., 2019 indicated that exposure to two types of BCC interventions (advocacy and policy dialogues, interpersonal communication and community mobilization, and mass communication) had a significant effect on MMF of infants and young children compared to those who did not expose to intervention (OR = 1.69; 95% CI: 1.13, 2.52; P-value < 0.010) [89]. Kang et. al., 2017 found that interventions using a community-based participatory nutrition program had significant impact on the mean score of MMF (Mean difference = 1.04; 95% CI: 0.35, 1.73; P-value = 0.003) [88]. Except for one study, which did not do a test for significance [82], the remaining four studies did not show significant improvements in MMF [79 81 84 90].

Minimum acceptable diet

Out of three studies that measured MAD as an outcome, two studies used the WHO definition for assessing MAD [79 92], while the remaining study calculated dietary adequacy indicator using a composite dietary diversity and meal frequency scores [88]. Abiyu et. al., 2020 implemented BCC interventions (including nutritional education and counselling, and recipe demonstration) to improve MAD. The result from this study showed the significant effect of interventions on MAD (RR = 2.70; 95% CI: 1.13, 7.23; P-value = 0.036) [92]. Community based participatory nutrition program by Kang et. al., 2017 reported significant effect of interventions on the dietary adequacy score (Mean difference = 1.40; 95% CI: 0.49, 2.32; P-value = 0.003)[88]. However, a study by Kim et. al., 2019 documented that implementing four or more type BCC interventions (advocacy and policy dialogues, interpersonal communication and community mobilization, and mass communication) improved MAD (OR = 14.29; 95% CI: 2.23, 90.28; P-value < 0.010 for 4+ interventions)[79].

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Table 2: Summary of evidence from reviewed studies

IYCF outcomes	Studies by authors	Statistical analysis	Results
EIBF	Abdulahi et. al., 2021[91]	Percent change in proportion	• EIBF was 72.7% in intervention group, and 40.6% in control group (P-value <0.001)
	Buchanan et. al., 2012 [76]	Percent change in proportion	• EIBF increased from 31.0% to 77.0%
	Callaghan-K. et. al., 2016 [77]	Percent change in proportion	• EIBF significantly increased from 51.4% to 67.9% (P value <0.010)
	Karim et. al., 2013[85]	Percent change in proportion	• EIBF increased from 46.0% to 54.2% (P-value <0.001)
	Karim et. al., 2018[86]	Fixed-effect logistic regression	• EIBF at 30 minutes (OR = 1.27; 95% CI: 1.10, 1.46; P-value = 0.001)
	Kim et. al., 2016 [79]	Fixed-effect logistic regression	• Single type of intervention: EIBF (OR = 1.88; 95% CI: 1.08, 3.27; P-value < 0.050) • Two types of interventions: EIBF (OR = 2.07; 95% CI: 1.09, 3.91; P-value < 0.050) • Four or more interventions: EIBF (OR = 2.48; 95% CI: 1.09, 5.64; P-value < 0.050)
	Kung'u et. al., 2018 [80]	Fixed-effect logistic regression	• EIBF (OR = 1.70; 95% CI: 1.17, 2.46; P-value = 0.005)
	Wereta et. al., 2018[87]	Propensity score matching	• EIBF (ATE = 2.8; 95% CI: −4.7, 10.6; P-value = 0.468)
EBF	Abdulahi et. al., 2021[91]	Percent change in proportion	• EBF was 68.3% in intervention group, and 54.8% in control group (P-value <0.001)
	Baxter et. al., 2016 [74]	Percent change in proportion	• EBF increased from 69.4% to 75.1%, but not statistically significant
	Berti et. al., 2010 [75]	Percent change in proportion	• Higher improvement in exclusive breastfeeding
	Buchanan et. al., 2012 [76]	Percent change in proportion	• EBF increased from 73.0% to 95.0%
	Callaghan-Koru et. al., 2016 [77]	Percent change in proportion	• EBF increased from 86.0% to 95.8% (P value <0.010)
	Carnell et. al., 2014 [78]	Difference-in-difference	• No significant difference in the improvement of EBF was observed
	Karim et. al., 2013[85]	Percent change in proportion	• EBF at one month increased from 82.4% to 95.3% (P-value <0.001)
	Kim et. al., 2016 [79]	Fixed-effect logistic regression	• Single type of intervention: EBF (OR = 1.51; 95% CI: 0.95, 2.40; P-value>0.050) • Two types of interventions: EBF (OR = 1.39; 95% CI: 0.96, 2.01; P-value>0.050) • Four or more interventions: EBF (OR = 0.93; 95% CI: 0.86, 1.01; P-value>0.050)
	Kung'u et. al., 2018 [80]	Fixed-effect logistic regression	• EBF at six months (OR = 2.62; 95% CI: 0.95, 7.21; P-value = 0.062)
	Villadsen et. al., 2016 [83]	Fixed-effect logistic regression	• EBF at 1 month (OR = 2.30; 95% CI: 1.40, 3.60; P-value = 0.001)
Introduction of complementary foods	Abiyu et. al., 2020[93]	Generalized estimating equations (GEE) regression	• Timely initiation of complementary foods (RR = 2.60; 95% CI: 1.78, 5.86; P-value = 0.002)
	Buchanan et. al., 2012 [76]	Percent change in proportion	• Introduction of complementary foods increased from 38.0% to 98.0%
	Kim et. al., 2016 [79]	Percent change in proportion	• Introduction of complementary foods increased from 37.4% to 59.7% (P-value <0.001)
	Kim et. al., 2019 [89]	Percent change in proportion	• Introduction of complementary foods did not show any improvement
Continued breastfeeding at 1–year	Tariku et. al., 2015 [90]	ANOVA and Kruskal-Wallis test	• No significant difference in the improvement of continued breastfeeding
Dietary diversity	Abiyu et. al., 2020[92]	Generalized estimating equations (GEE) regression	• MDD (RR = 3.00; 95% CI: 1.34, 7.39; P-value = 0.012)
	Bahru et. al., 2020[84]	Marginal structural model	• No significant difference in dietary diversity score (β = −0.18; 95% CI: -0.41, 0.05)

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	Kang et. al., 2017 [88]	Mixed-effect logistic regression	• No significant difference in mean score between intervention and control group (Mean difference = 0.38; 95% CI: -0.22, 0.97; P-value = 0.220)
	Kim et. al., 2016 [79]	Fixed-effect logistic regression	• Single type of intervention: MDD (OR = 6.15; 95% CI: 0.80, 47.10; P-value > 0.050) • Two types of interventions: MDD (OR = 6.87; 95% CI: 1.27, 37.14; P-value < 0.050) • Four or more interventions: MDD (OR = 18.75; 95% CI: 2.32, 151.52; P-value < 0.010)
	Kim et. al., 2019 [89]	Fixed-effect logistic regression	• Single type intervention: MDD (OR = 1.05; 95% CI: 0.74, 1.48; P-value > 0.050) • Two type of interventions: MDD (OR = 1.82; 95% CI: 1.14, 2.89; P-value < 0.050) • Four or more interventions: MDD (OR = 3.15; 95% CI: 2.18, 4.57; P-value < 0.001)
	Muluye et. al., 2020 [94]	Independent student's t test	• Baseline: no significant difference between intervention and control group (23.0% vs 26.0%, P-value = 0.519) • End line: there was significant difference between intervention and control group (67.0% vs 33.0%, P-value = 0.034)
	Negash et. al., 2014 [82]	Independent sample t-test	• Dietary diversity increased over the intervention period (P-value > 0.050)
	Passarelli et. al., 2020[95]	Ordinary least square regression	• There was no significant difference between intervention and control groups
	Tariku et. al., 2015 [90]	ANOVA and Kruskal-Wallis test	• Intervention group showed significant improvement in MDD (P<0.001)
	Teshome et. al., 2020[96]	Percent change in proportion	• Baseline: no significant difference between intervention and control group (7.5% vs 3.8%) • End line: there was significant difference between intervention and control group (51.5% vs 43.3%; P-value = 0.001)
Meal frequency	Abiyu et. al., 2020[92]	Generalized estimating equations (GEE) regression	• MMF (RR= 2.40; 95% CI: 1.37, 4.29; P-value = 0.003)
	Bahru et. al., 2020[84]	Marginal structural model	• No significant effect on meal frequency (β = 0.31; 95% CI: -0.07, 0.54)
	Kang et. al., 2017 [88]	Mixed-effect logistic regression	• Significant difference in mean score between intervention and control group (Mean difference = 1.04; 95% CI: 0.35, 1.73; P-value = 0.003)
	Kim et. al., 2016 [79]	Fixed-effect logistic regression	• Single type of intervention: MMF (OR = 1.06; 95% CI: 0.99, 1.13; P-value > 0.050) • Two types of interventions: MMF (OR = 1.31; 95% CI: 0.97, 1.77; P-value > 0.050) • Four or more interventions: MMF (OR = 2.04; 95% CI: 0.92, 4.54; P-value < 0.010)
	Kim et. al., 2019 [89]	Fixed-effect logistic regression	• Single type of intervention: MMF (OR = 1.05; 95% CI: 0.90, 1.23; P-value > 0.050) • Two types of interventions: MMF (OR = 1.69; 95% CI: 1.13, 2.52; P-value < 0.010) • Four or more interventions: MMF (OR = 1.93; 95% CI: 1.44, 2.57; P-value < 0.001)
	Mulualem et. al., 2016 [81]	ANOVA	• MMF frequency decreased in both experimental and control groups
	Negash et. al., 2014 [82]	Independent sample t-test	• Meal frequency increased over the intervention period (P-value > 0.050)
	Tariku et. al., 2015 [90]	ANOVA and Kruskal-Wallis test	• No significance effect on MMF
Minimum acceptable diet	Abiyu et. al., 2020[92]	Generalized estimating equations (GEE) regression	• MAD (RR = 2.70; 95% CI: 1.13, 7.23; P-value = 0.036)

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	Kang et. al., 2017 [88]	Mixed-effect logistic regression	• Significant effect on composite feeding index of dietary diversity and meal frequency (Mean difference = 1.40; 95% CI: 0.49, 2.32; P-value = 0.003)
	Kim et. al., 2019 [89]	Fixed-effect logistic regression	• Single type of intervention: MAD (OR = 5.20; 95% CI: 0.82, 32.99; P-value > 0.050) • Two types of interventions: MAD (OR = 5.82; 95% CI: 0.81, 41.90; P-value > 0.050) • Four or more interventions: MAD (OR = 14.29; 95% CI: 2.23, 90.28; P-value < 0.010)

IYCF: infant and young child feeding; EIBF: early imitation of breastfeeding; EBF: exclusive breastfeeding; MDD: minimum dietary diversity; MMF: minimum meal frequency; MAD: minimum acceptable diet; HBM: health belief model

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Discussion

A total of 23 studies were included, and of these, 14 were quasi-experimental studies [74-87], and nine were cluster randomised trials [88-96]. Five studies for EBF [75 77 83 85 91] and six studies for EIBF [77 79 80 85 86 91] showed that community- and facility-based interventions delivered in the form of BCC (such as policy advocacy, interpersonal communication, community mobilisation, and mass media campaigns) and health service strengthening had an impact on infants. Two studies for the introduction of complementary foods [79 93], six studies for MDD [79 89 90 92 94 96] and three studies for MMF [88 89 92] indicated significant effects of BCC on infants and young children. Two eligible studies showed that combined interventions delivered in the form of interpersonal communication, nutrition-sensitive agricultural activities, community mobilisation, and mass media campaigns had more impacts on EIBF, MDD, and MMF[79 89].

Global evidence indicates that BCC interventions (including policy advocacy, interpersonal communication, community mobilisation and mass media campaigns) are essential in promoting IYCF practices. These initiatives also create a supportive environment that helps mothers to adopt and sustain positive child feeding behaviours [11 97]. The main goals of BCC intervention include improving behaviours of a facility- and community-level health workers, mothers and other caregivers, and members of their households [11 97]. Our review showed that BCC interventions in the form of policy advocacy, interpersonal communication, community mobilisation, and mass media campaigns improved EIBF, MDD, and MMF. Our finding is consistent with systematic reviews from lower and middle income countries (LMICs), which indicated that group nutritional education and counselling, family or social support, integrated mass media coverage, and community mobilization were effective in improving IYCF practices [5 11].

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Although BCC interventions promote skills and knowledge that are necessary for improving IYCF practices, studies from LMICs indicated that achieving the intended target is often challenging without complementary interventions that provide or facilitate material inputs [97-100]. This could be due to the fact that child feeding decisions not only depend on the knowledge and perception of the mother but also are influenced by a complex interplays of factors, including available resources, partner support, mother's time and close family support [97-101]. Affleck et. al. 2012 reported that a lack of money to provide diversified diet for infants and young children was a major barrier to improve appropriate IYCF behaviours among Bangladeshi women [99]. Thus, promoting material inputs (such as cash transfer and provision of food materials) and/or non-material inputs (skills and knowledge) are required in achieving appropriate and sustained IYCF behaviours.

Research has shown that exposure to more than one IYCF intervention, delivered through different platforms (e.g., community- or facility-based), was more successful in improving IYCF outcomes compared to a single form of intervention delivered at a single setting [5]. Our review indicated that exposure to more than two types of interventions was associated with improvement in EIBF, MDD, and MMF [79 89]. This result may be due to interventions delivered through multiple settings may have an increased intensity that can be translated to behavioural changes due to repeated contacts and adequate coverage of interventions [102]. Our review suggests that interventions that aim to improve IYCF behaviours in Ethiopia should be delivered through different and specific types of contexts (including household, health facility and community) and should be provided along the continuum of care from conception until a child's second birthday.

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In the past two decades, the Ethiopian government's commitment to improve IYCF has been consistent, and this started with the development of the National Strategy for IYCF in 2004 and followed by National Strategy for Nutrition in 2008 [24 103], in line with the WHO/UNICEF IYCF strategy [104]. In 2015, the Ethiopian government launched the Health Sector Transformation Plan to improve health outcomes, including IYCF practices[25]. Based on this initiative, many facility- and community-based interventions have been implemented to protect, promote and support appropriate IYCF behaviours [10 11]. Nevertheless, our review found that there has been no formal IYCF intervention conducted in Addis Ababa and Dire Dawa city administrations, and Harari, Benishangu1, Gambella, and Somali regions. Our review suggests an expansion of additional experimental studies from successful subnational interventions implemented in Amhara[89 92 93], Oromia[83 88 91], and SNNP[94 96] regions, to maximise the opportunities for the prevention of IYCF-related morbidity and mortality in those regions.

Additionally, our review did not find any studies which incorporated the use of mobile health (mHealth) as part of an intervention package for improving IYCF [105]. Future experimental studies that consider the use of mHealth techniques are required, especially that previously published studies have shown the positive impacts of interventions (in form of voice messages, SMS, and phone calls) on IYCF practices [106-108]. Finally, interventional studies that consider the role of fathers/partners as well as grandmothers in improving IYCF behaviours may be warranted in Ethiopia [101 109 110].

The present systematic review of IYCF outcomes from community- and facility-based interventions (e.g., BCC, health service strengthening, community mobilizations, and nutrition sensitive agricultural activities) has strengths and limitations. First, a meta-analysis was not conducted due to the wide heterogeneity of the design and outcome measures. Second, this

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review was limited by the incompleteness of information from the individual studies. Nevertheless, the study authors were contacted for information when it is necessary. Third, our search was limited to published studies in English, which exclude studies published with other languages. Fourth, lack of evidence and under-representation of some regions may limit the generalizability of our review. However, evidence from this review is potentially relevant for IYCF policy efforts in Ethiopia. Despite the above limitations, the application of multiple search strategies in consultation with the medical librarian and registration of the systematic review protocol can be considered as strengths of this review. Furthermore, future studies that consider the assessment of the full set of the updated IYCF indicators for all Ethiopian children would be warranted [111]. This approach would ensure that both subnational and national and IYCF estimates are accurate and reliable and ensure that interventions and policy decisions in resource allocation are targeted at the most at-risk population. It would also ensure that IYCF programme monitoring and evaluation are consistent across all levels of government [111].

Conclusion

Our systematic review showed that 13 out of 21 eligible studies that implemented interventions in the form of policy advocacy, health service strengthening, interpersonal communication, community mobilisation, and mass media campaigns reported improved EIBF, EBF, and/or MDD in Ethiopia. The impacts of current and/or future interventions on IYCF practices in Ethiopia will be greater when interventions are combined, introduced and implemented along the continuum of care from conception until a child's second birthday.

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List of abbreviations

ACGG: African Chicken Genetic Gains; ANC: Antenatal Care; ATONU: Agriculture to Nutrition; BCC: Behavioural Change Communication; BFHI: Baby-Friendly Hospital Initiative; CI: Confidence Interval; CINAHL: Cumulative Index to Nursing and Allied Health Literature; CRTs: Cluster Randomised Trials; EBF: Exclusive Breastfeeding; EIBF: Early Initiation of Breastfeeding; GEE: Generalized estimating equations; GNTs: Global Nutrition Targets; HBM: Health Belief Model; HSTP: Health Sector Transformation Plan; IYCF: Infant and Young Child Feeding; JBI: Joanna Briggs Institute; LMICs: Lower and Middle-Income Countries; MAD: Minimum Acceptable Diet; MDD: Minimum Dietary Diversity; MeSH: Medical Subject Headings; MDG: Millennium Development Goal; MMF: Minimum Meal Frequency; OR: Odds Ratio; ORS: Oral Rehydration Salt; PICO: Population Intervention Comparator Outcome; Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA); RCTs: Randomised Controlled Trials; ROB: Risk of Bia; Risk of bias in non-randomised studies - of interventions (ROBINS); SDGs: Sustainable Development Goals; SNNPR: Southern Nations Nationalities and Peoples Regions; SUMARI: System for the Unified Management, Assessment and Review of Information; UNICEF: United Nation Children's Fund; WHO: World Health Organization.

Declarations

Contributors

KYA conceptualised the study idea, performed the literature searching, selected the paper of interests, performed data extraction, and critically revised the manuscript. AP and AA critically revised the manuscript for intellectual content. KEA performed the risk of bias assessment, and critically revised the manuscript. FAO contributed to the conception of the research idea,

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selected the paper of interests, performed data extraction, and critically revised the manuscript. All authors read and approved the final manuscript.

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Ethics approval

Ethical approval was not required for this study as the study was based on retrieved and synthesised data from already published studies.

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Patient Consent for publications

"Not applicable"

Availability of data and materials

"Not applicable"

Competing interests

The authors declare that they have no competing interests

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Figure 1: Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flowchart for selection of eligible studies

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Supplementary appendix

Supplementary file 1: PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) 2009 Checklist.

Supplementary file 2: Search strategy developed for Embase (OVID) database

Supplementary file 3: Risk of bias assessment for cluster randomised trial studies

Supplementary file 4: Risk of bias assessment for quasi-experimental studies

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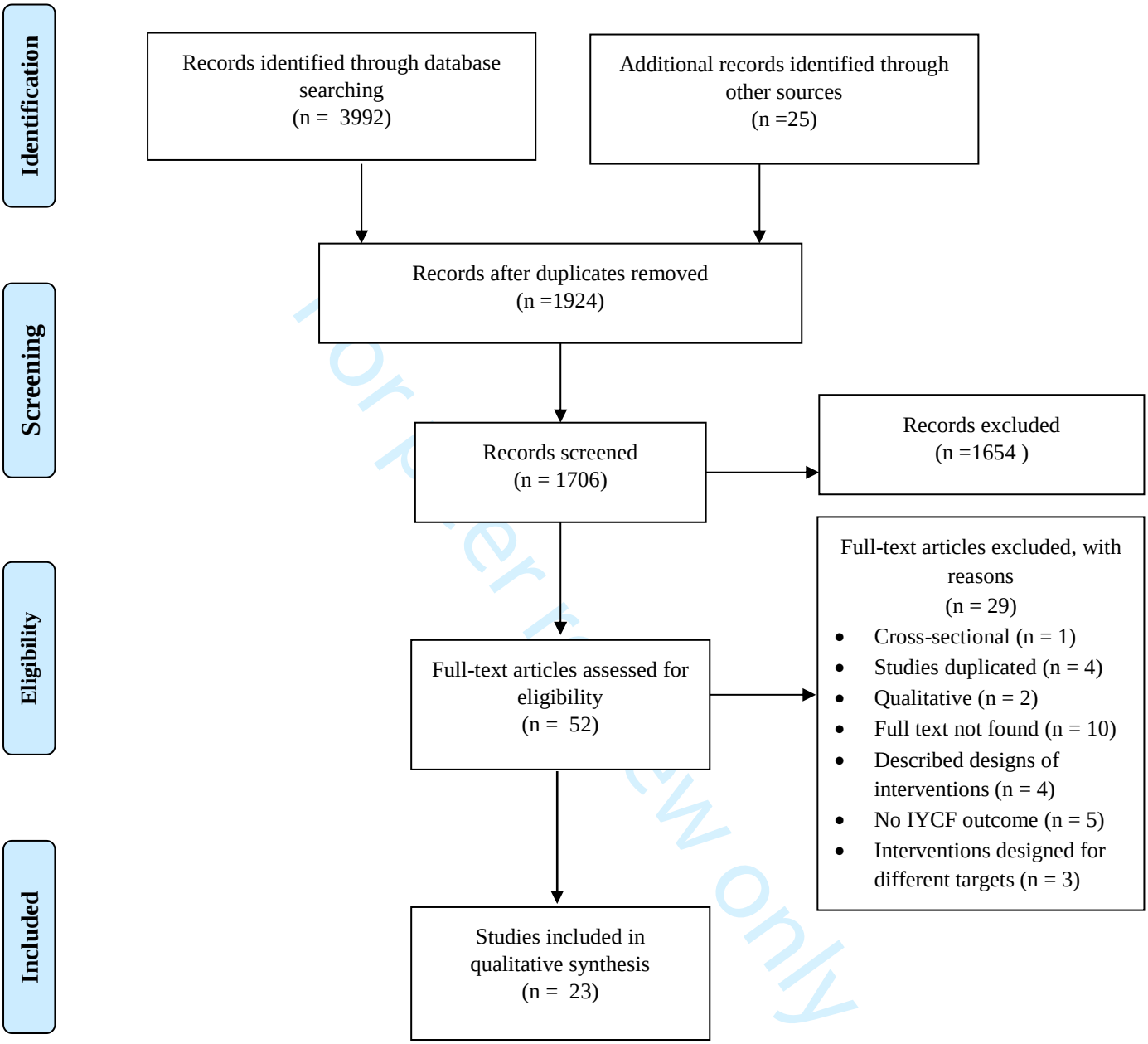


Figure 1: Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flowchart for selection of eligible studies [40]

PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2, 3,4
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	5, 6
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	6 Paragraph 2
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	6 Paragraph 3
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	7, 8
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	9 Paragraph 1
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	9, 10, 11, Supplementary file 2
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	11 Paragraph 2
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	12 Paragraph 1
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	12 Paragraph 1
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	12 Paragraph 2, Supplementary file 3 and 4
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	N/A

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PRISMA 2009 Checklist

Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	13 Paragraph 2

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	12 Paragraph 2, Supplementary file 3 and 4
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	N/A
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	13 Paragraph 4
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	14 Paragraph 2
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	14 Paragraph 3
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	14–26
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	N/A
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	14 Paragraph 3
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	N/A
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	27–29
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	29, 30
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	30
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	32 Paragraph 4

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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	2	child/ or boy/ or girl/ or toddler/	2128760	Advanced	Display Results | More	
	3	female/	10333062	Advanced	Display Results | More	
	4	mother/ or adolescent mother/ or mother child relation/	135425	Advanced	Display Results | More	
	5	caregiver/	89398	Advanced	Display Results | More	
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	7	1 or 2 or 3 or 4 or 5 or 6	13403899	Advanced	Display Results | More	
	8	breast feeding education/	2725	Advanced	Display Results | More	
	9	health education/ or childbirth education/ or nutrition education/ or parenting education/	112354	Advanced	Display Results | More	
	10	health literacy/	13474	Advanced	Display Results | More	
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	12	counseling/ or e-counseling/ or family counseling/ or nutritional counseling/ or parent counseling/ or peer counseling/	90024	Advanced	Display Results | More	
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	17	interpersonal communication/	175289	Advanced	Display Results | More	
	18	social change/	14822	Advanced	Display Results | More	
	19	social behavior/	89963	Advanced	Display Results | More	
	20	social network/ or online social network/	19132	Advanced	Display Results | More	
	21	peer group/	25754	Advanced	Display Results | More	
	22	advocacy group/	937	Advanced	Display Results | More	
	23	social marketing/	3922	Advanced	Display Results | More	
	24	public health campaign/ or public health message/	3212	Advanced	Display Results | More	
	25	mass communication/ or mass medium/ or public opinion/ or publication/ or radio/ or social media/ or telephone/ or television/ or text messaging/ or videoconferencing/ or wireless communication/	327907	Advanced	Display Results | More	
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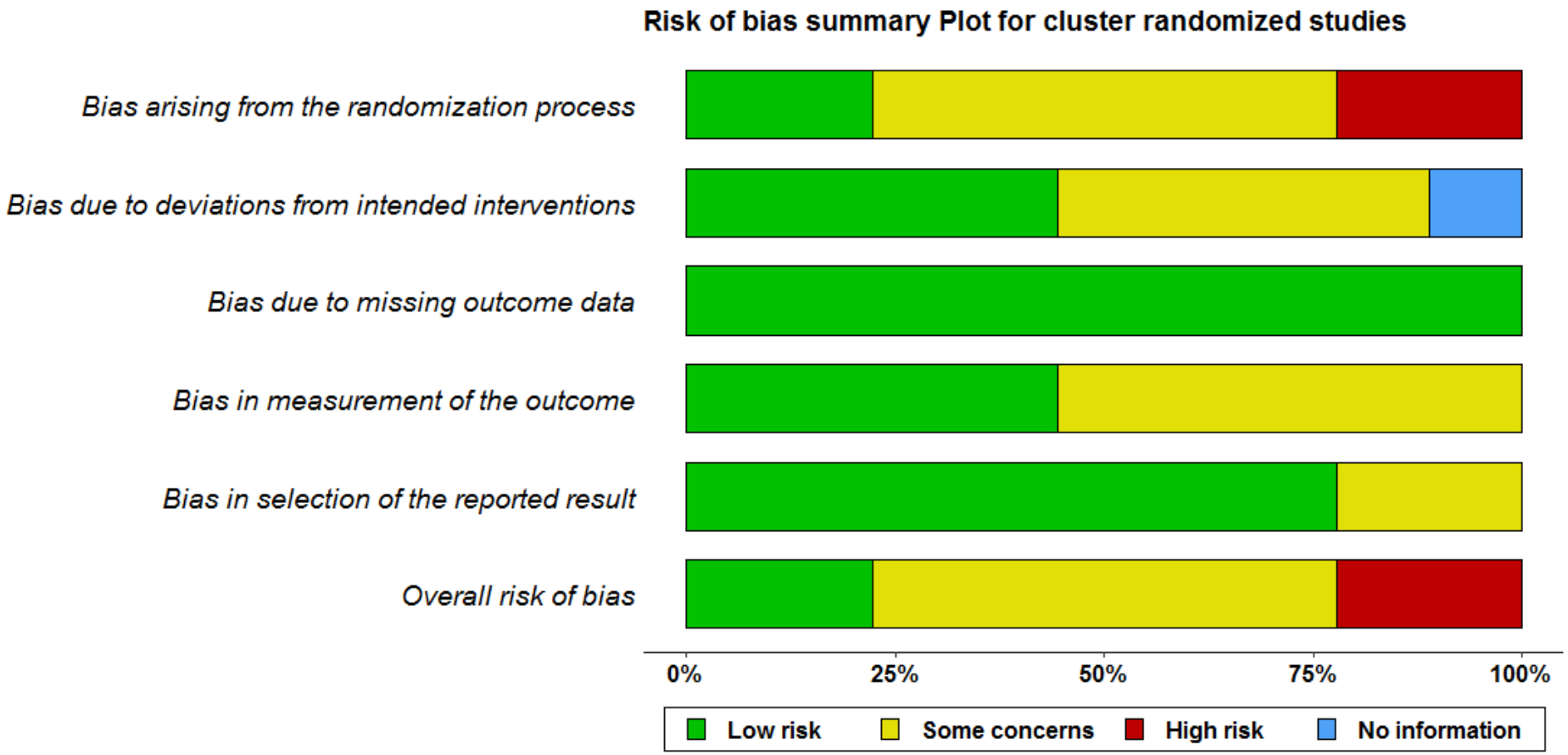
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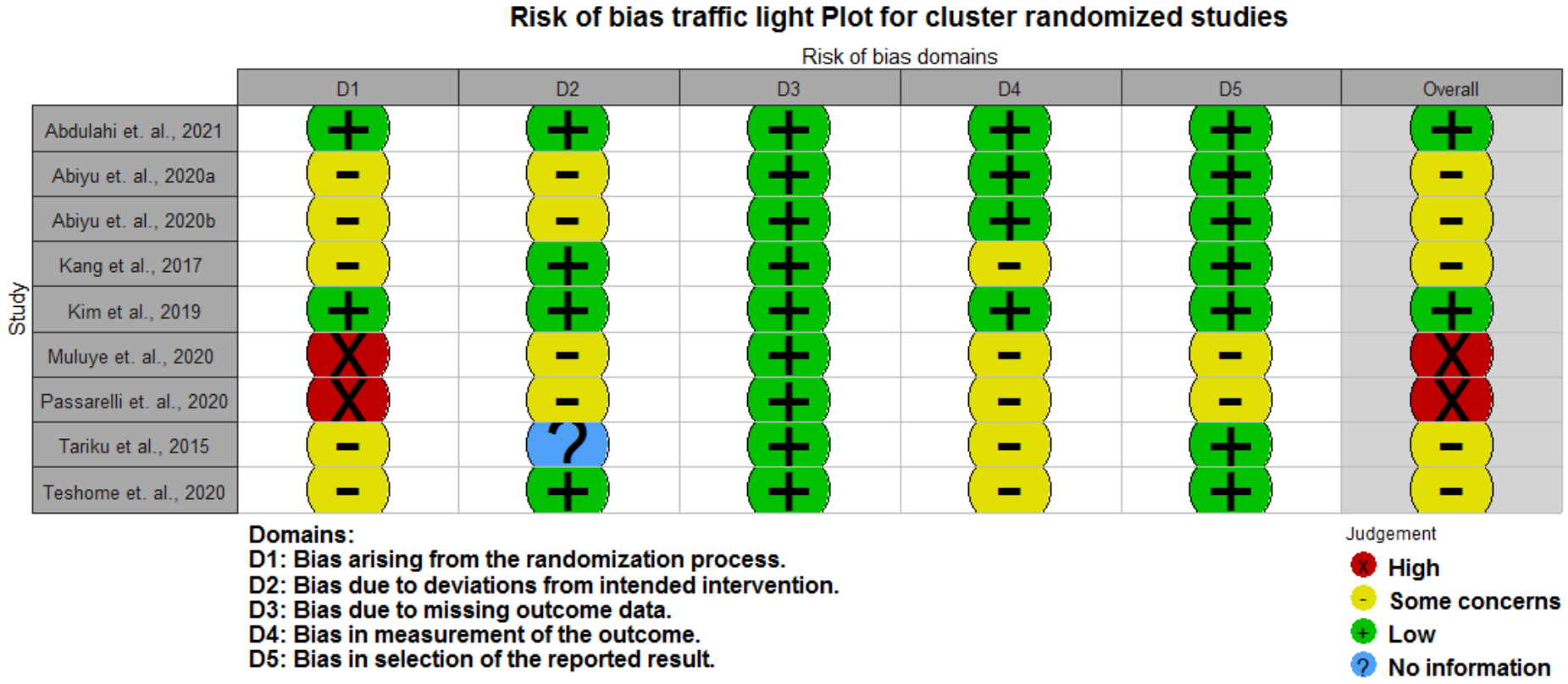
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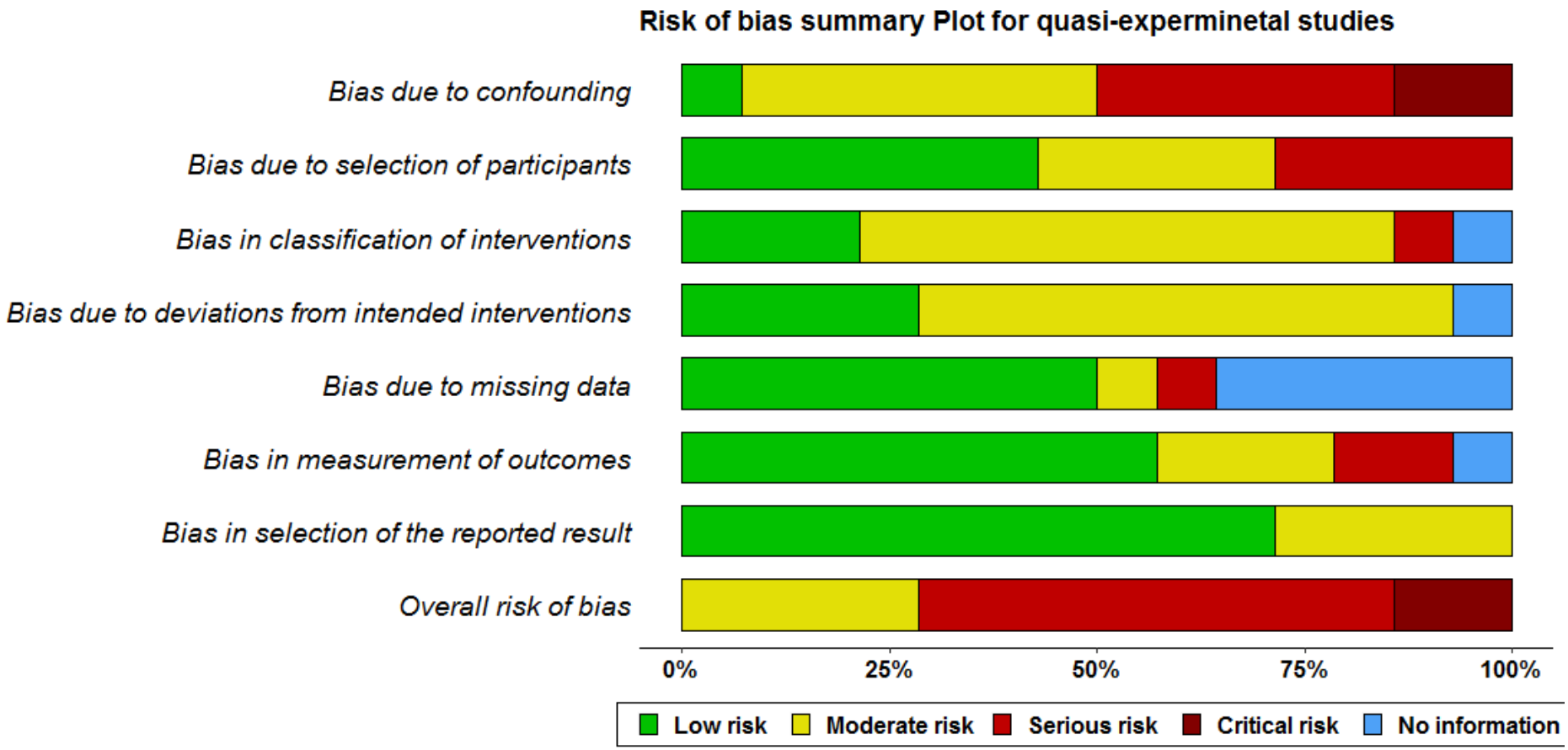
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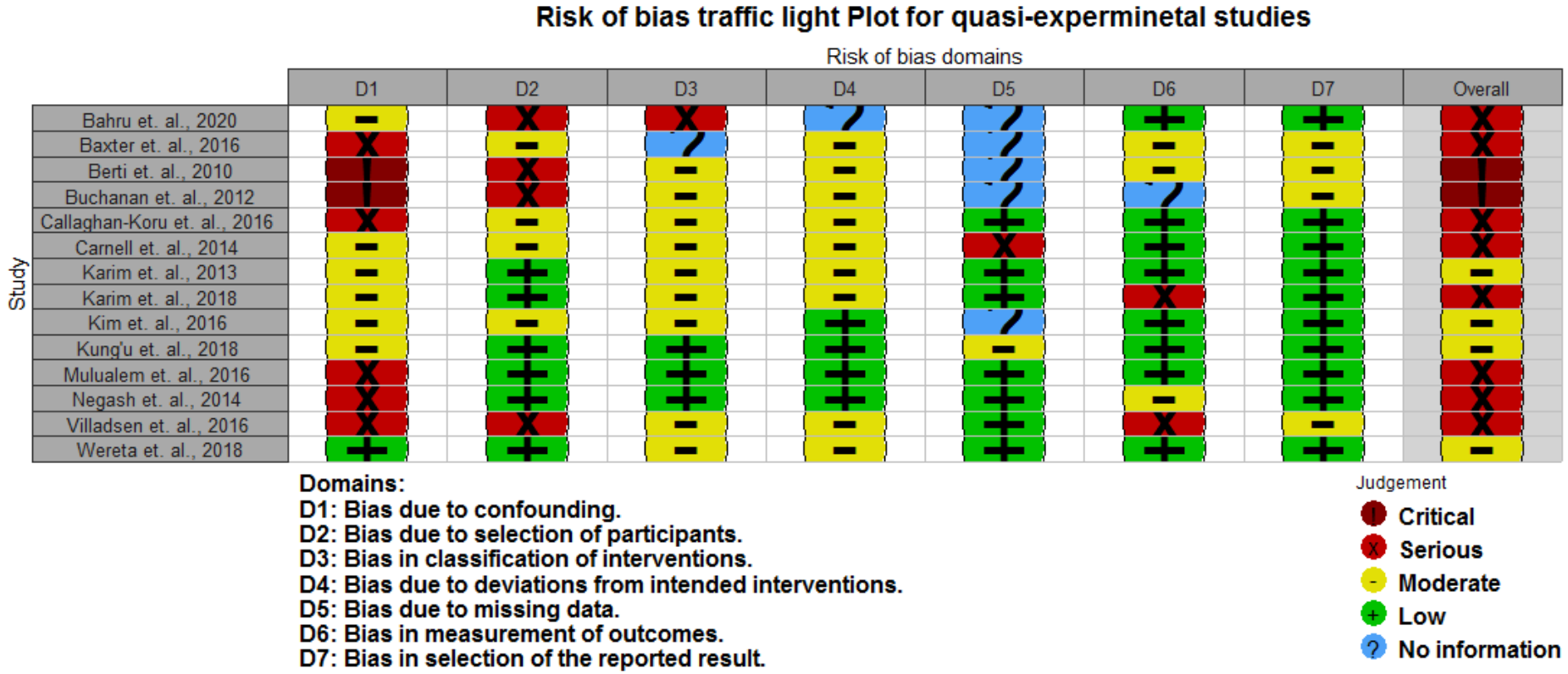
Risk of bias summary plot for cluster randomised studies: review authors' judgements about each risk of bias item as percentages



Risk of bias traffic light plot for cluster randomised studies: review authors' judgements about each risk of bias item for each studies



Risk of bias summary plot for quasi experimental studies: review authors' judgements about each risk of bias item presented as percentages



Risk of bias traffic light plot for quasi-experimental studies: review authors' judgements about each risk of bias item for each included studies

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Interventions to improve infant and young child feeding practices in Ethiopia: a systematic review

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Abstract

Objective: In Ethiopia, despite the implementation of several interventions to improve infant and young child feeding (IYCF) practices, no published studies have highlighted the most effective IYCF interventions in the country. This systematic review investigated the impacts of various interventions on IYCF in Ethiopia.

Source of information: A systematic search was conducted on seven computerized bibliographic databases (Embase, ProQuest Central, Scopus, PsycINFO, Web of Science, MEDLINE/PubMed and CINAHL) to locate experimental or quasi-experimental studies published between the year 2000 and May 2021.

Study eligibility criteria: Interventional studies that measured IYCF indicators (early initiation of breastfeeding [EIBF], exclusive breastfeeding [EBF], the introduction of complementary foods, minimum dietary diversity [MDD], minimum meal frequency [MMF] and minimum acceptable diet [MAD]) as outcome variables were included.

Study appraisal and synthesis: All included studies were examined for biases related to interventional studies (i.e., selection bias, performance bias, attrition bias, detection bias and reporting bias). Author reports of effect size measures were used to narratively report the findings of each study.

Results: Of the 23 eligible studies, 14 studies were quasi-experimental and nine studies were cluster randomised trials (CRTs). Eight quasi-experimental studies had a serious risk of bias, while two CRTs had a high risk of bias. Four studies for EBF and six studies for EIBF showed significant impacts of policy advocacy, health service strengthening, interpersonal

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communication, community mobilisation, and mass media campaigns. Six studies for MDD and three studies for MMF indicated significant effects of community-level and health facility complementary feeding promotions on infants and young children. Interventions that delivered in combination increased the impacts in improving EIBF, MDD and MMF compared to a single intervention.

Conclusion: Our review showed that 12 out of 21 eligible studies that implemented in the form of community-level and health facility interventions improved EIBF, EBF, and/or MDD in Ethiopia.

Keywords: Infant and young child feeding, breastfeeding, nutritional education, intervention, systematic review, Ethiopia

Protocol registration number: PROSPERO, CRD42020155519.

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Strengths and limitations of the study

- A meta-analysis was not conducted due to the wide heterogeneity of the study designs and outcome measures.
- This review was limited by the incompleteness of information from the individual studies (however, the study authors were contacted for information when necessary).
- Despite the limitations, the application of multiple search strategies in consultation with the medical librarian and registration of the systematic review protocol can be considered as strengths of this review.

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Introduction

Appropriate infant and young child feeding (IYCF) improves childhood growth (through reduced risk of diarrhoea and respiratory infections as well as improved nutrition) [1-7], increases the intellectual capacity of children [4 8], and reduces the risk of mothers experiencing diabetes mellitus, overweight and obesity [4 9]. Promoting appropriate IYCF using evidence-based interventions is essential to protect and support maternal knowledge, beliefs, and confidence in IYCF practices [10-13]. Community- and facility-based interventions that range from direct support and education to mothers and families on appropriate IYCF behaviours to training of health professionals can significantly improve IYCF behaviours [14-16].

Global efforts to improve maternal IYCF behaviours have increased in the past three decades. These efforts included the International Code of Marketing of Breast-milk Substitutes [17]; the Innocenti Declaration [18]; the Baby-Friendly Hospital Initiative (BFHI) [19]; the Millennium Development Goals (MDGs) [20]; and more recently, the Global Nutrition Targets 2025 [21] and the Sustainable Development Goals (SDGs) [22]. The World Health Organization and United Nations Children's Fund (WHO/UNICEF) have played major roles in driving these initiatives which aimed to protect, promote, and support breastfeeding and child nutrition [19]. For example, the WHO Global Nutrition Targets (WHO GNT, including Target-1 and -5) aim to reduce the global number of stunted children under-5 by at least 40% and increase the EBF rate to at least 50% by the year 2025, respectively. However, most subnational areas of many African countries (including Ethiopia) are not on track to achieve WHO GNT goal-5 [23].

In Ethiopia, while both government and non-governmental organisations have implemented various maternal and child health-related policy interventions over the last two decades to increase infant feeding [24-26], a recent study has revealed that early initiation of breastfeeding

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(EIBF) prevalence was 75.5% and that for exclusive breastfeeding (EBF) was 59.9% [27], with a wide heterogeneity across subnational areas of the country [23]. These rates are well below the Ethiopian Health Sector Transformation Plan (HSTP) target of 90% and 72%, respectively [25]. Additionally, the proportion of children who met minimum dietary diversity (MDD) and minimum meal frequency (MMF) was 7.1% and 43.6%, respectively [28]. In Ethiopia, more than five million children were stunted in 2016 [29], and this is a major public health issue that is largely attributable to inappropriate IYCF.

To date, no previously published studies have examined the most effective IYCF interventions in the Ethiopian context. A detailed understanding of the specific types of interventions that can be implemented to improve IYCF in Ethiopia is essential, to specifically guide the allocation of scarce resources and the provision of targeted strategies to increase childhood feeding. This information will also be helpful in scaling-up and/or improve upon current and future IYCF interventions in the country. Data from the assessment of relevant IYCF interventions would be crucial to stakeholders as Ethiopia implements national nutrition efforts (e.g., HSTP) [25] and global health programs such as the Sustainable Development Goal, SDG-2.2, to end all forms of malnutrition [22]. Accordingly, this systematic review investigated the impacts of community- and facility-based interventions in improving IYCF practices of mothers/caregivers with children under two years of age in Ethiopia.

Materials and methods

Our review adhered to the 2009 Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [30] **(Supplementary file 1)**, and the protocol for the review was registered in PROSPERO (International Prospective Register of Systematic Reviews, CRD42020155519) [31].

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Eligibility criteria

Studies were included if they meet the following criteria: (i) experimental or quasi-experimental study design (including randomized controlled trials [RCT], non-RCTs, before and after studies, and interrupted time-series studies) with or without comparison groups or cluster; (ii) conducted among pregnant women and mothers living with live children under 2 years to improve IYCF practices; (iii) used any type of IYCF interventions delivered between the antenatal period and 24 months postpartum; (iv) measured at least one core IYCF indicator such as EIBF, EBF, the introduction of solid, semi-solid and soft foods, continued breastfeeding at one year, MDD, MMF, and minimum acceptable diet (MAD); (v) were published in the English language; and (vi) conducted in Ethiopia from the year 2000 to date. We considered studies published over this period to demonstrate the impacts of the specific IYCF interventions in the context of Ethiopia to reflect the implementation of relevant global health programs, including the MDGs and the current SDGs [23].

The intervention strategies considered in this review focused on those performed at both facility- and community-levels. The interventions included nutritional education and counselling, interpersonal communication, mass media campaigns, nutrition-sensitive agricultural activities, group demonstration sessions, and strengthening of health facilities.

Studies conducted among mothers with health problems such as HIV/AIDS and children with cleft palate were excluded.

The IYCF practices included EIBF, EBF, the introduction of solid, semi-solid and soft foods, continued breastfeeding at one year, MDD, MMF, and MAD, measured according to the WHO definitions for assessing IYCF indicators [32]. These indicators were selected based on their effect on the health and development of infants and young children [11, 12, 35] and the evidence from previously published systematic reviews [11 33].

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- EIBF was defined as the proportion of children aged 0–23 months who commenced breastfeeding within the first hour of birth.
- EBF was defined as the proportion of infants 0–5 months of age who were fed no other food or drink, not even water, except breast milk (including milk expressed or from a wet nurse), but allows the infant to receive Oral Rehydration Salt (ORS), drops, and syrups (vitamins, minerals, and medicines).
- Continued breastfeeding at one year was defined as the proportion of children 12–15 months of age who are fed breast milk.
- Introduction of complementary foods (solid, semi-solid and soft foods) was defined as the proportion of infants 6–8 months of age who received solid, semi-solid or soft foods in the previous 24 hours, during the day and at night.
- MDD was defined as the proportion of children 6–23 months of age who received foods from four or more of the seven food groups. The seven food groups included grains, roots and tubers; legumes and nuts; dairy products (milk, yogurt, cheese); flesh foods (meat, fish, poultry and liver/organ meats); eggs; vitamin-A-rich fruits and vegetables; and other fruits and vegetables.
- MMF was defined as the proportion of breastfed and non-breastfed children 6–23 months of age, who received solid, semisolid or soft foods (including milk feeds for non-breastfed children) the minimum number of times or more (i.e., two times for breastfed infants aged 6–8 months, three times for breastfed children aged 9–23 months and four times for non-breastfed children aged 6–23 months, on the previous day).
- MAD was defined as the proportion of children 6–23 months of age who received both minimum dietary diversity and minimum meal frequency.

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Information sources and search strategy

Consistent with previously published systematic reviews [15 34], a three-stage search strategy was applied to locate both published and unpublished studies. In stage one, a manual search of previously published systematic reviews and the reference lists of all the eligible studies was performed to check for systematic reviews of interventions on IYCF in Ethiopia. In stage two, index terms (e.g., medical subject headings [MeSH] for Medline) and key terms were generated using the identified articles from the initial search for the full search strategy. In stage three, a full search strategy was conducted on seven computerized bibliographic databases (Embase [OVID], ProQuest Central, Scopus, PsycINFO [EBSCO], Web of Science, MEDLINE [OVID]/PubMed and Cumulative Index to Nursing and Allied Health Literature (CINAHL)/[EBSCO] to locate published studies, and secondary searching of OpenGrey and Grey Literature Report was conducted to locate unpublished works. The Population Intervention Comparator Outcome (PICO) criteria were used to devise the review question and related search terms. A combination of specified medical subject headings (MeSH) terms and keywords was drafted with some assistance from our institution-based medical librarian. The search strategy, including all identified index and keywords terms, was slightly adapted for each of the information sources. All electronic searches were conducted for eligible studies published between the year 2000 and May 2021. A full electronic search strategy developed for the Embase (OVID) is provided as **Supplementary files 2 and 3**.

The overall list of search terms used included:

Term 1 (population): child, infant, newborn, baby, neonate, perinatal, postnatal, kid, toddler, young child, paediatric, mother, female, women, and caregiver

Term 2 (interventions):

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- ***Education and counselling***: breastfeeding promotion, breastfeeding support, breastfeeding education, health education, health promotion, nutrition education, food education, parent education, mother education, counselling, and nutritional counselling
- ***Social behavioural change communication***: health behaviour, health-related behaviour, behaviour change, communication, interpersonal communication, information education communication, behaviour change communication, social change, social movement, social mobilization, social behaviour, social network, peer group, advocacy, advocacy group, mass communication, mass media, print media, mobile phone, mHealth, eHealth, internet, radio, social media, television, text message, and social market
- ***Community-based approaches****:* baby-friendly community initiative, community program, community project, home visit, community health action, community health service, community health care, community intervention, community engagement, community leader, community mobilization, demonstration, cooking demonstration, community role-play, model breastfeeding community, community health worker, and health extension worker
- ***Facility-based approaches****:* maternal care, child health care, child health service, paediatric health care, neonatal care, newborn care, rooming-in-care, newborn nursery, essential nutrition action, caregiver contact, baby-friendly hospital initiative, BFHI, antenatal care, postnatal care, Kangaroo care, and skin-to-skin contact
- ***Intervention designs****:* effectiveness, impact, evaluation study, program evaluation, health care program, project, health project, experimental study, interventional study, quasi-experimental study, randomized controlled trial, clinical trial, cluster-randomized trial, time-series study, control, placebo, comparison, and usual care

Term 3 (Context): Ethiopia

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Term 4 (Outcomes): infant and young child feeding, breastfeeding, early initiation of breastfeeding, timely initiation of breastfeeding, exclusive breastfeeding, breast milk substitute, complementary food, complementary feeding, weaning, introduction of complementary food, baby food, infant feeding, child feeding, meal, dietary diversity, meal frequency, child nutrition, infant nutrition, nutritional supplement, dietary supplement, lactation, and nursing

Study selection

Following the full search, all articles retrieved from the search databases were exported to EndNote X9 (Clarivate Analytics, USA) for removing duplicates, screening, and selection. Titles and abstracts were screened by two reviewers (KYA and FAO) independently for assessment against the inclusion criteria. Potentially relevant articles that were selected in the first screening phase were retrieved in full and their citation details imported into the Joanna Briggs Institute System for the Unified Management, Assessment and Review of Information (JBI SUMARI). Full texts of the included studies were read and checked by two independent reviewers (KYA and FAO) against the eligibility criteria. Studies that did not meet the eligibility criteria were excluded and reasons for the exclusion of studies were recorded and reported in the systematic review. Any disagreements between the two independent reviewers were resolved by consensus and arbitration by the third (KEA) author of the review. For studies with uncertainties regarding eligibility, the authors were contacted to seek additional information. A total of two contact attempts were made, and if no response was received, the articles were screened for eligibility based on the information available. The results of the study selection process are presented in a PRISMA flow diagram (Figure 1).

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Data collection process and data items

Relevant data were extracted from the eligible studies by KYA, and independently verified by FAO using piloted forms adapted from the Joanna Briggs Institute (JBI) reviewer's manual [35] and the Cochrane Pregnancy and Childbirth Group for Systematic Reviews [36]. Eligible studies were identified using the following information: first author, publication year, study design and setting, study participants, sample size, geographical region, intervention components, location of intervention delivery, outcome measures, statistical analysis, and results. Corresponding authors of eligible studies were contacted for information that was not clear in the data extraction. A total of two contact attempts were made, and if no response was received, only the information available was used.

Risk of bias Assessment

All included studies were examined for the five types of biases of interventional studies: (i) selection bias; (ii) performance bias; (iii) attrition bias; (iv) detection bias and; (v) reporting bias. KYA and KEA independently conducted the risk of bias assessment using the revised Cochrane risk of bias tool for clustered randomized trials (ROB 2.0) [37 38] and the risk of bias in non-randomised studies - of interventions (ROBINS-I) tool [39] for assessing the risk of bias in quasi-experimental studies. Cluster randomised trials were assessed for five domains (i.e., randomization and identification or recruitment bias, deviations from intended interventions, missing data, measurement of outcomes, and selection of reported result), and quasi-experimental studies were assessed for seven domains (i.e., confounding, selection of participants, classification of interventions, deviations from intended interventions, missing data, measurement of outcomes, and selection of reported result). For cluster randomised trials, the overall risk of bias judgement included low risk, some concerns and high risk [37 38], and for the quasi-experimental studies, the overall risk of bias outcome included low, moderate,

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serious, and critical risk of bias [39]. There was no disagreement between reviewers in the risk of bias assessment, no third arbitrator was needed.

Data synthesis

Author reports of effect size measures were used in reporting the findings of each study. Effect size measures included a comparison between experimental and control groups (e.g., odds ratio [OR]), as well as the difference in proportion between pre- and post-outcome measures (e.g., change in proportion). Where available, 95% confidence interval (CI) and P-value were obtained from the eligible studies. Based on published methodological studies [40 41], 95% CI, and P-value were calculated using the available information where the full text of the eligible studies did not report these estimates. Narrative methods including text, table, and figure were also used as a tool for data presentation.

Patient and public involvement

There was no direct patient or public involvement in this review

Results

Characteristics of eligible studies

A total of 4017 articles were retrieved from the database and hand searching. Of these, 1924 were retained in the Endnote library after removing duplicates. Titles and abstracts of the articles were screened, and 1654 articles were removed. Full-text eligibility checks of the remaining 52 articles excluded 29 studies. The reasons for exclusion of studies included: 4 studies were duplicates[42-44], 1 study was cross-sectional[45], 2 studies used qualitative evaluation [46 47], 10 studies were abstracts [48-57], 4 studies were based on the design and implementation of

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interventions[58-61], 5 studies did not measure IYCF outcomes [62-70], and 3 studies were based on interventions designed for different target groups [71-73] **(Figure 1).**

Among the 23 included studies, 14 were quasi-experimental studies [74-87], while the remaining nine were cluster randomized trials [88-96]. At least one interventional study was conducted in Southern Nations Nationalities and Peoples' Region (SNNPR) [77-79 81 82 84-86 90], Oromia[74 77 78 83-88 91 95], Amhara [76-78 84-87 89 92 93 95], Tigray [77 79 84-87 95], and Afar [80] regions, but we did not find published interventional studies conducted in either Addis Ababa and Dire Dawa city administrations, Harari, Benishangu1, Gambella, or Somali regions.

Risk of bias in included studies

Supplementary files 4 and 5 present the risk of bias assessment of the included studies. Nine cluster randomised trials were assessed for risk of bias using the Cochrane RoB 2.0 tool. Overall, studies by Muluye et. al., 2020 and Passarelli et. al., 2020 judged as high risk of bias[94 95], while studies by Abdulahi et. al., 2021 and Kim et. al., 2019 had a low risk of bias [89 91]. Two cluster randomised studies had a high risk of bias in the domains for randomisation [94 95]. Out of 14 quasi-experimental studies assessed using the ROBINS-1 tool, eight studies had a serious risk of bias [74 77 78 81-84 86], and four studies had a moderate risk of bias [79 80 85 87]. The remaining two studies were judged as critical risk of bias [75 76]. Seven out of fourteen studies had a serious or critical risk of bias in the confounder adjustment [74-78 81-83], while eight out of fourteen studies had a low risk of bias in outcome measurement [77-81 84 85 87].

Evidence from reviewed studies

In the eligible studies, n = 23, the majority of the interventions were provided in the form of behaviour change communication (BCC) using nutrition education [81 82 92 93 95], community

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mobilisation [76 78 87 89], health education materials [74 80 82 83 91], and mass media communication [79 89]. Other interventions were also provided in the form of peer/group support and training of mothers [91-93], promotion of chicken husbandry and home gardening [95], productive safety net program (in-cash/food rations) [84], strengthening of health care system [78 80 85-87], training and supervision of health care providers [76 78 83], and food preparation and practical demonstration sessions [81 82 88 91 94 96]. Additionally, two studies used the health belief model (HBM) as a mode of intervention delivery for BCC [81 90] **(Table 1)**.

Sixteen studies used control groups for comparison with experimental groups [78 80-84 87-96], while seven studies did not use control or comparison group[74-77 79 85 86]. Two studies used two experimental arms as an intervention group [90 95] **(Table 1)**.

Impact of interventions on IYCF practices in Ethiopia

EIBF

Tables 1 and **2** are presented to summarise evidence from eligible studies. Eight studies reported the effect of interventions to improve EIBF [76 77 79 80 85-87 91]. A health facility level intervention in the Afar region that integrated nutrition information into health professional guidelines, and also provided family education and counselling booklets resulted in a significant improvement in EIBF in the experimental group compared to the control group (OR = 1.70; 95% CI: 1.17, 2.46; P-value = 0.005) [80]. Kim et al., 2016 showed that interventions delivered through advocacy and policy dialogues, interpersonal communication and community mobilization, and mass communication improved EIBF [79]. This study also showed that the association between exposure to two or more types of interventions and improvement in EIBF was stronger compared to exposure to a single type of intervention (OR = 2.07; 95%

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CI: 1.09, 3.91; P-value < 0.05 for multiple intervention vs OR = 1.88; 95% CI: 1.08, 3.27; P-value < 0.05 for single intervention) [79].

Karim et. al., 2018 indicated that supporting community health workers decision-making using community health data improved EIBF (OR = 1.27; 95% CI: 1.10, 1.46; P-value = 0.001) [86]. However, Wereta et. al., 2018 reported no significant effect of community-based maternal and child health service quality improvement project on EIBF (average treatment effect [ATE] = 2.8; 95% CI: −4.7, 10.6; P-value = 0.468)[87].

Buchanan et. al., 2012 [76], Callaghan-K. et. al., 2016 [77], and Karim et. al., 2013 [85] found an increase in the proportions of EIBF (from 31.0% to 77.0%, 51.4% to 67.9%, and 46.0% to 54.2%, respectively), following the implementation of BCC and community-level health programs including breastfeeding promotions and family health services.

EBF

Ten studies reported the effect of interventions on EBF in Ethiopia [74-80 83 85 91]. Of these, eight studies measured EBF for up to six months [74-80 91], while two studies measured EBF for up to one month [83 85]. Villadsen et. al., 2016 found that antenatal care strengthening for pregnant women significantly improved EBF in health facilities under intervention compared to health facilities without intervention (OR = 2.30; 95% CI: 1.40, 3.60; P-value = 0.005) [83]. Integrating nutrition into the health workers' guidelines, and providing family education and counselling booklet (OR = 2.62; 95% CI: 0.95, 7.21; P-value = 0.062) [80], and exposure to BCC intervention (i.e., advocacy and policy dialogues, interpersonal communication and community mobilization, and mass communication) [OR = 1.51; 95% CI: 0.95, 2.40; P-value > 0.050] [79] did not affect EBF behaviours.

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Seven studies showed that there were increases in EBF during the intervention period [74-78 85 91], but out of these studies, two indicated that improvements in EBF were not statistically significant [74 78]. Abdulahi et. al., 2021 reported a significant difference in the proportion of EBF between the intervention (68.3%) and control groups (54.8%) [P-value < 0.001], following the interventions of peer support, breastfeeding education, and practical demonstration sessions on positioning and attachment [91]. Studies by Callaghan et. al., 2016 (P-value < 0.010) [77] and Karim et. al., 2013 (P-value < 0.001) [85] documented that implementing community-based and health facility interventions improved EBF, but the remaining two studies on EBF did not report the statistical test of significance[74 78].

Introduction of solid, semi-solid and soft foods

Four studies reported the effect of different types of BCC interventions on the introduction of complementary foods [76 79 89 93]. BCC interventions using complementary feeding messages, training of mothers in groups, counselling and support at home level, and facilitating participatory family discussion were effective in improving the timely initiation of complementary foods (RR = 2.60; 95% CI: 1.78, 5.86; P-value = 0.002) [93]. Kim et. al. investigated the effect of exposure to BCC interventions (advocacy and policy dialogues, interpersonal communication and community mobilization, and mass communication) on the introduction of solid, semi-solid and soft foods in two separate arms. The first arm of the study found an increase in the proportion of infants who were introduced to solid, semi-solid and soft foods (from 37.4% to 59.7%, P-value <0.001) [79]. In contrast, the second arm of the study did not find a significant improvement (from 58.7% to 61.3%, P-value >0.05) [89]. The remaining study by Buchanan et. al., 2012 showed improvement in proportion, but no test of significance was reported [76].

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Continued breastfeeding at one year

Only one study measured the impact of interventions on continued breastfeeding at one year. Tariku et. al., 2015 showed that IYCF messages using the health belief model (HBM) did not impact the continued breastfeeding rate at one year [90].

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Table 1: Summary of eligible studies

Authors	Design	Participants	Region	Intervention components	Location of intervention delivery	Outcome measures
Abdulahi et. al., 2021[91]	Cluster-randomized trial	Experimental group • Before intervention = 249 • After intervention = 212 Control group (routine care) • Before intervention = 219 • After intervention = 197	Oromia	Breastfeeding education and peer support • Peer support to encourage mothers for health facility birthing • Breastfeeding education including educational materials and • Practical demonstrations on proper positioning and latching	Community based	EIBF EBF
Abiyu et. al., 2020[92]	Cluster-randomized trial	Experimental group • Before intervention = 306 • After intervention = 272 Control group (routine care) • Before intervention = 306 • After intervention = 282	Amhara	Complementary feeding behaviour change communication • Complementary feeding messages • Group training of mothers • Home visits (for counselling and support) • Participatory family level discussion	Community based	MDD MMF MAD
Abiyu et. al., 2020[93]	Cluster-randomized trial	Experimental group • Before intervention = 306 • After intervention = 272 Control group (routine care) • Before intervention = 306 • After intervention = 282	Amhara	Complementary feeding behaviour change communication • Complementary feeding messages • Group training of mothers • Home visits (for counselling and support) • Participatory family level discussion	Community based	Timely initiation of complementary foods
Bahru et. al., 2020[84]	Quasi-experimental	Experimental group • Before intervention = 482 • After intervention = 275 Control group (routine care) • Before intervention = 740 • After intervention = 882	Tigray Oromia Amhara SNNPR	Productive safety net program (PSNP) • Offers employment opportunities for households with able-bodied members to work on community asset building projects and earn a wage either in cash or in-kind (food) • Direct support to households whose breadwinners are the elderly or disabled and hence could not take part in labour-intensive activities	Health facility and community- based	Dietary diversity Meal frequency
Baxter et. al., 2016 [74]	Quasi-experimental	Experimental group: • Before intervention = 173 • After intervention = 269	Oromia	EBF promotion strategies including • Home visits, • Educational materials, and • Training of influential community and family members)	Health facility and community- based	EBF
Berti et. al., 2010 [75]	Quasi-experimental	No information	No information	• Breastfeeding communication through community-based interventions for behaviour change	Community-based	EBF
Buchanan et. al., 2012 [76]	Quasi-experimental	No information	No information	• Behavioural change communication • Community mobilization • Capacity building	Health facility-based and community-based	EIBF EBF Introduction of complementary foods
Callaghan-Koru et. al., 2016 [77]	Quasi-experimental	Experimental group: • Before intervention=218 • After intervention=215	Tigray Oromia Amhara	• An integrated package of facility-based kangaroo mother care • Community-based promotion of EBF	Health facility and community based	EIBF EBF

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			SNNPR			
Carnell et. al., 2014 [78]	Quasi-experimental	Experimental group: • Before intervention = 1016 • After intervention = 911 Control group: • Before intervention = 1074 • After intervention = 903	Oromia Amhara SNNPR	• Strengthen the health system • Improve health workers' performance • Engage the community	Health facility and community based	EBF
Kang et. al., 2017 [88]	Cluster-randomized trial	• Experimental group: 570 • Control group: 629	Oromia	• Community-based participatory nutrition promotion program • Daily group nutrition sessions • Follow-up visits	Community-based	Meal frequency Dietary diversity Dietary adequacy
Karim et. al., 2013[85]	Quasi-experimental	Experimental group: • Before intervention = 1481 • After intervention = 1494	Tigray Oromia Amhara SNNPR	Supporting Ethiopian health extension program in: • Health education and demonstration for families on hygiene and environmental sanitation • Family health (Maternal and child health, reproductive health, immunization, and nutrition) services	Community-based	EIBF EBF at one month
Karim et. al., 2018[86]	Quasi-experimental	Experimental group: • Before intervention = 2124 • After intervention = 2113	Tigray Oromia Amhara SNNPR	Supporting Ethiopian health extension program using community-based data for decision-making • Family Conversation, a forum conducted at the household of the pregnant women • Birth notification to ensure early postnatal care • Establish household level surveillance system • Identify and address barriers to access maternal and newborn health services using generated data	Health facility and community based	EIBF within 30 min of childbirth
Kim et. al., 2016 [79]	Quasi-experimental	Experimental group: • Before intervention = 1404 • After intervention = 1404	SNNPR Tigray	BCC strategies including: • Advocacy and policy dialogues • Interpersonal communication and community mobilization • Mass communication	Community-based	EIBF EBF Introduction of complementary foods MDD MMF MAD
Kim et. al., 2019 [89]	Cluster-randomized trial	Experimental group: • Before intervention = 1328 • After intervention = 1360 Control group: • Before intervention = 1318 • After intervention = 1360	Amhara	BCC strategies including: • Interpersonal communication • Nutrition sensitive agricultural activities • community mobilization • Mass communication	Community-based	Introduction of complementary foods MDD MMF
Kung'u et. al., 2018 [80]	Quasi-experimental	Experimental group: • Before intervention = 2570 • After intervention = 2905 Control group: • Before intervention =1670	Afar	• Behaviour change interventions • Integrating nutrition to the frontline workers' guideline • Family education and counselling booklet	Health facility	EIBF EBF at six months

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		• After intervention = 1620				
Mulualem et. al., 2016 [81]	Quasi-experimental	Experimental group: • Before intervention = 80 • After intervention = 80 Control group: • Before intervention = 80 • After intervention = 80	SNNPR	Nutrition education based on the HBM model • Individual and group sessions, and • Recipe demonstration	Community-based	Meal frequency
Muluye et. al., 2020 [94]	Cluster randomized control trial	Experimental group: • Before intervention = 100 • After intervention = 100 Control group: • Before intervention = 100 • After intervention = 100	SNNPR	Education and practical demonstration on • complementary food recipes and preparation, • amount and frequency of feeding, • personal hygiene and sanitation, and • dietary diversification	Facility-based	MDD
Negash et. al., 2014 [82]	Quasi-experimental	Experimental group: • Before intervention = 100 • After intervention = 81 Control group: • Before intervention = 92 • After intervention = 76	SNNPR	• Nutrition education using visual materials • Demonstration of food preparation	Community-based	Dietary diversity Meal frequency
Passarelli et. al., 2020[95]	Cluster-randomized trial	ACGG intervention group: • Before intervention = 255 • After intervention = 255 ACGG/ATONU intervention group: • Before intervention = 263 • After intervention = 265 Control group: • Before intervention = 311 • After intervention = 311	Tigray Oromia Amhara SNNPR	ACGG • Each farmer received 25 vaccinated chicks that were ~6-week old • Farmers implemented chicken husbandry according to their individual preferences ACGG/ATONU • Additional nutrition sensitive intervention such as promotion of home gardening, provision of fruit and vegetable seeds for the gardens, and behavioural change communication	Community-based	Dietary diversity (7-days)
Teshome et. al., 2020[96]	Cluster-randomized trial	Experimental group: • Before intervention = 386 • After intervention = 307 Control group: • Before intervention = 386 • After intervention = 314	SNNPR	• Nutritional education and counseling on balanced diet, and food groups and benefits of pulse • Recipe demonstrations on preparation of porridge for complementary feeding	Community-based	MDD
Tariku et. al., 2015 [90]	Cluster-rando[91-93]mized trial	HBM intervention group: • Before intervention = 60 • After intervention = 54 Traditional nutrition education group: • Before intervention = 60 • After intervention = 56 Control group:	SNNPR	• Complementary feeding messages using the HBM model	Community-based	Continued breastfeeding MDD MMF

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		• Before intervention = 60 • After intervention = 56				
Villadsen et. al., 2016 [83]	Quasi-experimental	Experimental group: • Before intervention =775 • After intervention = 1434 Control group: • Before intervention = 350 • After intervention = 422	Oromia	• Training of health professionals • Health education materials • Supervisions of ANC providers	Health facility	EBF at 1 month
Wereta et. al., 2018[87]	Quasi-experimental	Experimental group • Before intervention = 408 • After intervention = 990 Control group (routine care) • Before intervention = 408 • After intervention = 990	Tigray Oromia Amhara SNNPR	• Increase access and quality of ANC, delivery and perinatal outcomes • Community engagement in planning (e.g., joint action plan to address these barriers and gaps) and implementation • Enhance interactions between the communities, HEWs, health care providers and local health officials	Health facility and community based	EIBF

EIBF: early imitation of breastfeeding; EBF: exclusive breastfeeding; MDD: minimum dietary diversity; MMF: minimum meal frequency; MAD: minimum acceptable diet; BCC: behaviour change communication; HBM: health belief model; SNNPR: Southern Nations Nationalities and Peoples Region; ACGG: African Chicken Genetic Gains; ATONU: Agriculture to Nutrition.

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Minimum dietary diversity

Ten studies reported the effect of interventions on MDD[79 82 84 88-90 92 94-96]. Of these, six were measured according to the WHO definition for MDD [79 89 90 92 94 96], while three studies used any four food groups as a cut-off point to measure MDD [82 84 88]. A study by Abiyu et. al., 2020 using BCC interventions including complementary feeding messages, mothers group training, home-based counselling and support, and participatory family discussion improved MDD for infants and young children (RR = 3.00; 95% CI: 1.34, 7.39; P-value = 0.012)[92]. Two studies by Kim et. al. revealed the significant impacts of exposure to more than one type of BCC intervention (advocacy and policy dialogues, interpersonal communication and community mobilization, and mass communication) on MDD, but these studies showed that exposure to a single type of intervention did not have an impact on MDD [79 89]. Tariku et. al., 2015 showed interventions involving complementary feeding messages using the HBM model significantly increased the proportion of MDD (from 37.5% to 67.9%; P-value <0.05) [90].

End line survey results following the intervention of nutritional education and counselling, and recipe demonstrations by Muluye et. al., 2020 (67.0% vs 33.0%, P-value = 0.034)[94] and Teshome et. al., 2020 (51.5% vs 43.3%; P-value = 0.001)[96] showed significant differences in the proportion of MDD between intervention and control groups. The remaining four studies indicated that the proportion of MDD increased over the intervention period, but the effect was not significant [82 84 88 95].

Minimum meal frequency

MMF was included as an outcome variable in eight interventional studies[79 81 82 84 88-90 92]. Four studies used the WHO definition for assessing MMF [79 89 90 92]. Complementary feeding BCC interventions using nutritional education and counselling, and recipe demonstration sessions improved MMF practice of mothers (RR = 2.40; 95% CI: 1.37, 4.29; P-value = 0.003)[92]. Kim

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et. al., 2019 indicated that exposure to two types of BCC interventions (advocacy and policy dialogues, interpersonal communication and community mobilization, and mass communication) had a significant effect on MMF of infants and young children compared to those who did not expose to intervention (OR = 1.69; 95% CI: 1.13, 2.52; P-value < 0.010) [89]. Kang et. al., 2017 found that interventions using a community-based participatory nutrition program had significant impact on the mean score of MMF (Mean difference = 1.04; 95% CI: 0.35, 1.73; P-value = 0.003) [88]. Except for one study, which did not do a test for significance [82], the remaining four studies did not show significant improvements in MMF [79 81 84 90].

Minimum acceptable diet

Out of three studies that measured MAD as an outcome, two studies used the WHO definition for assessing MAD [79 92], while the remaining study calculated dietary adequacy indicator using a composite dietary diversity and meal frequency scores [88]. Abiyu et. al., 2020 implemented BCC interventions (including nutritional education and counselling, and recipe demonstration) to improve MAD. The result from this study showed the significant effect of interventions on MAD (RR = 2.70; 95% CI: 1.13, 7.23; P-value = 0.036) [92]. Community based participatory nutrition program by Kang et. al., 2017 reported significant effect of interventions on the dietary adequacy score (Mean difference = 1.40; 95% CI: 0.49, 2.32; P-value = 0.003)[88]. However, a study by Kim et. al., 2019 documented that implementing four or more type BCC interventions (advocacy and policy dialogues, interpersonal communication and community mobilization, and mass communication) improved MAD (OR = 14.29; 95% CI: 2.23, 90.28; P-value < 0.010 for 4+ interventions)[79].

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476 **Table 2: Summary of evidence from reviewed studies**

IYCF outcomes	Studies by authors	Statistical analysis	Results
EIBF	Abdulahi et. al., 2021[91]	Percent change in proportion	• EIBF was 72.7% in intervention group, and 40.6% in control group (P-value <0.001)
	Buchanan et. al., 2012 [76]	Percent change in proportion	• EIBF increased from 31.0% to 77.0%
	Callaghan-K. et. al., 2016 [77]	Percent change in proportion	• EIBF significantly increased from 51.4% to 67.9% (P value <0.010)
	Karim et. al., 2013[85]	Percent change in proportion	• EIBF increased from 46.0% to 54.2% (P-value <0.001)
	Karim et. al., 2018[86]	Fixed-effect logistic regression	• EIBF at 30 minutes (OR = 1.27; 95% CI: 1.10, 1.46; P-value = 0.001)
	Kim et. al., 2016 [79]	Fixed-effect logistic regression	• Single type of intervention: EIBF (OR = 1.88; 95% CI: 1.08, 3.27; P-value < 0.050) • Two types of interventions: EIBF (OR = 2.07; 95% CI: 1.09, 3.91; P-value < 0.050) • Four or more interventions: EIBF (OR = 2.48; 95% CI: 1.09, 5.64; P-value < 0.050)
	Kung'u et. al., 2018 [80]	Fixed-effect logistic regression	• EIBF (OR = 1.70; 95% CI: 1.17, 2.46; P-value = 0.005)
	Wereta et. al., 2018[87]	Propensity score matching	• EIBF (ATE = 2.8; 95% CI: −4.7, 10.6; P-value = 0.468)
EBF	Abdulahi et. al., 2021[91]	Percent change in proportion	• EBF was 68.3% in intervention group, and 54.8% in control group (P-value <0.001)
	Baxter et. al., 2016 [74]	Percent change in proportion	• EBF increased from 69.4% to 75.1%, but not statistically significant
	Berti et. al., 2010 [75]	Percent change in proportion	• Higher improvement in exclusive breastfeeding
	Buchanan et. al., 2012 [76]	Percent change in proportion	• EBF increased from 73.0% to 95.0%
	Callaghan-Koru et. al., 2016 [77]	Percent change in proportion	• EBF increased from 86.0% to 95.8% (P value <0.010)
	Carnell et. al., 2014 [78]	Difference-in-difference	• No significant difference in the improvement of EBF was observed
	Karim et. al., 2013[85]	Percent change in proportion	• EBF at one month increased from 82.4% to 95.3% (P-value <0.001)
	Kim et. al., 2016 [79]	Fixed-effect logistic regression	• Single type of intervention: EBF (OR = 1.51; 95% CI: 0.95, 2.40; P-value>0.050) • Two types of interventions: EBF (OR = 1.39; 95% CI: 0.96, 2.01; P-value>0.050) • Four or more interventions: EBF (OR = 0.93; 95% CI: 0.86, 1.01; P-value>0.050)
	Kung'u et. al., 2018 [80]	Fixed-effect logistic regression	• EBF at six months (OR = 2.62; 95% CI: 0.95, 7.21; P-value = 0.062)
	Villadsen et. al., 2016 [83]	Fixed-effect logistic regression	• EBF at 1 month (OR = 2.30; 95% CI: 1.40, 3.60; P-value = 0.001)
Introduction of complementary foods	Abiyu et. al., 2020[93]	Generalized estimating equations (GEE) regression	• Timely initiation of complementary foods (RR = 2.60; 95% CI: 1.78, 5.86; P-value = 0.002)
	Buchanan et. al., 2012 [76]	Percent change in proportion	• Introduction of complementary foods increased from 38.0% to 98.0%
	Kim et. al., 2016 [79]	Percent change in proportion	• Introduction of complementary foods increased from 37.4% to 59.7% (P-value <0.001)
	Kim et. al., 2019 [89]	Percent change in proportion	• Introduction of complementary foods did not show any improvement
Continued breastfeeding at 1–year	Tariku et. al., 2015 [90]	ANOVA and Kruskal-Wallis test	• No significant difference in the improvement of continued breastfeeding
Dietary diversity	Abiyu et. al., 2020[92]	Generalized estimating equations (GEE) regression	• MDD (RR = 3.00; 95% CI: 1.34, 7.39; P-value = 0.012)
	Bahru et. al., 2020[84]	Marginal structural model	• No significant difference in dietary diversity score (β = −0.18; 95% CI: -0.41, 0.05)

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	Kang et. al., 2017 [88]	Mixed-effect logistic regression	• No significant difference in mean score between intervention and control group (Mean difference = 0.38; 95% CI: -0.22, 0.97; P-value = 0.220)
	Kim et. al., 2016 [79]	Fixed-effect logistic regression	• Single type of intervention: MDD (OR = 6.15; 95% CI: 0.80, 47.10; P-value > 0.050) • Two types of interventions: MDD (OR = 6.87; 95% CI: 1.27, 37.14; P-value < 0.050) • Four or more interventions: MDD (OR = 18.75; 95% CI: 2.32, 151.52; P-value < 0.010)
	Kim et. al., 2019 [89]	Fixed-effect logistic regression	• Single type intervention: MDD (OR = 1.05; 95% CI: 0.74, 1.48; P-value > 0.050) • Two types of interventions: MDD (OR = 1.82; 95% CI: 1.14, 2.89; P-value < 0.050) • Four or more interventions: MDD (OR = 3.15; 95% CI: 2.18, 4.57; P-value < 0.001)
	Muluye et. al., 2020 [94]	Independent student's t-test	• Baseline: no significant difference between intervention and control group (23.0% vs 26.0%, P-value = 0.519) • End line: there was significant difference between intervention and control group (67.0% vs 33.0%, P-value = 0.034)
	Negash et. al., 2014 [82]	Independent sample t-test	• Dietary diversity increased over the intervention period (P-value > 0.050)
	Passarelli et. al., 2020[95]	Ordinary least square regression	• There was no significant difference between intervention and control groups
	Tariku et. al., 2015 [90]	ANOVA and Kruskal-Wallis test	• Intervention group showed significant improvement in MDD (P<0.001)
	Teshome et. al., 2020[96]	Percent change in proportion	• Baseline: no significant difference between intervention and control group (7.5% vs 3.8%) • End line: there was significant difference between intervention and control group (51.5% vs 43.3%; P-value = 0.001)
Meal frequency	Abiyu et. al., 2020[92]	Generalized estimating equations (GEE) regression	• MMF (RR= 2.40; 95% CI: 1.37, 4.29; P-value = 0.003)
	Bahru et. al., 2020[84]	Marginal structural model	• No significant effect on meal frequency (β = 0.31; 95% CI: -0.07, 0.54)
	Kang et. al., 2017 [88]	Mixed-effect logistic regression	• Significant difference in mean score between intervention and control group (Mean difference = 1.04; 95% CI: 0.35, 1.73; P-value = 0.003)
	Kim et. al., 2016 [79]	Fixed-effect logistic regression	• Single type of intervention: MMF (OR = 1.06; 95% CI: 0.99, 1.13; P-value > 0.050) • Two types of interventions: MMF (OR = 1.31; 95% CI: 0.97, 1.77; P-value > 0.050) • Four or more interventions: MMF (OR = 2.04; 95% CI: 0.92, 4.54; P-value < 0.010)
	Kim et. al., 2019 [89]	Fixed-effect logistic regression	• Single type of intervention: MMF (OR = 1.05; 95% CI: 0.90, 1.23; P-value > 0.050) • Two types of interventions: MMF (OR = 1.69; 95% CI: 1.13, 2.52; P-value < 0.010) • Four or more interventions: MMF (OR = 1.93; 95% CI: 1.44, 2.57; P-value < 0.001)
	Mulualem et. al., 2016 [81]	ANOVA	• MMF frequency decreased in both experimental and control groups
	Negash et. al., 2014 [82]	Independent sample t-test	• Meal frequency increased over the intervention period (P-value > 0.050)
	Tariku et. al., 2015 [90]	ANOVA and Kruskal-Wallis test	• No significance effect on MMF
Minimum acceptable diet	Abiyu et. al., 2020[92]	Generalized estimating equations (GEE) regression	• MAD (RR = 2.70; 95% CI: 1.13, 7.23; P-value = 0.036)

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	Kang et. al., 2017 [88]	Mixed-effect logistic regression	• Significant effect on composite feeding index of dietary diversity and meal frequency (Mean difference = 1.40; 95% CI: 0.49, 2.32; P-value = 0.003)
	Kim et. al., 2019 [89]	Fixed-effect logistic regression	• Single type of intervention: MAD (OR = 5.20; 95% CI: 0.82, 32.99; P-value > 0.050) • Two types of interventions: MAD (OR = 5.82; 95% CI: 0.81, 41.90; P-value > 0.050) • Four or more interventions: MAD (OR = 14.29; 95% CI: 2.23, 90.28; P-value < 0.010)

IYCF: infant and young child feeding; EIBF: early imitation of breastfeeding; EBF: exclusive breastfeeding; MDD: minimum dietary diversity; MMF: minimum meal frequency; MAD: minimum acceptable diet; HBM: health belief model

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Discussion

A total of 23 studies were included, and of these, 14 were quasi-experimental studies [74-87], and nine were cluster randomised trials [88-96]. Four studies for EBF [77 83 85 91] and six studies for EIBF [77 79 80 85 86 91] showed that community- and facility-based interventions delivered in the form of BCC (such as policy advocacy, interpersonal communication, community mobilisation, and mass media campaigns) and health service strengthening had an impact on infants. Two studies for the introduction of complementary foods [79 93], six studies for MDD [79 89 90 92 94 96] and three studies for MMF [88 89 92] indicated significant effects of BCC on infants and young children. Two eligible studies showed that combined interventions delivered in the form of interpersonal communication, nutrition-sensitive agricultural activities, community mobilisation, and mass media campaigns had more impacts on EIBF, MDD, and MMF[79 89].

Global evidence indicates that BCC interventions (including policy advocacy, interpersonal communication, community mobilisation and mass media campaigns) are essential in promoting IYCF practices. These initiatives also create a supportive environment that helps mothers to adopt and sustain positive child feeding behaviours [11 97]. The main goals of BCC intervention include improving behaviours of facility- and community-level health workers, mothers and other caregivers, and members of their households [11 97]. Our review showed that BCC interventions in the form of policy advocacy, interpersonal communication, community mobilisation, and mass media campaigns improved EIBF, MDD, and MMF. Our finding is consistent with systematic reviews from lower- and middle-income countries (LMICs), which indicated that group nutritional education and counselling, family or social support, integrated mass media coverage, and community mobilization were effective in improving IYCF practices [5 11].

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Although BCC interventions promote skills and knowledge that are necessary for improving IYCF practices, studies from LMICs indicated that achieving the intended target is often challenging without complementary interventions that provide or facilitate material inputs [97-100]. This could be due to the fact that child feeding decisions not only depend on the knowledge and perception of the mother but also are influenced by a complex interplay of factors, including available resources, partner support, mother's time and close family support [97-101]. Affleck et. al. 2012 reported that a lack of money to provide diversified diet for infants and young children was a major barrier to improve appropriate IYCF behaviours among Bangladeshi women [99]. Thus, promoting material inputs (such as cash transfer and provision of food materials) and/or non-material inputs (skills and knowledge) are required in achieving appropriate and sustained IYCF behaviours.

Research has shown that exposure to more than one IYCF intervention, delivered through different platforms (e.g., community- or facility-based), was more successful in improving IYCF outcomes compared to a single form of intervention delivered at a single setting [5]. Our review indicated that exposure to more than two types of interventions was associated with improvement in EIBF, MDD, and MMF [79 89]. This result may be due to interventions delivered through multiple settings may have an increased intensity that can be translated to behavioural changes due to repeated contacts and adequate coverage of interventions [102]. Our review suggests that interventions that aim to improve IYCF behaviours in Ethiopia should be delivered through different and specific types of contexts (including household, health facility and community) and should be provided along the continuum of care from conception until a child's second birthday.

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In the past two decades, the Ethiopian government's commitment to improving IYCF has been consistent, and this started with the development of the National Strategy for IYCF in 2004 and followed by National Strategy for Nutrition in 2008 [24 103], in line with the WHO/UNICEF IYCF strategy [104]. In 2015, the Ethiopian government launched the Health Sector Transformation Plan to improve health outcomes, including IYCF practices[25]. Based on this initiative, many facility- and community-based interventions have been implemented to protect, promote and support appropriate IYCF behaviours [10 11]. Nevertheless, our review found that there has been no formal IYCF intervention conducted in Addis Ababa and Dire Dawa city administrations, and Harari, Benishangu1, Gambella, and Somali regions. Our review suggests an expansion of additional experimental studies from successful subnational interventions implemented in Amhara[89 92 93], Oromia[83 88 91], and SNNP[94 96] regions, to maximise the opportunities for the prevention of IYCF-related morbidity and mortality in those regions.

Additionally, our review did not find any studies which incorporated the use of mobile health (mHealth) as part of an intervention package for improving IYCF [105]. Future experimental studies that consider the use of mHealth techniques are required, especially that previously published studies have shown the positive impacts of interventions (in form of voice messages, SMS, and phone calls) on IYCF practices [106-108]. Finally, interventional studies that consider the role of fathers/partners as well as grandmothers in improving IYCF behaviours may be warranted in Ethiopia [101 109 110].

The present systematic review of IYCF outcomes from community- and facility-based interventions (e.g., BCC, health service strengthening, community mobilizations, and nutrition sensitive agricultural activities) has strengths and limitations. First, a meta-analysis was not conducted due to the wide heterogeneity of the design and outcome measures. Second, this

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review was limited by the incompleteness of information from the individual studies. Nevertheless, the study authors were contacted for information when it is necessary. Third, our search was limited to published studies in English, which exclude studies published with other languages. Fourth, lack of evidence and under-representation of some regions may limit the generalizability of our review. However, evidence from this review is potentially relevant for IYCF policy efforts in Ethiopia. Despite the above limitations, the application of multiple search strategies in consultation with the medical librarian and registration of the systematic review protocol can be considered as strengths of this review. Furthermore, future studies that consider the assessment of the full set of the updated IYCF indicators for all Ethiopian children would be warranted [111]. This approach would ensure that both subnational and national and IYCF estimates are accurate and reliable and ensure that interventions and policy decisions in resource allocation are targeted at the most at-risk population. It would also ensure that IYCF programme monitoring and evaluation are consistent across all levels of government [111].

Conclusion

Our systematic review showed that 12 out of 21 eligible studies that implemented interventions in the form of policy advocacy, health service strengthening, interpersonal communication, community mobilisation, and mass media campaigns reported improved EIBF, EBF, and/or MDD in Ethiopia. The impacts of current and/or future interventions on IYCF practices in Ethiopia will be greater when interventions are combined, introduced and implemented along the continuum of care from conception until a child's second birthday.

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List of abbreviations

ACGG: African Chicken Genetic Gains; ANC: Antenatal Care; ATONU: Agriculture to Nutrition; BCC: Behavioural Change Communication; BFHI: Baby-Friendly Hospital Initiative; CI: Confidence Interval; CINAHL: Cumulative Index to Nursing and Allied Health Literature; CRTs: Cluster Randomised Trials; EBF: Exclusive Breastfeeding; EIBF: Early Initiation of Breastfeeding; GEE: Generalized estimating equations; GNTs: Global Nutrition Targets; HBM: Health Belief Model; HSTP: Health Sector Transformation Plan; IYCF: Infant and Young Child Feeding; JBI: Joanna Briggs Institute; LMICs: Lower and Middle-Income Countries; MAD: Minimum Acceptable Diet; MDD: Minimum Dietary Diversity; MeSH: Medical Subject Headings; MDG: Millennium Development Goal; MMF: Minimum Meal Frequency; OR: Odds Ratio; ORS: Oral Rehydration Salt; PICO: Population Intervention Comparator Outcome; Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA); RCTs: Randomised Controlled Trials; ROB: Risk of Bia; Risk of bias in non-randomised studies - of interventions (ROBINS); SDGs: Sustainable Development Goals; SNNPR: Southern Nations Nationalities and Peoples Regions; SUMARI: System for the Unified Management, Assessment and Review of Information; UNICEF: United Nation Children's Fund; WHO: World Health Organization.

Declarations

Contributors

KYA conceptualised the study idea, performed the literature searching, selected the paper of interests, performed data extraction, and critically revised the manuscript. AP and AA critically revised the manuscript for intellectual content. KEA performed the risk of bias assessment, and critically revised the manuscript. FAO contributed to the conception of the research idea,

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selected the paper of interests, performed data extraction, and critically revised the manuscript. All authors read and approved the final manuscript.

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Ethics approval

Ethical approval was not required for this study as the study was based on retrieved and synthesised data from already published studies.

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Competing interests

The authors declare that they have no competing interests

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Figure 1: Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flowchart for selection of eligible studies

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Supplementary appendix

Supplementary file 1: PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) 2009 Checklist.

Supplementary file 2: Search strategy developed for Embase (OVID) database

Supplementary file 3: Electronic copy of the search strategy developed for Embase (OVID) database

Supplementary file 4: Risk of bias assessment for cluster randomised trial studies

Supplementary file 5: Risk of bias assessment for quasi-experimental studies

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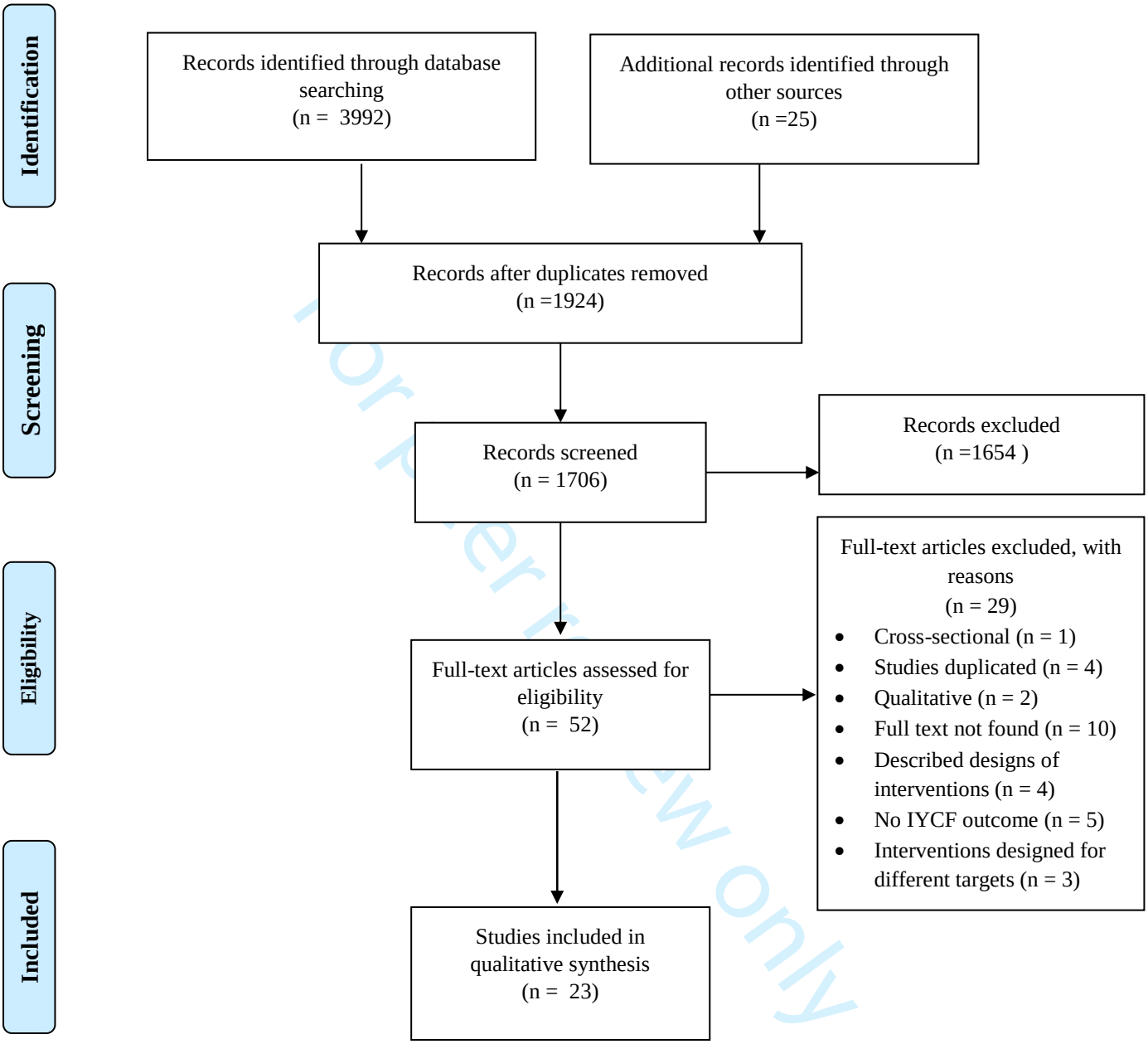


Figure 1: Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flowchart for selection of eligible studies [40]

PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2, 3,4
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	5, 6
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	6 Paragraph 2
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	6 Paragraph 3
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	7, 8
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	9 Paragraph 1
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	9, 10, 11, Supplementary file 2
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	11 Paragraph 2
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	12 Paragraph 1
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	12 Paragraph 1
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	12 Paragraph 2, Supplementary file 3 and 4
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	N/A

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PRISMA 2009 Checklist

Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	13 Paragraph 2

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	12 Paragraph 2, Supplementary file 3 and 4
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	N/A
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	13 Paragraph 4
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	14 Paragraph 2
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	14 Paragraph 3
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	14–27
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	N/A
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	14 Paragraph 3
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	N/A
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	28-30
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	30, 31
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	31
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	33 Paragraph 4

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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For more information, visit: www.prisma-statement.org

Search strategy developed for Embase (OVID) database

1. infant/ or baby/ or newborn/
2. child/ or boy/ or girl/ or toddler/
3. female/
4. mother/ or adolescent mother/ or mother child relation/
5. caregiver/
6. (infan* or newborn* or new-born* or bab* or neonat* or perinat* or postnat* or kid* or toddler* or child* or young child* or p?ediatric* or mother* or female* or wom#n or caregiver* or care giver* or carer* or maternal).ti,ab.
7. 1 or 2 or 3 or 4 or 5 or 6
8. breast feeding education/
9. health education/ or childbirth education/ or nutrition education/ or parenting education/
10. health literacy/
11. health promotion/ or public health campaign/
12. counseling/ or e-counseling/ or family counseling/ or nutritional counseling/ or parent counseling/ or peer counseling/
13. ((breastfeed* adj2 (promot* or support* or education)) or (breast feed* adj2 (education or promot* or support)) or (health adj2 (education or literacy or promotion)) or ((nutrition* or food* or parent*or mother* or maternal) adj2 education) or ((nutrition* or parent or telephone) adj2 counsel*)).ti,ab.
14. 8 or 9 or 10 or 11 or 12 or 13
15. health behavior/ or health belief/ or health belief model/
16. behavior change/
17. interpersonal communication/
18. social change/
19. social behavior/
20. social network/ or online social network/
21. peer group/
22. advocacy group/
23. social marketing/
24. public health campaign/ or public health message/
25. mass communication/ or mass medium/ or public opinion/ or publication/ or radio/ or social media/ or telephone/ or television/ or text messaging/ or videoconferencing/ or wireless communication/
26. internet/ or web-based intervention/
27. mobile phone/ or smartphone/
28. ((health adj2 (behavio?r or promot* behavio?r or related behavio?r)) or (behavio?r* adj2 (change or centered or change communication)) or communication or interpersonal communication or (information education and communication) or IEC or BCC or (social adj2 (change* or movement or mobiliz* or behavio?r or network* or market* or media)) or peer group* or advocacy or advocacy group* or (mass adj2 (communication or media))

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or print media or (mobile adj2 (phone or health)) or mHealth or eHealth or electronic health or internet or radio or television or text messag* or SMS).ti,ab.

29. 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24 or 25 or 26 or 27 or 28
30. home visit/
31. community care/ or community integration/ or community program/ or food assistance/ or relief work/
32. preventive health service/
33. health auxiliary/
34. (home visit or (community adj2 (program* or project* or intervention*or engagement* or leader or mobilization or gathering* or care service* or health action* or health service* or health care or role-play or health worker* or health aid* or health practitioner*)) or baby friendly community initiative* or baby-friendly community initiative* or demonstration* or cooking demonstration* or model breastfeeding communit* or auxiliary health worker* or health extension worker*).ti,ab.
35. 30 or 31 or 32 or 33 or 34
36. child health care/ or maternal child health care/
37. early childhood intervention/
38. maternal care/ or rooming in/
39. newborn care/ or kangaroo care/ or newborn assessment/ or newborn monitoring/
40. prenatal care/
41. perinatal care/
42. postnatal care/
43. (((matern* or antenatal or prenatal or peri-natal or delivery or birth or home or neonatal or newborn or postnatal or postpartum or Kangaroo) adj2 care) or ((matern* or child or p?ediatric) adj2 health service*) or rooming-in-care or newborn nursery or essential nutrition action* or (home adj2 (visit* or help)) or homecare or caregiver* contact or baby-friendly or baby-friendly hospital initiative or BFHI or ANC or PNC or skin-to-skin contact or skin to skin contact or kangarooing or kangaroo mother care).ti,ab.
44. 36 or 37 or 38 or 39 or 40 or 41 or 42 or 43
45. comparative effectiveness/
46. evaluation study/
47. health program/
48. program evaluation/ or "program cost effectiveness"/ or program effectiveness/ or program impact/
49. experimental study/
50. quasi experimental study/
51. randomized controlled trial/
52. clinical trial/
53. controlled clinical trial/
54. intervention study/
55. (effective* or impact* or evaluation or program* evaluation or project* or health project* or ((health or health care or healthcare) adj2 program*) or ((evaluation or experimental or interventional or quasi-experimental or quasi experimental or time-series or time series or random* controlled or cluster random*) adj2 stud*) or random* controlled design or

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random* controlled trial or RCT or random* clinical trial or cluster random* trial or cluster random* design or control* or placebo or comparison or ((usual or routine or standard) adj2 care)).ti,ab.

56. 45 or 46 or 47 or 48 or 49 or 50 or 51 or 52 or 53 or 54 or 55
57. 14 or 29 or 35 or 44 or 56
58. breast feeding/ or breast milk expression/
59. infant feeding/ or bottle feeding/
60. complementary feeding/
61. child nutrition/
62. weaning/
63. feeding/
64. infant nutrition/
65. baby food/
66. nutritional parameters/ or diet composition/ or estimated daily intake/ or food composition/ or food quality/ or food quantity/ or meal frequency/
67. diet/ or healthy diet/
68. supplementation/ or diet supplementation/
69. breast milk/
70. (breastfe* or early initiation of breast feed* or initiation of breast feed* or timely initiation of breast feed* or exclusive breast feed* or EBF or ((breast or complementary or child or infant or young child) adj2 fe*) or wean* or breast milk substitute or introduction of complementary food* or ((solid or soft or complementary or baby) adj2 food*) or IYCF or meal* or diet* diversity or meal* frequen* or MDD or MMF or MAD or ((child or infant) adj2 nutrition) or ((nutrient or nutrition* or diet*) adj2 supplement*) or ((breast or human or woman or mother or maternal) adj2 milk) or lactation or nursing).ti,ab.
71. 58 or 59 or 60 or 61 or 62 or 63 or 64 or 65 or 66 or 67 or 68 or 69 or 70
72. Ethiopia/
73. limit 72 to (human and english language and yr="2000 - 2021")
74. Ethiopia.ti,ab.
75. limit 74 to (human and english language and yr="2000 - 2021")
76. 73 or 75
77. 7 and 57 and 71 and 76

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▼ **Search History** (77) View Saved

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	1	infant/ or baby/ or newborn/	1201159	Advanced	Display Results | More	
	2	child/ or boy/ or girl/ or toddler/	2128760	Advanced	Display Results | More	
	3	female/	10333062	Advanced	Display Results | More	
	4	mother/ or adolescent mother/ or mother child relation/	135425	Advanced	Display Results | More	
	5	caregiver/	89398	Advanced	Display Results | More	
	6	(infan* or newborn* or new-born* or bab* or neonat* or perinat* or postnat* or kid* or toddler* or child* or young child* or p?ediatric* or mother* or female* or wom#n or caregiver* or care giver* or carer* or maternal).ti,ab.	6667975	Advanced	Display Results | More	
	7	1 or 2 or 3 or 4 or 5 or 6	13403899	Advanced	Display Results | More	
	8	breast feeding education/	2725	Advanced	Display Results | More	
	9	health education/ or childbirth education/ or nutrition education/ or parenting education/	112354	Advanced	Display Results | More	
	10	health literacy/	13474	Advanced	Display Results | More	
	11	health promotion/ or public health campaign/	103110	Advanced	Display Results | More	
	12	counseling/ or e-counseling/ or family counseling/ or nutritional counseling/ or parent counseling/ or peer counseling/	90024	Advanced	Display Results | More	
	13	((breastfeed* adj2 (promot* or support* or education)) or (breast feed* adj2 (education or promot* or support)) or (health adj2 (education or literacy or promotion)) or ((nutrition* or food* or parent*or mother* or maternal) adj2 education) or ((nutrition* or parent or telephone) adj2 counsel*)).ti,ab.	127005	Advanced	Display Results | More	
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	15	health behavior/ or health belief/ or health belief model/	82136	Advanced	Display Results | More	
	16	behavior change/	40062	Advanced	Display Results | More	
	17	interpersonal communication/	175289	Advanced	Display Results | More	
	18	social change/	14822	Advanced	Display Results | More	
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	20	social network/ or online social network/	19132	Advanced	Display Results | More	
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	22	advocacy group/	937	Advanced	Display Results | More	
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	25	mass communication/ or mass medium/ or public opinion/ or publication/ or radio/ or social media/ or telephone/ or television/ or text messaging/ or videoconferencing/ or wireless communication/	327907	Advanced	Display Results | More	
	26	internet/ or web-based intervention/	114097	Advanced	Display Results | More	
	27	mobile phone/ or smartphone/	33761	Advanced	Display Results | More	
	28	((health adj2 (behavio?r or promot* behavio?r or related behavio?r)) or (behavio?r* adj2 (change or centered or change communication)) or communication or interpersonal communication or (information education and communication) or	667696	Advanced	Display Results | More	

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		IEC or BCC or (social adj2 (change* or movement or mobiliz* or behavio?r or network* or market* or media)) or peer group* or advocacy or advocacy group* or (mass adj2 (communication or media)) or print media or (mobile adj2 (phone or health)) or mHealth or eHealth or electronic health or internet or radio or television or text messag* or SMS).ti,ab.				
☐	29	15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24 or 25 or 26 or 27 or 28	1270028	Advanced	Display Results | More	
☐	30	home visit/	3581	Advanced	Display Results | More	
☐	31	community care/ or community integration/ or community program/ or food assistance/ or relief work/	68061	Advanced	Display Results | More	
☐	32	preventive health service/	29982	Advanced	Display Results | More	
☐	33	health auxiliary/	7769	Advanced	Display Results | More	
☐	34	(home visit or (community adj2 (program* or project* or intervention*or engagement* or leader or mobilization or gathering* or care service* or health action* or health service* or health care or role-play or health worker* or health aid* or health practitioner*)) or baby friendly community initiative* or baby-friendly community initiative* or demonstration* or cooking demonstration* or model breastfeeding communit* or auxiliary health worker* or health extension worker*).ti,ab.	167476	Advanced	Display Results | More	
☐	35	30 or 31 or 32 or 33 or 34	261548	Advanced	Display Results | More	
☐	36	child health care/ or maternal child health care/	39892	Advanced	Display Results | More	
☐	37	early childhood intervention/	2914	Advanced	Display Results | More	
☐	38	maternal care/ or rooming in/	19707	Advanced	Display Results | More	
☐	39	newborn care/ or kangaroo care/ or newborn assessment/ or newborn monitoring/	16524	Advanced	Display Results | More	
☐	40	prenatal care/	44245	Advanced	Display Results | More	
☐	41	perinatal care/	14726	Advanced	Display Results | More	
☐	42	postnatal care/	7895	Advanced	Display Results | More	
☐	43	(((matern* or antenatal or prenatal or peri-natal or delivery or birth or home or neonatal or newborn or postnatal or postpartum or Kangaroo) adj2 care) or ((matern* or child or p?ediatric) adj2 health service*) or rooming-in-care or newborn nursery or essential nutrition action* or (home adj2 (visit* or help)) or homecare or caregiver* contact or baby-friendly or baby-friendly hospital initiative or BFHI or ANC or PNC or skin-to-skin contact or skin to skin contact or kangarooing or kangaroo mother care).ti,ab.	165401	Advanced	Display Results | More	
☐	44	36 or 37 or 38 or 39 or 40 or 41 or 42 or 43	262760	Advanced	Display Results | More	
☐	45	comparative effectiveness/	102083	Advanced	Display Results | More	
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☐	55	(effective* or impact* or evaluation or program* evaluation or project* or health project* or ((health or health care or healthcare) adj2 program*) or ((evaluation or experimental or interventional or	1445361	Advanced	Display Results | More	

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		quasi-experimental or quasi experimental or time-series or time series or random* controlled or cluster random*) adj2 stud*) or random* controlled design or random* controlled trial or RCT or random* clinical trial or cluster random* trial or cluster random* design or control* or placebo or comparison or ((usual or routine or standard) adj2 care)).ti,ab.				
☐	56	45 or 46 or 47 or 48 or 49 or 50 or 51 or 52 or 53 or 54 or 55	12110292	Advanced	Display Results | More	
☐	57	14 or 29 or 35 or 44 or 56	13299979	Advanced	Display Results | More	
☐	58	breast feeding/ or breast milk expression/	58544	Advanced	Display Results | More	
☐	59	infant feeding/ or bottle feeding/	12428	Advanced	Display Results | More	
☐	60	complementary feeding/	1220	Advanced	Display Results | More	
☐	61	child nutrition/	17800	Advanced	Display Results | More	
☐	62	weaning/	29610	Advanced	Display Results | More	
☐	63	feeding/	63170	Advanced	Display Results | More	
☐	64	infant nutrition/	11539	Advanced	Display Results | More	
☐	65	baby food/	5685	Advanced	Display Results | More	
☐	66	nutritional parameters/ or diet composition/ or estimated daily intake/ or food composition/ or food quality/ or food quantity/ or meal frequency/	38101	Advanced	Display Results | More	
☐	67	diet/ or healthy diet/	276591	Advanced	Display Results | More	
☐	68	supplementation/ or diet supplementation/	123384	Advanced	Display Results | More	
☐	69	breast milk/	31847	Advanced	Display Results | More	
☐	70	(breastfe* or early initiation of breast feed* or initiation of breast feed* or timely initiation of breast feed* or exclusive breast feed* or EBF or ((breast or complementary or child or infant or young child) adj2 fe*) or wean* or breast milk substitute or introduction of complementary food* or ((solid or soft or complementary or baby) adj2 food*) or IYCF or meal* or diet* diversity or meal* frequen* or MDD or MMF or MAD or ((child or infant) adj2 nutrition) or ((nutrient or nutrition* or diet*) adj2 supplement*) or ((breast or human or woman or mother or maternal) adj2 milk) or lactation or nursing).ti,ab.	726729	Advanced	Display Results | More	
☐	71	58 or 59 or 60 or 61 or 62 or 63 or 64 or 65 or 66 or 67 or 68 or 69 or 70	1154054	Advanced	Display Results | More	
☐	72	Ethiopia/	20234	Advanced	Display Results | More	
☐	73	limit 72 to (human and english language and yr="2000 - 2021")	14867	Advanced	Display Results | More	
☐	74	Ethiopia.ti,ab.	19712	Advanced	Display Results | More	
☐	75	limit 74 to (human and english language and yr="2000 - 2021")	14567	Advanced	Display Results | More	
☐	76	73 or 75	16095	Advanced	Display Results | More	
☐	77	7 and 57 and 71 and 76	846	Advanced	Display Results | More	

Save | Remove | Combine with: AND | OR

Save All | Edit | Create RSS | View Saved | Email All Search History | Copy Search History Link | Copy Search History Details

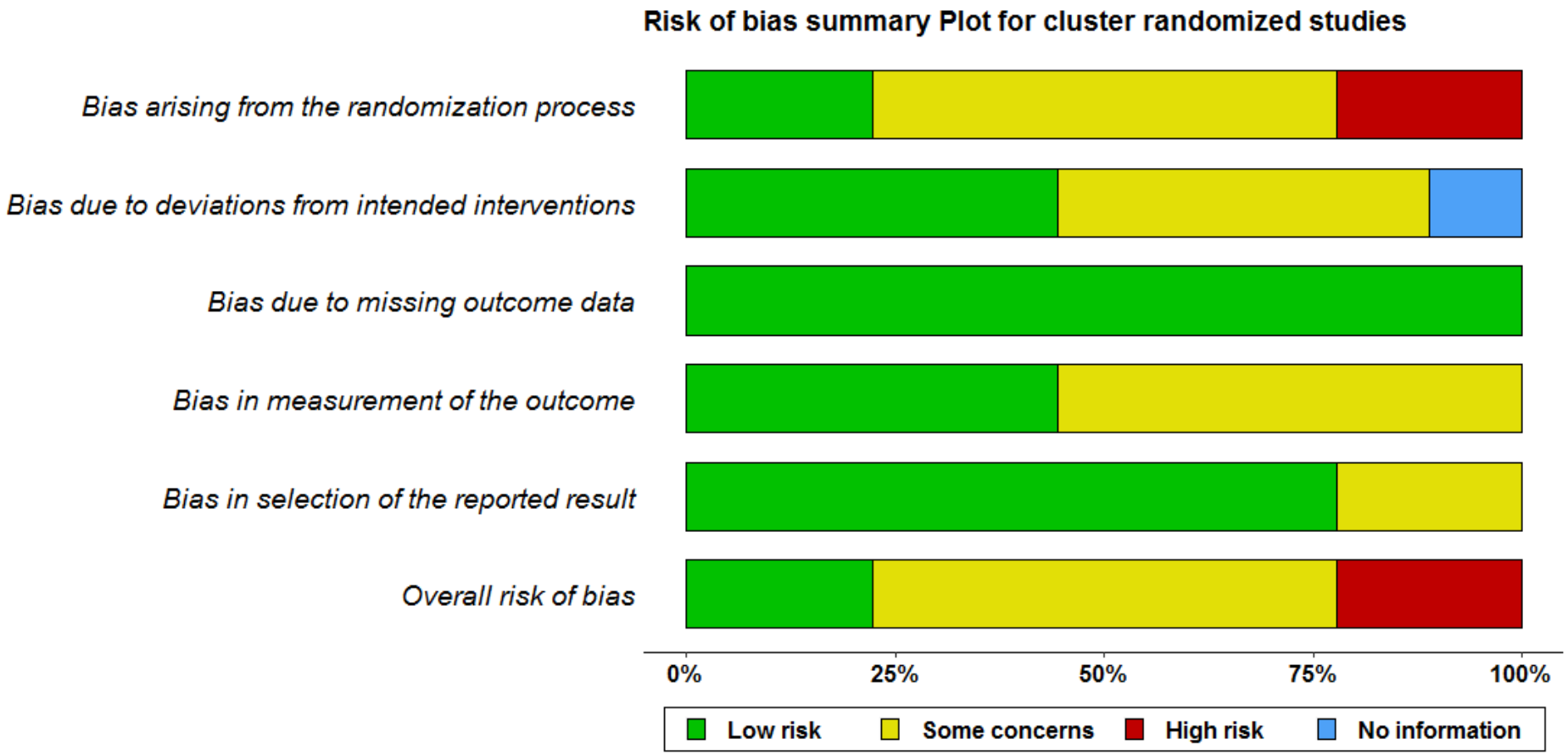
Basic Search | Find Citation | Search Tools | Search Fields | **Advanced Search** | Multi-Field Search

1 Resource selected | Hide | Change

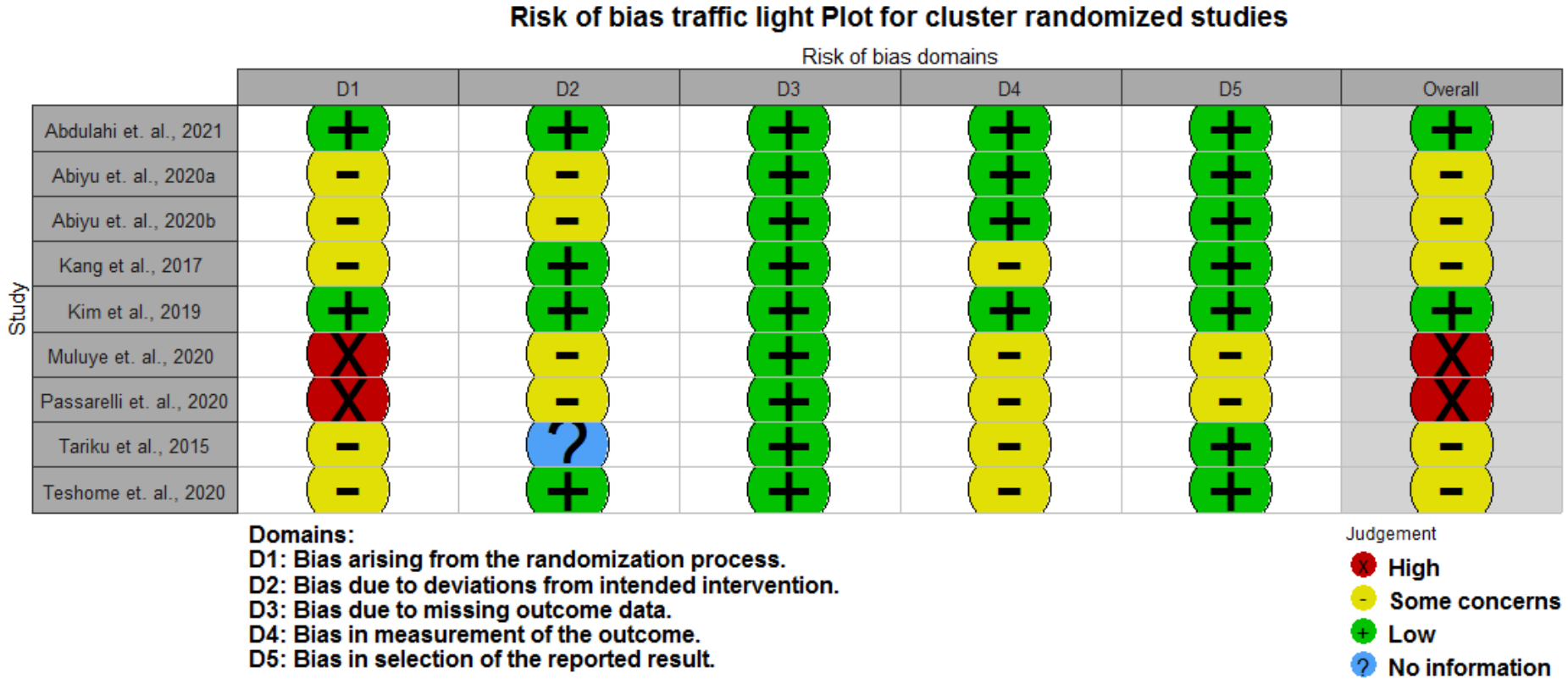
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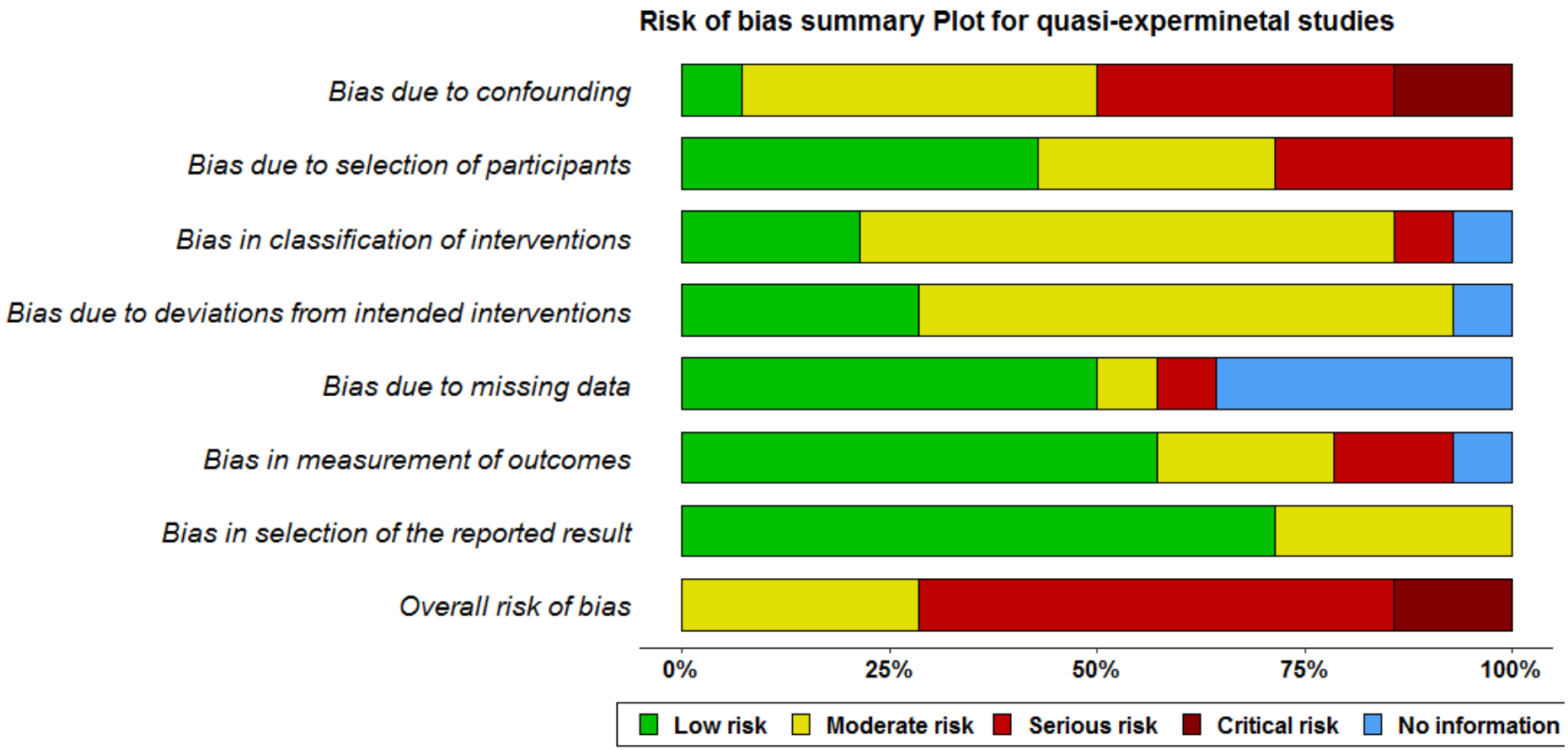
Enter keyword or phrase (* or $ for truncation) ◉ **Keyword** ○ Author ○ Title ○ Journal Search



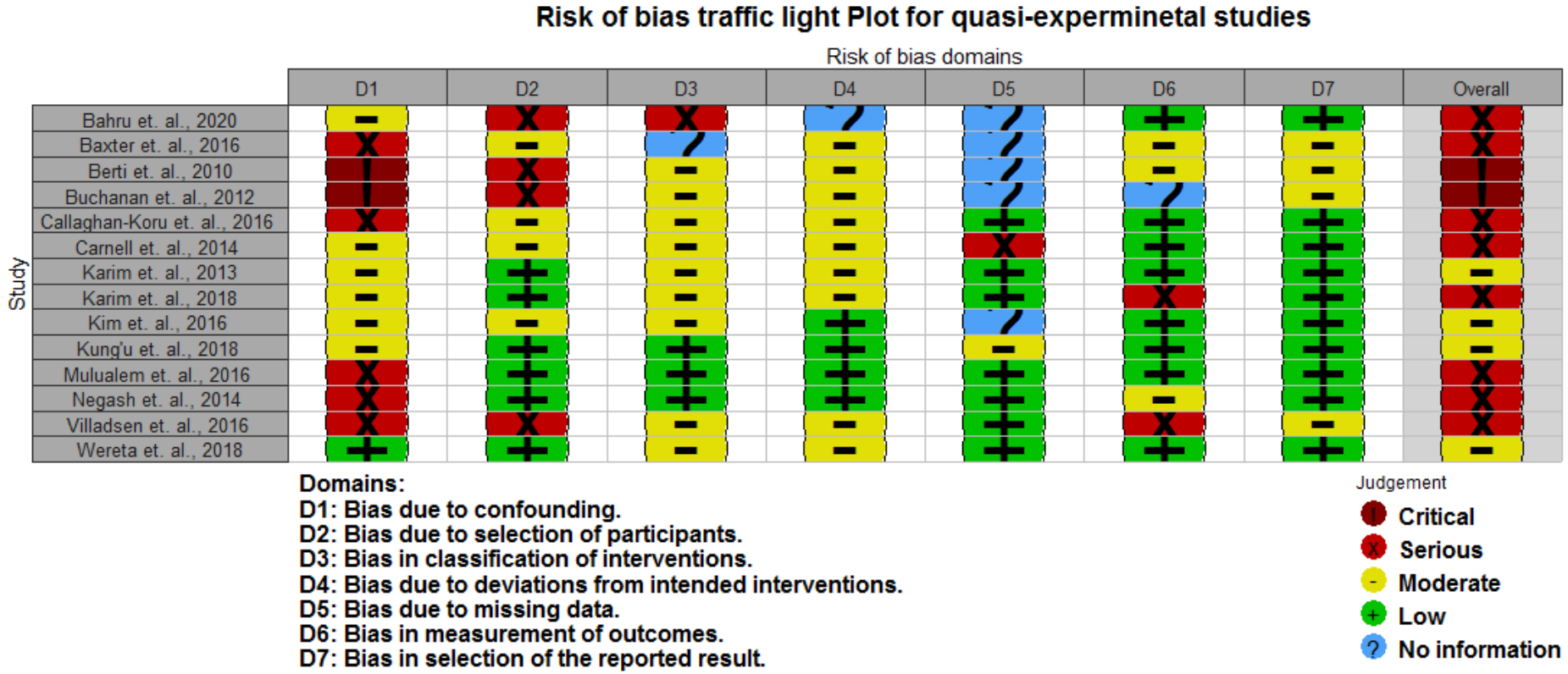
Risk of bias summary plot for cluster randomised studies: review authors' judgements about each risk of bias item as percentages



Risk of bias traffic light plot for cluster randomised studies: review authors' judgements about each risk of bias item for each studies



Risk of bias summary plot for quasi experimental studies: review authors' judgements about each risk of bias item presented as percentages



Risk of bias traffic light plot for quasi-experimental studies: review authors' judgements about each risk of bias item for each included studies